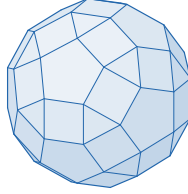


Non-Rupertness of the Rhombicosidodecahedron

Bence Hervay



Abstract

Can a solid pass through a straight tunnel drilled through an identical copy of itself? Although it may seem counterintuitive, it turns out that a cube can pass through itself in this way: John Wallis showed this in 1693, settling a wager of Prince Rupert of the Rhine, after whom the property is called *Rupert's property*. All five Platonic solids turned out to share it, and for a long time it seemed that every convex polyhedron might. Passages were found for many highly regular solids, but some resisted extensive randomised searches. Steininger and Yurkevich conjectured that the rhombicosidodecahedron (RID), a highly regular Archimedean solid, does not pass through itself, and a recent major result of theirs is a first polyhedron without Rupert's property, the Noperthedron, constructed for the purpose. We prove their conjecture for the RID using a computer-assisted branch-and-bound elimination, which covers the entire symmetry-reduced configuration space with 192,696 regions, each verified with exact arithmetic in $\mathbb{Q}(\sqrt{5})$ to contain no passage. The RID is thereby **the first Archimedean solid proved not to be Rupert**, and the first such polyhedron that was not constructed for the purpose. Our main novel addition is the *zoom lemma*. The real obstacle is formed by the configurations in which the two copies touch without either passing or sticking out: no strict inequality holds there, so subdivision alone can never eliminate their neighbourhoods. Near a set of configurations known to contain no passage, the zoom lemma divides a necessary inequality exactly by the distance from that set, so that one exact sign test eliminates a whole neighbourhood, touching configurations included; the same lemma handles every such case of the proof.

Contents

1	Introduction	2
1.1	Rupert's property	2
1.2	Previous approaches and the remaining difficulty	3
1.3	Overview of the proof	6
2	Geometric formulation	7
2.1	The rhombicosidodecahedron	7
2.2	Projection containment	7
2.3	Parametrisation of the configuration space	8
2.4	The margin function and <i>fit</i> / <i>poke</i> / <i>touch</i>	10
3	Symmetry-reduced domain	12
3.1	Symmetries preserve the projection problem	12
3.2	Rotation and reflection reduction	12
3.3	The reduced domain	15
4	Branch and bound elimination in the reduced domain	16
4.1	Boxes, subdivision, and elimination	16
4.2	Component hierarchy	17
4.3	<i>Domain</i> component: skipping boxes outside the reduced domain	18
5	Reduction to exact arithmetic	20
5.1	Exact representation of field elements with integers	20
5.2	Bernstein coefficients	21
6	Global component	23
6.1	Eliminating <i>poke</i> configurations	23
6.2	Robustly detecting potential Rupertness	27
6.3	Range and insufficiency for <i>touch</i> configurations	28

7	<i>Local component</i>	30
7.1	Geometry of <i>aligned</i> configurations	30
7.2	Eliminating <i>aligned</i> configurations	31
7.3	The zoom lemma	32
7.4	Face zooms and the <i>Local</i> component	34
7.5	Range and exotic viewing directions	34
8	<i>Square and pentagon configurations</i>	36
8.1	Eliminating the <i>square</i> configuration	39
8.2	Eliminating the <i>pentagon</i> configuration	40
8.3	A neighbourhood of all aligned configurations	42
9	<i>Arc configurations</i>	42
9.1	Geometry of <i>non-aligned touch</i> configurations	42
9.2	Eliminating <i>arc</i> configurations	45
10	<i>Endpoint and crossing configurations</i>	46
10.1	Eliminating the <i>endpoint</i> configuration	46
10.2	Eliminating the <i>crossing</i> configuration	47
10.3	The arcs together	50
11	<i>Generating and verifying the certificate</i>	50
11.1	Complete coverage and the main theorem	50
11.2	Implementation, execution, and reproducibility	51
12	<i>Remarks and further directions</i>	52
12.1	Relation to previous work	52
12.2	Extensions and possible automation	53
12.3	History of this work	53
A	<i>Symmetry data and checks</i>	54
A.1	Quaternion families and the full symmetry group	54
A.2	Reflection normals and the viewing cone	54
A.3	Selected quaternions for the rotation comparisons	55
B	<i>Zoom covers</i>	55

1 Introduction

1.1 Rupert's property

Can a cube pass through a hole in another cube of the same size? The earliest known written account of this problem is by Wallis [1693, pp. 470–471], who records that Prince Rupert of the Rhine wagered that it could. Wallis gave a geometric solution to this apparently impossible puzzle: suitably orienting the two cubes allows one to slide, without turning, through a straight tunnel cut in the other, with positive clearance. Thus the cube has *Rupert's property*, or, equivalently, the cube is *Rupert*. We call the moving body the *plug* and the body through which it moves the *hole*.

It turns out that even a somewhat larger cube can pass through. The limiting ratio of the plug's edge length to the hole's is

$$\frac{3\sqrt{2}}{4} \approx 1.060660172.$$

The construction giving this ratio was found by Pieter Nieuwland and published posthumously by van Swinden [1816, pp. 608–610]. Its optimality when the plug moves along a straight line without turning was proved by Bezdek et al. [2021]. This limiting scale factor is called the cube's *Nieuwland number*; the same notion applies to other polyhedra.

The same question applies to any convex polyhedron: can a congruent plug pass through it in this way? Jerrard et al. [2017] suggested that every convex polyhedron might be Rupert, and Chai et al. [2018] stated this as a conjecture the following year. A few years later, Steininger and Yurkevich [2023] conjectured that the rhombicosidodecahedron (RID), a highly regular and symmetric Archimedean solid with 60 vertices, was an exception. A polyhedron without Rupert's property is called *Nopert*, a term coined by Tom Murphy VII, also known as Tom 7, combining *Nope* and *Rupert* [Murphy, 2025, p. 352]. Steininger and Yurkevich [2026] subsequently constructed the first convex polyhedron proved to be Nopert, the *Nopertedron*, but their proof did not settle the RID [Steininger and Yurkevich, 2026, Section 9.1]. Here we settle the RID's case with new geometric arguments that handle its exotic cases.

Theorem 1.1. *The rhombicosidodecahedron is Nopert.*

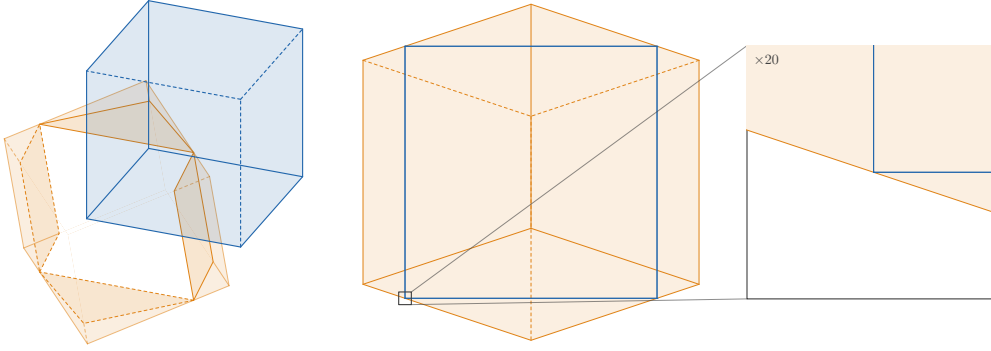


Figure 1: Nieuwland's passage for the cube. Left: a unit cube (orange, faint) with the tunnel cut out along the direction $(2, 2, -1)$, and a cube of edge $3\sqrt{2}/4$ (blue) leaving it; only four wedges of the unit cube remain. Middle: both shadows along the direction of motion, a square inside a hexagon, the hole's shadow tinted. Right: a corner of the square, enlarged: it lies on the boundary of the hexagon, so no larger cube passes. In this and the later drawings, in space and in projection, the edges on the far side of each solid are dashed.

We call a choice of the plug's and hole's orientations, their relative displacement and the direction of motion a *configuration*. The proof is computer-assisted. We divide the configuration space into 192,696 regions, each resolved using exact arithmetic calculations. The final argument is given in Section 11.1.

The key idea. Most configurations are easy to rule out: some vertex of the plug sticks out of the tunnel, and it still does after any small change of the configuration, so a whole region around the configuration can be eliminated at once. The obstacle is formed by the configurations in which the plug touches the walls of the tunnel without sticking out. The simplest example is a plug turned exactly like the hole: the two copies then line up, and so do their shadows. Every strict inequality that could exclude a fit vanishes at such a configuration, so no region containing it can be eliminated by testing inequalities, however finely the regions are subdivided. Yet the plug does stick out once it is turned slightly. Suppose, for instance, that turning it by a small angle $x \geq 0$ moves some vertex outward by the amount $p(x, y) = x(1 + x - y)$, where $y \in [0, 1/2]$ stands for the viewing direction. A sign test of p must fail on any region containing $x = 0$, where p vanishes. The factor x , however, is explicit: the quotient $p/x = 1 + x - y$ is at least $1/2$ throughout, one sign test shows it, and the configurations with $x = 0$ are already known not to be fits.

We turn this observation into a single lemma, the *zoom lemma* (Lemma 7.3). Around a set of configurations known to contain no fit, a *zoom* is a polynomial change of coordinates that magnifies the neighbourhood of that set, so that the distance from it becomes a coordinate that can be divided out exactly. After the division, one exact sign test settles a whole region, including the touching configurations themselves. The Local and Exotic components are instances of this lemma, whose zooms differ only in the set around which they magnify and in how many distances they divide out; the Domain and Global tests are the same idea with nothing divided out. The four *components* of the proof, each a method for eliminating whole regions of configurations, are listed in Table 1.

Component	Eliminates regions	Divides out	Section
Domain	outside the symmetry-reduced domain	nothing	4.3
Global	where some plug vertex sticks out	nothing	6
Local	around the plug turned like the hole	the size of the rotation	7
Exotic	around the other touching configurations	one or two distances	8–10

Table 1: The four components of the proof.

1.2 Previous approaches and the remaining difficulty

When the plug can pass through the hole with positive clearance, we call this a *fit* configuration. To prove that a polyhedron is Rupert, even a single fit configuration is enough, though finding one can be

difficult. Examples include the construction of Fredriksson [2024] for the triakis tetrahedron and the proof of Renshaw [2026b] in the Lean proof assistant that Tom 7’s candidate #214 is Rupert. However, to prove that a polyhedron is Nopert, we must rule out every possible fit configuration across the entire continuous space of orientations, displacements and directions of motion. An unsuccessful search over finitely many configurations, such as those sampled by the randomized algorithms of Steininger and Yurkevich [2023], however extensive, is insufficient to rule out all infinitely many configurations.

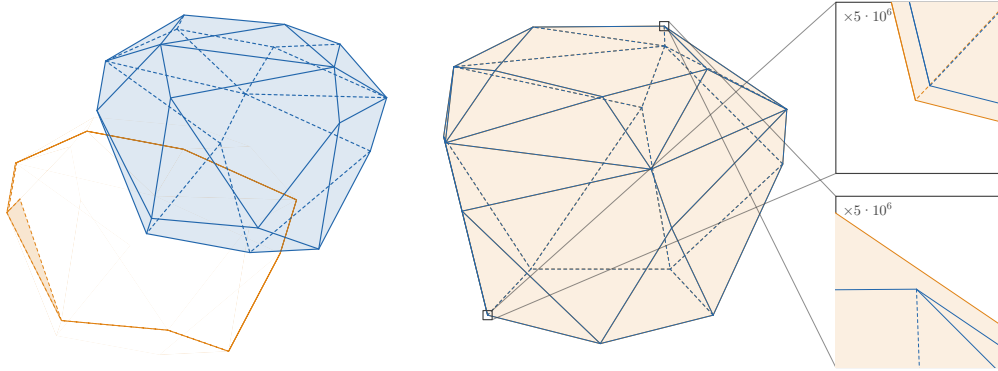


Figure 2: Tom 7’s candidate #214 passes through itself, in the configuration of Renshaw [2026b]. Left: the tunnel removes all of the hole (orange) but a thin rim; the plug (blue) leaves it. Middle: the two shadows, indistinguishable at this scale. Right: the two places of least clearance, enlarged five million times; the gap is about $1.8 \cdot 10^{-8}$, for a solid of circumradius 1.

Constructive results and numerical searches. Jerrard et al. [2017] completed the proof that all five Platonic solids are Rupert, building on earlier results for the cube, tetrahedron and octahedron. Chai et al. [2018] examined the Archimedean solids and proved Rupertness for eight of the thirteen. Subsequent work brought the total to ten, leaving the snub cube, the RID and the snub dodecahedron unresolved at that stage [Steininger and Yurkevich, 2023].

Computational methods enabled more extensive searches for fit configurations. Steininger and Yurkevich [2023] developed randomized searches for such configurations and improved lower bounds for Nieuwland numbers by perturbing known examples. Fredriksson [2024] used nonlinear optimization of the viewing directions from many starting points, and Gosain and Grimmer [2025] combined nonsmooth local optimization with high-precision calculations of scale factors.

The triakis tetrahedron, a Catalan solid with eight vertices, illustrates how a fit configuration can escape a large search. It was among the solids left unresolved by Steininger and Yurkevich [2023] in their randomized searches. Fredriksson [2024] then found a configuration giving a reported lower bound of 1.000004 for its Nieuwland number, corresponding to a clearance of only about four parts per million in linear scale. This shows how little room a fit configuration may have and gives a concrete example of a configuration missed by earlier randomized searches. More extensive searches and higher numerical precision can improve the chances of finding such configurations, but failure to find one still does not prove nonexistence.

Proving non-Rupertness. Steininger and Yurkevich [2026] prove that the Nopertshedron is Nopert by combining arguments eliminating whole regions of configurations with special treatment of boundary cases. They also explain why their local theorem is insufficient for the RID, identifying an exotic viewing direction and stating that its neighbourhood can be treated using polynomial inequalities, while reporting further difficult regions [Steininger and Yurkevich, 2026, Section 9.1].

The RID was a promising candidate for this research because of its highly regular structure and many symmetries, including central symmetry: reflecting it through its centre leaves the RID unchanged. Central symmetry also simplifies the problem, as explained in Section 2.2. For such a regular solid, an extremely narrow, overlooked region of fit configurations seemed less plausible to us. Despite this intuition, we needed a way to recognise and rigorously verify a fit configuration if one existed contrary to our expectations. We explain how to handle this possibility in Section 6.2.

Steininger and Yurkevich [2023] also give an algorithm guaranteed to decide whether a polyhedron with algebraic coordinates is Rupert, which they describe as too computationally expensive for practical

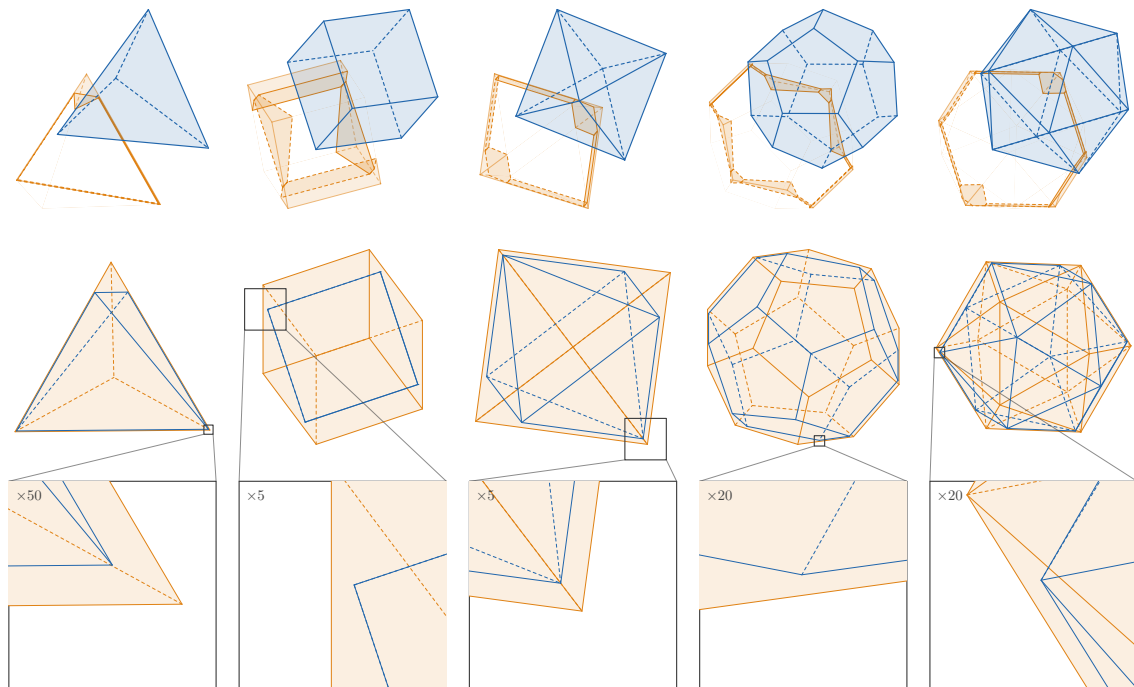


Figure 3: Passages of the tetrahedron, cube, octahedron, dodecahedron and icosahedron, found by a numerical search. Top: the hole (orange, faint) with the tunnel cut out, and the plug (blue) leaving it. Middle: the shadows along the direction of motion. Bottom: the place of least clearance, enlarged. The plugs could be enlarged by factors of about 1.0145, 1.0607, 1.0607, 1.0108 and 1.0108 and still pass; only the tetrahedron, which is not centrally symmetric, needs its plug's shadow shifted.

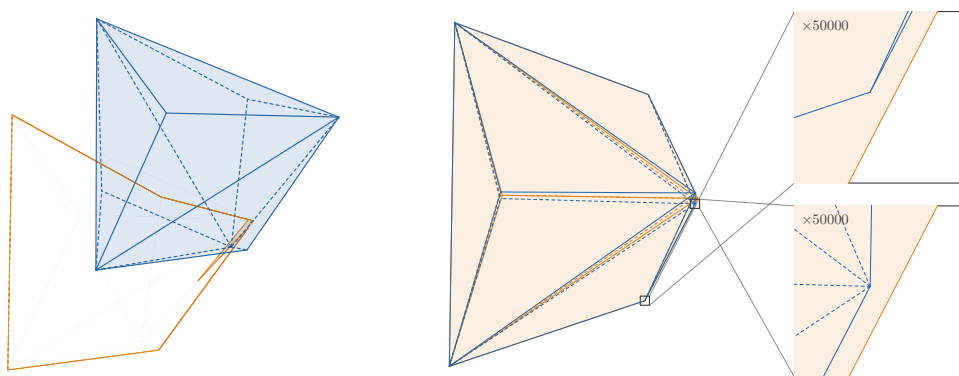


Figure 4: The triakis tetrahedron passes through itself [Fredriksson, 2024]. Drawn as in Figure 2, with the two places of least clearance enlarged 50,000 times; the gap is about $3.3 \cdot 10^{-6}$ at circumradius $5\sqrt{6}/8 \approx 1.53$. The parameters are Fredriksson's, refined slightly by a local numerical minimisation that changes none of them by more than $2.3 \cdot 10^{-5}$.

use. Our aim is a feasible computer-assisted proof for the RID that addresses the special configurations where ordinary elimination methods break down.

1.3 Overview of the proof

For *translative motion*, meaning motion without turning, the problem has a useful two-dimensional description. Project the plug and hole perpendicularly onto a plane normal to the direction of motion, viewing their projections as filled *shadows*. A fit configuration is one in which the plug’s shadow lies entirely in the interior of the hole’s shadow. For centrally symmetric convex bodies, relative translations can be removed: if any relative displacement gives fit, coincident projected centres give fit as well [Steininger and Yurkevich, 2023]. We give the argument in Section 2.2 so that this reduction requires no background from the Rupert literature.

We may then fix the hole and describe a configuration by two parameters for the viewing direction and three for the plug’s relative rotation. Our coordinates make the relevant expressions rational functions of these five parameters. Clearing denominators of known positive sign reduces the comparisons to polynomial inequalities. In contrast, the angular parametrisation in the Noperthedron proof of Steininger and Yurkevich [2026, Section 6.1] requires rigorously bounded approximations to sine and cosine. The RID’s symmetries allow us to restrict attention to a small bounded region of these coordinates while retaining at least one representative of every remaining configuration. The parametrisation and the proof that these representatives suffice are developed in Sections 2 and 3.

Eliminating configuration regions. Within that region, the procedure attempts to eliminate an entire parameter region at once. Only representatives in the symmetry-reduced domain need to be eliminated; regions entirely outside it can be skipped. We call each method for certifying such an elimination a *component*. When no available component succeeds, the region is bisected and both halves are retained for further examination. Failure of a test is inconclusive: it neither establishes a fit configuration nor gives a reason to discard the region. Together, the eliminated regions and their checked reasons form a *certificate*. Section 4 explains the subdivision and what is required of such a certificate; the exact calculations behind the checks are developed in Section 5.

Completing the certificate. The remaining difficulty is to make subdivision terminate with a complete certificate. We develop components successively to handle configurations that the preceding ones leave unresolved. First, if part of the plug’s shadow protrudes beyond the hole’s shadow, we call this a *poke* configuration. A sufficiently small change preserves that protrusion, allowing us to eliminate a whole surrounding region. The *Global* component of Section 6 eliminates these cases.

The difficulty arises when the plug’s shadow is contained in the hole’s, but their boundaries meet: we call this a *touch* configuration. For example, in *aligned configurations*, the plug and hole have the same physical orientation, allowing for the RID’s symmetries. The centred bodies then coincide, as do their shadows. There is no protrusion to detect. Further subdivision leaves at least one region containing the same touch configuration, so testing for protrusion alone cannot complete the elimination. We give formal definitions of fit, touch and poke in Section 2.4.

Around aligned configurations, we instead study how the plug’s vertices move when the plug begins to rotate. The inequality expressing outward motion contains the size of the rotation as an explicit factor, which vanishes exactly at the aligned configurations, as in the example above. Section 7 develops the *Local* component from this observation and states the zoom lemma in general (Section 7.3). At two exotic viewing directions, the outward motion itself vanishes to first order, and Section 8 divides out a second distance there.

Surprisingly, even after the aligned configurations have been handled, non-aligned touch configurations remain—what seems to us to be a rare feature of the RID. These families lie in two planes of configurations that contain no fit, because in one direction the plug reaches exactly as far as the hole. The same lemma, now dividing by the distance from these planes, handles them, again with two distances at the end of one family and where the two families cross (Sections 9 and 10). These covers, together with those of Section 8, form the *Exotic* component. With it, the collection of four components eliminates the entire symmetry-reduced configuration space.

All accepted inequalities are checked with exact arithmetic. The final certificate records a reason for each eliminated region. Its verification has two distinct tasks: check that every reason is valid for all required configurations in its region, and check that the regions together cover all those configurations.

Together these establish [Theorem 1.1](#). We unify these requirements and describe the implementation and computation details in [Section 11](#).

2 Geometric formulation

2.1 The rhombicosidodecahedron

The RID is an Archimedean solid: all its faces are regular polygons—20 equilateral triangles, 30 squares and 12 regular pentagons—and it is *vertex-transitive*, meaning that its symmetries can carry any vertex to any other. At each of its 60 vertices, the faces meet in the order triangle, square, pentagon, square; altogether it has 120 edges [[McCooey, n.d.](#)]. It has icosahedral symmetry, which we will use to reduce the configuration domain in [Section 3](#).

For the calculations, we need exact coordinates. The golden ratio $\varphi = (1 + \sqrt{5})/2$, familiar from regular pentagons, makes these particularly simple. Let V consist of all independent sign choices and cyclic coordinate permutations of

$$(1, 1, \varphi^3), \quad (\varphi^2, \varphi, 2\varphi), \quad (2 + \varphi, 0, \varphi^2). \quad (2.1)$$

Here cyclic permutation means replacing (a, b, c) by (b, c, a) or (c, a, b) . The three families contain respectively 24, 24 and 12 distinct points, giving the 60 vertices. Their convex hull, which we denote by $\mathcal{R}$, is the filled RID with edge length two; these are also the coordinates used by [Steininger and Yurkevich \[2023\]](#). Using edge length two keeps the coordinates simple. Rescaling both the plug and hole by the same factor does not change whether they fit.

The sign choices make $V = -V$ and hence $\mathcal{R} = -\mathcal{R}$, expressing central symmetry about the origin. The first family includes the eight corners of $[-1, 1] \times [-1, 1] \times [-\varphi^3, \varphi^3]$, so the origin lies in the interior of $\mathcal{R}$. This will let us compare the two shadows from a common interior point.

The shape also appears outside polyhedral geometry: a modified RID forms the connector balls in Zometool.¹ In the connector’s idealized polyhedral model, the triangular faces have edges φ times as long as those of the pentagonal faces, and rectangles with this same side ratio replace the squares [[Hart and Picciotto, 2001](#), pp. 51, 197].

The golden ratio is present throughout this geometry: the 1,770 pairs of vertices lie at 17 distinct distances, and all but two of their squares are irrational numbers of the form $a + b\sqrt{5}$; the two exceptions are the edge and the diagonal of a square face.

At unit edge length, the regular RID has volume $(60 + 29\sqrt{5})/3 \approx 41.6153$ [[McCooey, n.d.](#)], which rounds to 42.²

2.2 Projection containment

Fix a vector $\mathbf{u} \in \mathbb{R}^3 \setminus \{\mathbf{0}\}$ along the direction of motion. Orthogonal projection onto the plane $\mathbf{u}^\perp$ is the map $\pi_{\mathbf{u}} : \mathbb{R}^3 \rightarrow \mathbf{u}^\perp$ defined by

$$\pi_{\mathbf{u}}(\mathbf{v}) = \mathbf{v} - \frac{\mathbf{v} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}} \mathbf{u}.$$

The image of a body under this map is its *shadow*, including the interior; the boundary of the shadow is its *silhouette*. Since the projection is linear, the shadow of $\mathcal{R}$ is the convex hull of the projected vertices $\pi_{\mathbf{u}}(\mathbf{v})$, for $\mathbf{v} \in V$.

Suppose initially that the hole is $R_H \mathcal{R}$ and the plug is $R_P \mathcal{R} + \mathbf{a}$, where $R_H, R_P \in \text{SO}(3)$ are their rotations about the origin and $\mathbf{a} \in \mathbb{R}^3$ is a displacement. Here $\text{SO}(3)$ is the group of real 3×3 orthogonal matrices with determinant 1. Translating the plug along $\mathbf{u}$ does not change its shadow. Over the whole line of motion, the plug sweeps out the cylinder consisting of all lines parallel to $\mathbf{u}$ through its shadow. The plug can pass through the hole with positive clearance, giving a fit configuration, exactly when

$$\pi_{\mathbf{u}}(R_P \mathcal{R}) + \pi_{\mathbf{u}}(\mathbf{a}) \subseteq \text{int}_{\mathbf{u}^\perp} \pi_{\mathbf{u}}(R_H \mathcal{R}), \quad (2.2)$$

where the interior is taken in the two-dimensional screen plane [[Steininger and Yurkevich, 2023](#), Proposition 1]. Indeed, strict containment leaves room for a slightly wider cross-section around the plug’s shadow, still inside the hole’s shadow, so the plug can move through with positive clearance. Conversely, a fit requires the entire plug shadow to lie in the interior of the hole shadow. This is also called a *strict*

¹The author’s childhood experience of building with Zometool provided an additional motivation for choosing the RID.

²Also the answer to life, the universe and everything in *The Hitchhiker’s Guide to the Galaxy* [[Adams, 1979](#)].

fit; without qualification, fit always has this meaning. A *weak fit* requires only ordinary containment, allowing the plug's shadow to meet the boundary of the hole's shadow. Such contact is permitted but not required, so weak fit includes both fit and touch configurations. Displacement along $\mathbf{u}$ only changes the plug's starting position, so only the planar displacement $\pi_{\mathbf{u}}(\mathbf{a})$ enters this comparison.

Removing translations. For centrally symmetric bodies, the planar displacement can be discarded. The following short argument is the centering reduction of [Steininger and Yurkevich \[2023, Proposition 2\]](#); it applies equally to weak fit and to a uniformly scaled plug.

Lemma 2.1 (Centering). *Let $P = -P$ and $H = -H$ be convex planar bodies centred at the origin. If a translate $P + \mathbf{b}$ is contained in H , then P is contained in H . If $P + \mathbf{b}$ is contained in $\text{int } H$, then P is contained in $\text{int } H$.*

Proof. Reflecting $P + \mathbf{b} \subseteq H$ through the origin gives $P - \mathbf{b} \subseteq H$. For each $\mathbf{p} \in P$, the points $\mathbf{p} + \mathbf{b}$ and $\mathbf{p} - \mathbf{b}$ belong to H , so their midpoint $\mathbf{p}$ belongs to H by convexity. The same argument works in $\text{int } H$, which is also convex and centrally symmetric. $\square$

Both projected copies of $\mathcal{R}$ are centrally symmetric. Thus, for any fixed orientations and viewing direction, a translated fit exists exactly when the centred plug fits. Replacing P by λP in [Lemma 2.1](#) gives the same conclusion for each scale $\lambda > 0$. We may consequently restrict the search, and the optimization of scale, to coincident projected centres. A displacement can still turn a fit configuration into a touch or poke configuration; the reduction preserves the existence of a fit.

Fixing the hole. Finally, rotating the whole configuration by R_H^{-1} fixes the hole as $\mathcal{R}$ and replaces the plug rotation by $R_H^{-1}R_P$, with the viewing direction rotated along with them. In this fixed-hole frame, we continue to write $\mathbf{u}$ for the viewing vector and R_P for the plug's rotation. From now on, a configuration c is centred and consists of a viewing line together with the plug's rotation relative to this fixed hole. The respective hole and plug shadows are:

$$H_c = \pi_{\mathbf{u}}(\mathcal{R}), \quad P_c = \pi_{\mathbf{u}}(R_P \mathcal{R}).$$

The sign and length of $\mathbf{u}$ do not matter: only its line matters. We are left with two degrees of freedom for the viewing direction and three for the relative rotation. For a polyhedron that is not centrally symmetric, the component of the plug's displacement relative to the hole perpendicular to the viewing direction must generally also be retained. This adds two degrees of freedom, giving seven in total.

2.3 Parametrisation of the configuration space

We choose coordinates that make the projected vertices rational functions of five real parameters. This will allow us to compare their positions using polynomial inequalities that we verify with exact arithmetic. We represent the viewing line by a vector with positive z -coordinate, scaled to

$$\mathbf{u} = (s, t, 1), \quad (s, t) \in \mathbb{R}^2.$$

Thus s and t determine the viewing line. The RID has sufficiently many symmetries that we can choose such a representative with bounded s, t , without needing any viewing direction in the xy -plane. We prove this reduction in [Section 3](#).

Relative rotation. For the relative rotation, we use a nonzero quaternion $q = (\alpha, \mathbf{a}) \neq (0, \mathbf{0})$, consisting of a scalar part $\alpha \in \mathbb{R}$ and a vector part $\mathbf{a} \in \mathbb{R}^3$. We write $\|\cdot\|$ for Euclidean length. A rotation by angle $\theta \in [0, \pi]$ around an axis $\mathbf{n} \in \mathbb{R}^3$ with $\|\mathbf{n}\| = 1$ is represented by the unit quaternion

$$q = (\alpha, \mathbf{a}) = (\cos(\theta/2), \mathbf{n} \sin(\theta/2)).$$

Here $\alpha^2 + \|\mathbf{a}\|^2 = 1$. Nonzero real scalar multiples represent the same rotation. Whenever $\alpha \neq 0$, we may therefore divide by it and write

$$q = (1, \mathbf{r}), \quad \mathbf{r} = \frac{\mathbf{a}}{\alpha} = (r_1, r_2, r_3) \in \mathbb{R}^3.$$

The vector $\mathbf{r}$ is a coordinate for the rotation, not the rotation itself. For $\mathbf{r} \neq \mathbf{0}$, its direction is the rotation axis and its length is $\tan(\theta/2)$ for the angle $0 < \theta < \pi$. The corresponding rotation acts on a spatial vector $\mathbf{v}$ by

$$R_P(\mathbf{r})\mathbf{v} = \frac{(1 - \mathbf{r} \cdot \mathbf{r})\mathbf{v} + 2\mathbf{r}(\mathbf{r} \cdot \mathbf{v}) + 2\mathbf{r} \times \mathbf{v}}{1 + \mathbf{r} \cdot \mathbf{r}}. \quad (2.3)$$

For example, the component parallel to $\mathbf{r}$ stays fixed; on the perpendicular plane the two coefficients are $(1 - \|\mathbf{r}\|^2)/(1 + \|\mathbf{r}\|^2) = \cos \theta$ and $2\|\mathbf{r}\|/(1 + \|\mathbf{r}\|^2) = \sin \theta$. This verifies the axis and angle and fixes the sign of the rotation. At $\mathbf{r} = \mathbf{0}$ the formula gives the identity matrix. In this fixed-hole frame, a configuration is aligned exactly when the relative rotation is a symmetry of $\mathcal{R}$. We will denote a configuration by $c = [s, t, \mathbf{r}] \in \mathbb{R}^2 \times \mathbb{R}^3$, or equivalently $[s, t, (r_1, r_2, r_3)] \in \mathbb{R}^5$, wherever these coordinates are available.

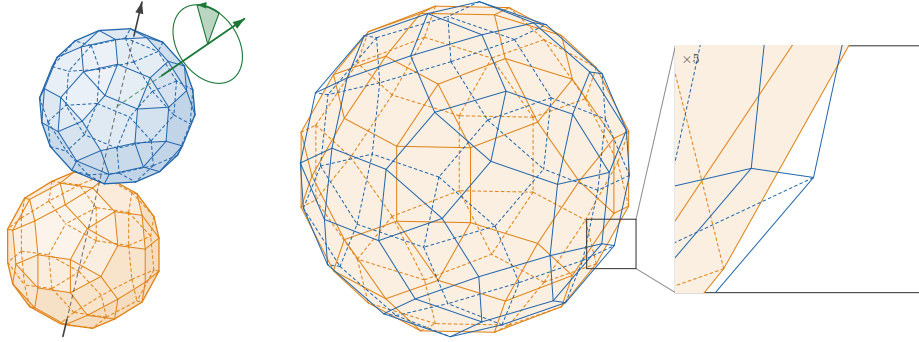


Figure 5: A configuration $c = [s, t, \mathbf{r}]$ with $s = 1/5$, $t = 1/10$ and $\mathbf{r} = (1/4, -1/12, 5/24)$. Left: the hole (orange) and the plug (blue), moved apart along the viewing direction $\mathbf{u} = (s, t, 1)$ (black arrow) so that both can be seen. The plug is turned about the axis $\mathbf{n} = \mathbf{r}/\|\mathbf{r}\|$ (green arrow, from the plug's centre) by the angle $\theta = 2 \arctan \|\mathbf{r}\| \approx 37^\circ$, shaded on a circle about the axis; dashed lines run behind or inside the solids. Middle: the two shadows along $\mathbf{u}$. Right: the plug's shadow sticks out of the hole's (enlarged five times), so this configuration is a poke.

Screen coordinates. There is a similarly simple choice of coordinates on the screen. Define $L_{s,t} : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ by

$$L_{s,t}(v_1, v_2, v_3) = (v_1 - sv_3, v_2 - tv_3). \quad (2.4)$$

Its kernel is precisely the viewing line spanned by $\mathbf{u} = (s, t, 1)$. Consequently, its restriction T to $\mathbf{u}^\perp$ is an invertible linear map from that plane to $\mathbb{R}^2$, and $L_{s,t} = T\pi_{\mathbf{u}}$. In other words, it gives coordinates for the same orthogonal shadow. Although these coordinates can distort lengths and angles, applying the same invertible map to both shadows preserves containment and containment in the interior. It also commutes with uniform scaling, so every comparison of a scaled plug shadow with the hole shadow is preserved.

Using these screen coordinates, we can write the hole and plug shadows as:

$$H_c = L_{s,t}\mathcal{R} = \text{conv}\{L_{s,t}\mathbf{v} : \mathbf{v} \in V\}, \quad P_c = L_{s,t}R_P(\mathbf{r})\mathcal{R} = \text{conv}\{L_{s,t}R_P(\mathbf{r})\mathbf{v} : \mathbf{v} \in V\}. \quad (2.5)$$

Projected hole vertices have coordinates affine in s, t , while projected plug vertices have rational coordinates with the common denominator $d = 1 + \|\mathbf{r}\|^2 \geq 1$. Their coefficients belong to the quadratic field

$$\mathbb{Q}(\sqrt{5}) = \{a + b\sqrt{5} : a, b \in \mathbb{Q}\},$$

because the vertices have coordinates in this field. Each coefficient is represented exactly by the two rational numbers a and b . Multiplying a comparison by d , or by a positive power of d , preserves its sign. This is how the rational expressions give polynomial inequalities without requiring bounds on trigonometric or other transcendental functions. The parameters themselves remain real; the field describes the coefficients of these expressions.

This parametrisation does not represent rotations by 180° (half-turns), whose representing quaternions have scalar part zero. The reduction in Section 3 also provides bounded relative-rotation representatives, including for these rotations. Until that reduction is proved, a statement about all configurations refers to the geometric configurations of Section 2.2, not merely those represented by these parameters.

2.4 The margin function and *fit* / *poke* / *touch*

To measure how close a configuration is to fitting, we compare the two shadows on rays from their common origin. This gives a relative scale: we ask how far the plug shadow extends compared with the hole shadow, rather than measuring an absolute distance between their silhouettes.

For a compact convex shadow C containing the origin in its interior and a unit planar vector ω , define its *radial extent* by

$$\rho_C(\omega) = \max\{\lambda \in \mathbb{R}_{\geq 0} : \lambda\omega \in C\}.$$

Each ray meets C in exactly the segment from the origin to $\rho_C(\omega)\omega$. This extent is positive and finite, and it varies continuously with ω . To see the continuity, a limit of ray endpoints still belongs to C , by compactness. In the other direction, any point strictly between the origin and an endpoint is interior, by convexity and the interior disk about zero; a small change of direction keeps that point inside. Together these observations prevent either an upward or a downward jump in radial extent.

Definition 2.2 (Configuration margin). For a centred configuration c , with hole shadow H_c and plug shadow P_c , define

$$M_{\mathcal{R}}(c) = \max_{\|\omega\|=1} \frac{\rho_{P_c}(\omega)}{\rho_{H_c}(\omega)}. \quad (2.6)$$

The maximum exists because the unit circle is compact and the denominator is positive. The largest ratio controls containment of the whole shadow: it is the smallest factor by which the hole shadow must be scaled to contain the plug shadow.

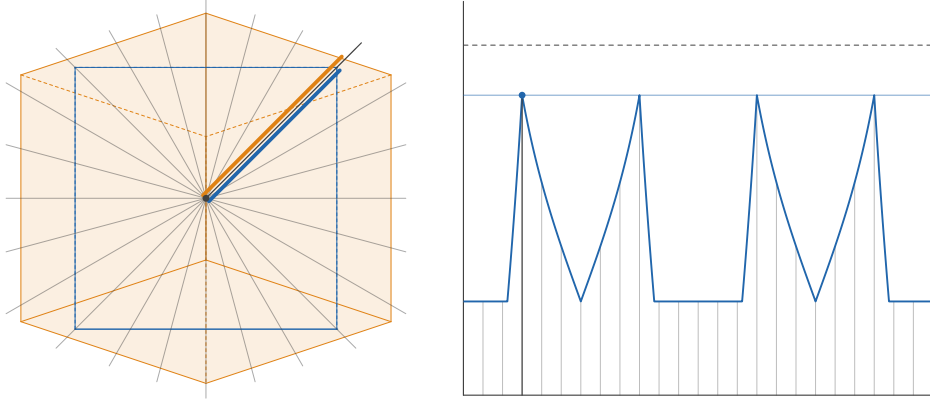


Figure 6: The margin of two congruent cubes $\mathcal{C}$ in the orientations of Figure 1. Left: the hole's shadow (orange) and the plug's (blue) about their common centre, with rays every 15° ; along the ray where the plug's radial extent is largest relative to the hole's, the two extents are drawn side by side. Right: that ratio over a full turn of directions, the grey lines marking the rays on the left. Its maximum, marked, is the margin $M_{\mathcal{C}}(c) = 2\sqrt{2}/3 \approx 0.943$; it lies below the dashed level 1, so c is a fit.

Lemma 2.3 (Margin and containment). For every centred configuration c and every $\lambda \in \mathbb{R}_{>0}$,

$$P_c \subseteq \lambda H_c \iff M_{\mathcal{R}}(c) \leq \lambda, \quad P_c \subseteq \text{int}(\lambda H_c) \iff M_{\mathcal{R}}(c) < \lambda.$$

The margin is unchanged by applying the same invertible linear screen map to both shadows.

Proof. Containment holds exactly when the plug shadow's radial extent is at most λ times the hole shadow's in every direction. For interior containment, every radial comparison is strict. Since the maximum ratio is attained, these comparisons are equivalent to $M_{\mathcal{R}}(c) < \lambda$. Finally, an invertible linear map preserves each relation $P_c \subseteq \lambda H_c$, and hence preserves the least possible λ . $\square$

In particular, the rational screen of Equation (2.4) gives the same margin as orthogonal screen coordinates. We can now express the three configuration types introduced in Section 1 as predicates:

$$\begin{aligned} \text{fit}(c) &\iff M_{\mathcal{R}}(c) < 1, \\ \text{touch}(c) &\iff M_{\mathcal{R}}(c) = 1, \\ \text{poke}(c) &\iff M_{\mathcal{R}}(c) > 1. \end{aligned} \quad (2.7)$$

A touch configuration has a weak fit with at least one boundary point shared by the two shadows. In the direction from the common origin to such a point, the radial extents of the hole and plug shadows match. In particular, aligned configurations have identical shadows and margin one, since their radial extents match for all ω . Note that the silhouettes do not necessarily have to fully coincide in a touch configuration.

Scale and the Nieuwland number. Scaling the plug by $\lambda > 0$ multiplies every numerator in Equation (2.6) by λ . It fits weakly exactly when $\lambda M_{\mathcal{R}}(c) \leq 1$, and strictly exactly when $\lambda M_{\mathcal{R}}(c) < 1$. Thus $1/M_{\mathcal{R}}(c)$ is the maximum scale for weak fit and the supremum of the scales for strict fit.

Optimising the margin over the full configuration space means taking its infimum, which is the reciprocal of the Nieuwland number [Steininger and Yurkevich, 2023, Definition 3]:

$$\nu(\mathcal{R}) = \sup_c \frac{1}{M_{\mathcal{R}}(c)} = \frac{1}{\inf_c M_{\mathcal{R}}(c)}. \quad (2.8)$$

Here c ranges over all centred geometric configurations. The centering lemma shows that allowing translations does not improve this supremum. It is finite: if $\mathcal{R}$ contains a ball of radius $a > 0$ and is contained in one of radius b , both centred at zero, then the orthogonal shadows contain radius- a disks and lie in radius- b disks, giving $a/b \leq M_{\mathcal{R}}(c) \leq b/a$. Aligned configurations give $\nu(\mathcal{R}) \geq 1$. Consequently $\mathcal{R}$ is *Rupert exactly* when $\nu(\mathcal{R}) > 1$, and *Nopert exactly* when $\nu(\mathcal{R}) = 1$. A barely Rupert body has a best margin just below one, or equivalently a Nieuwland number just above one.

The examples in Section 1.2 show how little clearance known fit configurations can have. For the triakis tetrahedron, Fredriksson [2024] reported a lower bound of 1.000004, which Gosain and Grimmer [2025] refined numerically to 1.000004055715, still only about four parts per million above one. For Tom 7's candidate #214, the Lean formalisation of Renshaw [2026b] proves fit, and its accompanying source reports a shadow clearance of approximately 1.8×10^{-8} in the published coordinate units [Renshaw, 2026a,b]. This last value measures a distance rather than a scale factor.

Continuity of the margin. After the compact representative domain is established in Section 3, continuity will also justify attaining the minimum margin there.

A useful geometric picture is that small changes to a configuration produce small changes to its margin, even when the choice of projected vertices and edges forming the silhouette changes discontinuously. For example, a projected vertex can move onto or away from the boundary as the viewing direction changes, while the shadow and its margin continue to vary continuously.

Proposition 2.4. *The margin $M_{\mathcal{R}}$ is continuous on centred configurations. The sets of fit and poke configurations are open, and the set of touch configurations is closed.*

Proof. Work near any given configuration. Choose fixed spatial axes so that the viewing line has a representative with positive z -coordinate, and use the rational screen Equation (2.4) throughout a neighbourhood. Let the entries of the actual relative rotation matrix vary continuously; this includes half-turns. Each projected vertex then varies continuously, whether or not it lies on the silhouette.

Compare configurations c and c' , and let δ be the largest displacement between corresponding projected vertices of either body. Every point of a shadow is a convex combination of its projected vertices. Keeping the same coefficients in the other configuration gives a point within distance δ , in both directions. All four shadows contain a common disk of some radius $a > 0$ about zero: $\mathcal{R}$ and its rotations contain a radius- a ball, and $L_{s,t}(v_1, v_2, 0) = (v_1, v_2)$.

Put $\lambda = 1 + \delta/a$. For any corresponding pair of shadows C, C' , the preceding distance bound implies $C \subseteq \lambda C'$. Indeed, write a point of C as $\mathbf{p} + \mathbf{z}$, with $\mathbf{p} \in C'$ and $\|\mathbf{z}\| \leq \delta$. If $\delta > 0$, then

$$\frac{\mathbf{p} + \mathbf{z}}{\lambda} = \frac{1}{\lambda} \mathbf{p} + \frac{\delta/a}{\lambda} \frac{a}{\delta} \mathbf{z} \in C'$$

by convexity and the interior disk; if $\delta = 0$, the inclusion is immediate. The reverse inclusion $C' \subseteq \lambda C$ follows in the same way. One factor of λ compares the plug shadows and another compares the hole shadows:

$$P_{c'} \subseteq \lambda P_c \subseteq \lambda M_{\mathcal{R}}(c) H_c \subseteq \lambda^2 M_{\mathcal{R}}(c) H_{c'}.$$

Using Lemma 2.3 and then reversing the roles of c and c' gives

$$\lambda^{-2} M_{\mathcal{R}}(c) \leq M_{\mathcal{R}}(c') \leq \lambda^2 M_{\mathcal{R}}(c).$$

As c' tends to c , the finitely many vertex displacements tend to zero, so λ tends to one. This proves continuity. The openness and closedness statements follow from the comparisons with one in [Equation \(2.7\)](#). $\square$

3 Symmetry-reduced domain

The RID's symmetries introduce redundancy into the configuration space. We can use this redundancy to restrict the parameter ranges for the viewing direction and relative rotation to a small bounded region, while retaining at least one representative of every configuration. This reduction also justifies using the parametrisation of [Section 2.3](#): viewing directions and relative rotations that it does not represent directly have equivalent representatives within the chosen parameter ranges.

3.1 Symmetries preserve the projection problem

A *body symmetry* is an orthogonal linear map taking $\mathcal{R}$ to itself. Orientation-preserving body symmetries have determinant 1 and are rotations; orientation-reversing symmetries have determinant -1 . An orientation-reversing symmetry that fixes a plane pointwise is a *reflection*. Let $\mathcal{G}$ be the group of rotation symmetries. Applying one of these to the plug before its relative rotation does not change the plug at all: for $G_P \in \mathcal{G}$, $R_P G_P \mathcal{R} = R_P \mathcal{R}$. Applying a hole symmetry $G_H \in \mathcal{G}$ to the whole configuration also leaves the hole fixed as $\mathcal{R}$, while rotating the viewing direction and plug together.

These two operations replace the viewing vector and relative rotation by

$$(\mathbf{u}, R_P) \mapsto (G_H \mathbf{u}, G_H R_P G_P), \quad G_H, G_P \in \mathcal{G}.$$

Orthogonal projection respects this common change of frame:

$$\pi_{G_H \mathbf{u}}(G_H \mathbf{v}) = G_H \pi_{\mathbf{u}}(\mathbf{v}).$$

Consequently, the two new shadows are the images of the old shadows under the same planar isometry. Every containment comparison, including after scaling either shadow by $\lambda > 0$, is preserved. By [Lemma 2.3](#), the margin is unchanged, as are the truth values of the fit, touch and poke predicates.

We can also use the orientation-reversing symmetries. Since $\mathcal{R} = -\mathcal{R}$, any such symmetry G_H has $-G_H \in \mathcal{G}$. Applying $-G_H$ instead keeps the plug's transformation a rotation, and its viewing vector $-G_H \mathbf{u}$ represents the same line as $G_H \mathbf{u}$. Thus all orthogonal body symmetries are available when choosing the viewing line. They form $\mathcal{G} \cup (-\mathcal{G})$.

Choosing the sign of the viewing vector also lets us make its z -coordinate positive whenever it is nonzero, as assumed in the parametrisation of [Section 2.3](#). Viewing lines in the xy -plane will be handled by the reduction in [Section 3.2](#).

The finite symmetry group. The RID has icosahedral symmetry, so $\mathcal{G}$ contains 60 rotations and $-\mathcal{G}$ contains 60 orientation-reversing symmetries, including 15 planar reflections. Let $\mathcal{Q}$ be the set of unit quaternions representing the rotations in $\mathcal{G}$, with both signs g and $-g$ included for each rotation. Thus $\mathcal{Q}$ has 120 members, whose coordinates belong to $\mathbb{Q}(\sqrt{5})$. It includes all eight signed coordinate-unit quaternions: one coordinate is 1 or -1 and the other three are zero. The two signs represent the same rotation; composing that rotation with central inversion gives its orientation-reversing counterpart. The quaternion families and exact checks establishing these finite facts and the full symmetry count are given in [Appendix A.1](#).

3.2 Rotation and reflection reduction

We first restrict the viewing line, then keep its representative vector $\mathbf{u}$ fixed while choosing the relative rotation. The reflection planes divide the viewing directions into equivalent regions. We will select a representative in one closed triangular region, including its boundary.

Reflections arise from the members of $\mathcal{Q}$ whose scalar part is zero. A unit quaternion $(0, \mathbf{n})$ represents the half-turn $2\mathbf{n}\mathbf{n}^\top - I$. Its negative as a spatial matrix is the reflection

$$S_{\mathbf{n}} = I - 2\mathbf{n}\mathbf{n}^\top,$$

which is a body symmetry by central inversion. There are 15 such normal lines. Choose $\mathbf{h} = (1, 2, 9)$ and orient each unit normal so that $\mathbf{h} \cdot \mathbf{n} > 0$; none is perpendicular to $\mathbf{h}$. This auxiliary vector chooses which side of each reflection plane we retain.

Among the images of a nonzero viewing vector under all 120 orthogonal symmetries, choose one, still denoted by $\mathbf{u}$, that maximises $\mathbf{h} \cdot \mathbf{u}$. If $\mathbf{n} \cdot \mathbf{u} < 0$ for any of the oriented normals, its reflection would give

$$\mathbf{h} \cdot (S_{\mathbf{n}}\mathbf{u} - \mathbf{u}) = -2(\mathbf{h} \cdot \mathbf{n})(\mathbf{n} \cdot \mathbf{u}) > 0,$$

contradicting maximality. Thus the chosen vector $\mathbf{u}$ lies on the retained side of every reflection plane.

The viewing triangle. Writing $\mathbf{u} = (u_1, u_2, u_3)$, the intersection of these 15 closed halfspaces is exactly the cone

$$u_1 \geq 0, \quad u_2 \geq 0, \quad u_3 \geq \varphi u_1 + \varphi^2 u_2. \quad (3.1)$$

The equivalence between these halfspace conditions and the three cone inequalities is proved in [Appendix A.2](#).

A nonzero vector in the cone has positive z -coordinate, so we can divide by u_3 and write $\mathbf{u} = (s, t, 1)$. The retained parameter pairs (s, t) form the closed triangle

$$s \geq 0, \quad t \geq 0, \quad \varphi s + \varphi^2 t \leq 1. \quad (3.2)$$

In particular, $s \leq 1/\varphi < 2/3$ and $t \leq 1/\varphi^2 < 2/5$. Every viewing line has a representative here, including lines initially in the xy -plane. Ties in the maximisation cause no difficulty: we keep all equality boundaries.

Choosing a rotation representative. For the relative rotation, the geometric aim is to choose an equivalent description with as small a rotation angle as possible. Besides the plug's symmetries, there is one more operation that preserves its shadow. Let $J_{\mathbf{u}}$ be the half-turn about the viewing line. On the perpendicular screen it changes every vector's sign, so

$$\pi_{\mathbf{u}} J_{\mathbf{u}} = -\pi_{\mathbf{u}}.$$

The plug is centrally symmetric, and consequently

$$\pi_{\mathbf{u}} J_{\mathbf{u}} R_P \mathcal{R} = -\pi_{\mathbf{u}} R_P \mathcal{R} = \pi_{\mathbf{u}} R_P \mathcal{R}. \quad (3.3)$$

We may therefore apply this half-turn to the plug while keeping the hole and viewing vector $\mathbf{u}$ fixed. It need not be a symmetry of the plug: it preserves the plug's shadow, which is what the containment comparison uses.

Quaternion multiplication expresses both operations simply. With scalar parts $\alpha, \beta \in \mathbb{R}$ and vector parts $\mathbf{a}, \mathbf{b} \in \mathbb{R}^3$, its convention is

$$(\alpha, \mathbf{a})(\beta, \mathbf{b}) = (\alpha\beta - \mathbf{a} \cdot \mathbf{b}, \alpha\mathbf{b} + \beta\mathbf{a} + \mathbf{a} \times \mathbf{b}). \quad (3.4)$$

The product represents composition in the same order as rotation matrices: the right-hand rotation acts first. Since $\mathcal{G}$ is closed under composition and both quaternion signs are included, $\mathcal{Q}$ is closed under this product. Let p be a unit quaternion of the current relative rotation, and put

$$j = \frac{(0, \mathbf{u})}{\|\mathbf{u}\|}, \quad \|\mathbf{u}\|^2 = 1 + s^2 + t^2.$$

Then j represents $J_{\mathbf{u}}$ and $j^2 = -1$.

We choose a quaternion p_* with largest scalar part from the finite set

$$\{pg, jpg : g \in \mathcal{Q}\}. \quad (3.5)$$

Both quaternion signs occur. For a unit quaternion with nonnegative scalar part, that part is $\cos(\theta/2)$ for $0 \leq \theta \leq \pi$. Maximising it therefore minimises the relative rotation angle among these equivalent descriptions. The set in [Equation \(3.5\)](#) is unchanged by further right multiplication by a member of $\mathcal{Q}$, or by left multiplication by j : closure of $\mathcal{Q}$ gives the first assertion, and $j^2 = -1$, together with both signs, gives the second. Thus the chosen representative satisfies both sets of comparisons simultaneously.

Write $p_* = (\alpha, \mathbf{a})$. Its scalar part $\alpha = \text{scalar}(p_*)$ is at least $1/2$. Indeed, the signed coordinate-unit quaternions in $\mathcal{Q}$ can move any chosen coordinate of p , with either sign, to the scalar position. At least one of the four coordinates of a unit quaternion has absolute value at least $1/2$. We can therefore divide by $\alpha > 0$ to obtain

$$p_* = \alpha(1, \mathbf{r}), \quad \mathbf{r} = \frac{\mathbf{a}}{\alpha}.$$

This gives a representative even when the original relative rotation was a half-turn, represented by a quaternion with scalar part zero.

Bounds from body symmetries. The comparisons with plug symmetries acting on the right give $\text{scalar}((1, \mathbf{r})g) \leq 1$ for every $g \in \mathcal{Q}$. Use cyclic subscripts, with $r_4 = r_1$. Twelve particular group elements give the comparisons

$$\epsilon r_i + (-1 + \varphi)\delta r_{i+1} \leq 2 - \varphi, \quad i = 1, 2, 3, \quad \epsilon, \delta \in \{-1, 1\}. \quad (3.6)$$

These group elements and their substitutions are given in [Appendix A.3](#). Averaging the two inequalities with opposite signs of δ gives $\epsilon r_i \leq 2 - \varphi$. Using both signs of ϵ , we obtain $|r_i| \leq 2 - \varphi < 2/5$. These necessary comparisons provide the bounded rotation coordinates we need; we do not need to characterise all scalar-maximising representatives.

The fold inequalities. The comparisons involving the projected half-turn, which we call the *fold*, yield further restrictions; the next paragraph explains why they are needed. Since both g and $-g$ are available, maximality gives $|\text{scalar}(jp_*g)| \leq \alpha$. Define

$$\ell_g(c) = \text{scalar}((0, \mathbf{u})(1, \mathbf{r})g). \quad (3.7)$$

Substituting $p_* = \alpha(1, \mathbf{r})$ and cancelling $\alpha > 0$ gives $|\ell_g(c)| \leq \|\mathbf{u}\|$, or equivalently

$$\ell_g(c)^2 \leq 1 + s^2 + t^2. \quad (3.8)$$

Squaring loses no sign information because the preceding comparison is two-sided. Opposite quaternions give the same squared inequality, leaving 60 inequalities. Let $\mathcal{Q}_+$ contain one quaternion from each pair $g, -g$, choosing the one whose first nonzero coordinate is positive. It suffices to impose [Equation \(3.8\)](#) for $g \in \mathcal{Q}_+$. If $g = (\beta, \mathbf{b})$, the product formula gives

$$\ell_g(c) = -\beta \mathbf{u} \cdot \mathbf{r} - \mathbf{b} \cdot (\mathbf{u} + \mathbf{u} \times \mathbf{r}).$$

Thus these are polynomial inequalities in our five parameters, with coefficients in $\mathbb{Q}(\sqrt{5})$.

A second family with equal shadows. The fold matters near aligned configurations. The aligned configurations $\mathbf{r} = \mathbf{0}$ have exactly the hole's shadow, and so does a second family. Let Z be the half-turn symmetry about the z -axis applied to the plug, represented by $k = (0, 0, 0, 1)$. Since $Z\mathcal{R} = \mathcal{R}$, the plug $J_{\mathbf{u}}Z\mathcal{R}$ has exactly the hole's shadow by [Equation \(3.3\)](#). The product

$$-jk = \frac{(1, -t, s, 0)}{\|\mathbf{u}\|} \quad (3.9)$$

shows that this family has rotation coordinates $\mathbf{r} = (-t, s, 0)$. Its touch configurations approach $\mathbf{r} = \mathbf{0}$ as (s, t) approaches $(0, 0)$. For $g = k$, the fold inequality is

$$(1 + sr_2 - tr_1)^2 \leq 1 + s^2 + t^2.$$

On this family it simplifies to $\|\mathbf{u}\| \leq 1$, which holds only at $s = t = 0$. The fold removes the other representatives of this family; their identical shadows already have representatives with $\mathbf{r} = \mathbf{0}$ at the same parameter pair (s, t) . This separation will be useful when proving the local results around aligned configurations.

3.3 The reduced domain

The preceding bounds place a representative of every configuration in the five-dimensional rectangle

$$B_0 = \left[0, \frac{2}{3}\right] \times \left[0, \frac{2}{5}\right] \times \left[-\frac{2}{5}, \frac{2}{5}\right]^3. \quad (3.10)$$

The first two intervals describe s, t and the last three describe $\mathbf{r} = (r_1, r_2, r_3)$. Within this rectangle, define the reduced domain

$$D = \left\{ c = [s, t, \mathbf{r}] \in B_0 : \begin{array}{l} \varphi s + \varphi^2 t \leq 1, \\ \epsilon r_i + (-1 + \varphi)\delta r_{i+1} \leq 2 - \varphi \quad (i = 1, 2, 3; \epsilon, \delta \in \{-1, 1\}), \\ \ell_g(c)^2 \leq 1 + s^2 + t^2 \quad (g \in \mathcal{Q}_+) \end{array} \right\}. \quad (3.11)$$

These are the viewing-triangle condition, twelve relative-rotation comparisons and sixty fold comparisons: 73 inequalities in total, in addition to the interval bounds defining B_0 .

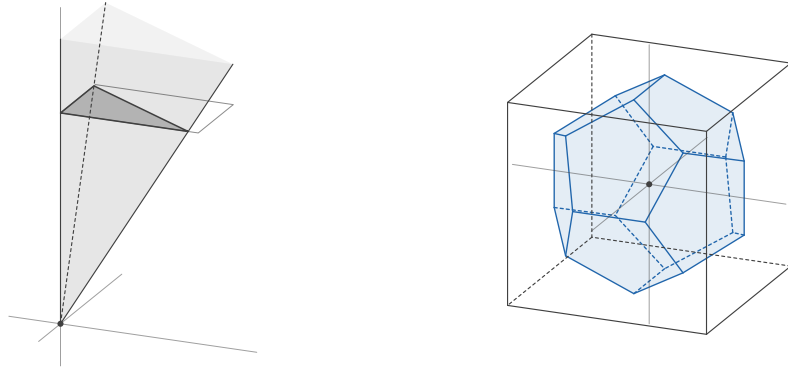


Figure 7: The reduced ranges, in the spatial axes of Figure 5. Left: the viewing directions kept form a cone (grey) through the viewing triangle $s, t \geq 0$, $\varphi s + \varphi^2 t \leq 1$ in the plane $z = 1$ (dark), which lies in the rectangle $[0, 2/3] \times [0, 2/5]$ of B_0 (thin). Right: the box $[-2/5, 2/5]^3$ of rotation vectors $\mathbf{r}$ (grey) and the part of it satisfying the twelve rotation comparisons (blue), a dodecahedron; the fold comparisons, which also involve s and t , cut it further.

Proposition 3.1 (Representative coverage). *Every centred geometric configuration has a representative in D with the same margin. In particular, a fit exists if and only if one exists in D .*

Proof. First apply a hole symmetry to the viewing line and the common frame to place the representative vector $\mathbf{u}$ in the cone Equation (3.1). An orientation-reversing symmetry is implemented by its negative, with the viewing vector's sign reversed, as in Section 3.1. The transformed vector has positive z -coordinate, so normalising it gives $\mathbf{u} = (s, t, 1)$ satisfying Equation (3.2) and the first two interval bounds of B_0 .

Keep (s, t) fixed and choose the maximal-scalar quaternion from Equation (3.5). Its positive scalar part permits the normalisation $(1, \mathbf{r})$, and its comparisons give Equations (3.6) and (3.8) and the remaining interval bounds of B_0 . Every operation preserves the margin, so this is the required representative. Conversely, each point of D specifies an actual viewing line and rotation by the formulas of Section 2.3. $\square$

All the inequalities are non-strict. In particular, the argument retains viewing directions lying in reflection planes, ties between rotation representatives and equality in the fold comparisons. A configuration can have more than one representative in D ; coverage is all that the proof needs. For a configuration initially allowing displacement, the centering lemma first preserves the existence of a fit, and the proposition then puts a centred fit in D .

Compactness and the minimum margin. The coverage proposition ensures that D is nonempty. Its polynomial inequalities define a closed subset of the compact rectangle B_0 , so D is compact. By [Proposition 2.4](#), the margin therefore attains its minimum on D . Representative coverage gives the promised bounded form of [Equation \(2.8\)](#):

$$\nu(\mathcal{R}) = \frac{1}{\min_{c \in D} M_{\mathcal{R}}(c)}. \quad (3.12)$$

The remaining task is to show that $M_{\mathcal{R}}(c) \geq 1$ throughout D . We will work from the simpler enclosing rectangle B_0 and use these inequalities to set aside regions entirely outside D . Their physical configurations already have representatives in D ; no classification of these as fit, touch or poke configurations is needed to set those regions aside.

4 Branch and bound elimination in the reduced domain

To rule out fit throughout D , we examine whole regions of configurations at a time. Each successful component must rule out fit for every configuration in the region that belongs to D . We subdivide unresolved regions, seeking a resolution high enough for at least one component to eliminate them. The proof rests on two facts about the result: every region was eliminated soundly, and the eliminated regions cover D . [Section 11](#) checks both for the final certificate and lists everything the proof relies on. The notions of range and completion in [Section 4.2](#) explain why the subdivision finishes and how the components were found; the proof itself does not use them.

4.1 Boxes, subdivision, and elimination

A *box* is a Cartesian product of closed parameter intervals,

$$B = \prod_{i=1}^5 [l_i, h_i] \subseteq B_0, \quad l_i, h_i \in \mathbb{Q}, \quad l_i \leq h_i,$$

in the coordinate order s, t, r_1, r_2, r_3 . Each point of B represents a configuration. Rational endpoints allow the box itself to be recorded exactly. The boxes created by our search have positive width in every coordinate, so they cover entire five-dimensional regions of configuration space.

We *eliminate* a box when we certify that it contains no fit configuration in D . In terms of the margin, the required conclusion is

$$M_{\mathcal{R}}(c) \geq 1 \quad \text{for all } c \in B \cap D. \quad (4.1)$$

A component takes a box and attempts to justify its elimination. If it succeeds, we store a *certificate record* identifying the box, the successful component and the data needed to verify its result. Otherwise, its answer is inconclusive. A component is *sound* if every accepted result establishes [Equation \(4.1\)](#). In particular, elimination allows touch configurations, and a box disjoint from D satisfies this condition without any geometric comparison.

Subdivision. We call a set of components used together for elimination a *collection*. The procedure starts with B_0 . For each unresolved box, it tries the members of the collection in a fixed order, stopping at the first successful one. If none succeeds, it bisects one of the intervals at its midpoint $m = (l_i + h_i)/2$, replacing $[l_i, h_i]$ by $[l_i, m]$ and $[m, h_i]$ while leaving the other four intervals unchanged. Both children are closed, axis-parallel boxes containing the four-dimensional dividing face, and their union is exactly the parent box.

The number of bisections from B_0 to a box is its *depth*. We bisect coordinates cyclically in the order s, t, r_1, r_2, r_3 , and process boxes in increasing depth. Boxes that remain unresolved are therefore refined adaptively to arbitrarily small widths in all five coordinates. The decision to subdivide depends on whether a box has been eliminated; the coordinate order does not.

Complete coverage. An important invariant is that the eliminated boxes together with the unresolved boxes always cover B_0 . This holds initially and is preserved both when a box is eliminated and when it is replaced by its two children. Consequently, if the procedure finishes with finitely many eliminated boxes and none unresolved, their certificate records form a complete certificate of non-Rupertness. Indeed, a fit

would have a representative in D by [Proposition 3.1](#). That representative would belong to an eliminated box, contradicting [Equation \(4.1\)](#). Shared boundaries introduced by the bisections cause no exception to this reasoning.

This explains what completion would prove; it does not yet show that the collection will resolve every box. Stopping at a time or depth limit leaves the unresolved regions to be treated. [Section 11](#) describes how these records and their coverage are checked.

For convex polyhedra with algebraic vertex coordinates, [Steininger and Yurkevich \[2023\]](#) give the theoretical decision procedure discussed in [Section 1.2](#). Their procedure reduces the problem to solving systems of polynomial equations and inequalities, but they regard the resulting computations as impractical. Our aim is to develop a collection that uses the geometric properties of the RID to make certificate generation feasible.

4.2 Component hierarchy

Even though components are applied to entire boxes, their usefulness and scope can be better understood by considering individual configurations. If a component works uniformly throughout some neighbourhood around a configuration, it will generally handle sufficiently small boxes there. We make this connection precise so that later descriptions of where a component applies have a common and unambiguous meaning.

Call any box obtainable by the cyclic midpoint subdivision of B_0 a *search box*, whether or not a particular run reaches it. Neighbourhoods below are relative to B_0 : at its boundary, only configurations inside B_0 need to be considered. Distances and diameters refer to Euclidean distance between the five parameter coordinates.

Lemma 4.1 (Search boxes in neighbourhoods). *Every configuration $c \in B_0$ belongs to a search box at every depth. For any neighbourhood U of c relative to B_0 , all sufficiently deep search boxes containing c lie entirely in U . In particular, every nonempty relatively open subset of B_0 contains a search box.*

Proof. The full subdivision at each depth covers B_0 , since the two closed children cover their parent. At depth d , each coordinate has been bisected at least $\lfloor d/5 \rfloor$ times, so every search box at that depth satisfies

$$\text{diam } B \leq 2^{-\lfloor d/5 \rfloor} \text{diam } B_0. \quad (4.2)$$

Choose $\varepsilon > 0$ such that every point of B_0 at distance less than ε from c belongs to U . For sufficiently large d , this diameter bound is less than ε , so every point of a search box containing c belongs to U . The argument also applies when c lies on a boundary of B_0 or a subdivision face. $\square$

Definition 4.2 (Range of a component). A configuration $c \in B_0$ belongs to the *range* of a component if it has a neighbourhood U in B_0 such that the component accepts every search box $B \subseteq U$. The range of a collection is defined in the same way, allowing any member to accept each such box.

Acceptance here means that the component's check succeeds, including any selection of the data needed for its certificate record. A box does not count as accepted just because a valid record exists; the component must find and verify one. Different boxes may use different data in their records, or different members of a collection. This definition describes an open region relative to B_0 .

In particular, accepting one closed box with c on its boundary does not establish that c lies in the component's range. Smaller boxes around c can extend across that boundary. The same issue arises where two proved regions meet: a box extending across both regions needs a component that can certify all of it. A component may check that the box is covered by a union of already proved regions, or another component may handle the whole box. Acceptance of only the box's centre is not enough.

Lemma 4.3 (Finite completion from local coverage). *Suppose the range of a fixed finite collection covers B_0 , all its members are sound, and every component call finishes. Then the increasing-depth search with cyclic midpoint subdivision terminates after finitely many boxes, provided no time or depth limit stops it.*

Proof. Around each configuration in B_0 , choose a ball of radius 2ε , with $\varepsilon > 0$, whose intersection with B_0 lies in an accepting neighbourhood. The balls with radii ε still cover B_0 . Compactness gives a finite subcover; let $\delta > 0$ be the smallest radius in that finite subcover.

Any search box of diameter less than δ lies in one of the corresponding larger balls. Indeed, choose a point of the box and a smaller ball containing it. Every other point of the box is within δ of that point, hence within twice the smaller ball's radius of its centre. Such a box is therefore accepted.

By Equation (4.2), at sufficiently large finite depth every search box has diameter less than δ , so every box at that depth is accepted without further subdivision. There are only finitely many boxes up to that depth, and processing each one takes finitely many component calls, all of which finish. The search therefore terminates. $\square$

Developing the collection. The same finite-cover reasoning applies to any compact subset of the collection's range: sufficiently small unresolved boxes cannot continue to meet that subset. Refinement can therefore help locate the configurations for which further components are needed. Small changes in the margin alone do not establish this property; we must show that the relevant component actually succeeds on the nearby boxes.

This gives a way to develop the collection successively. When refinement concentrates on configurations the current collection cannot handle, we examine their geometry and add a component to handle them. Each addition works alongside the preceding components. The need to treat aligned configurations was already apparent before running the Global component, which detects protrusion: at an aligned configuration the shadows coincide, so no projected plug vertex protrudes beyond the hole's shadow.

The final collection has four components, in the order in which we develop them: the *Domain* component (Section 4.3), which discards boxes outside D ; the *Global* component (Section 6), for poke configurations; the *Local* component (Section 7), for neighbourhoods of aligned configurations; and the *Exotic* component (Sections 8 to 10), with eight covers around the remaining touch configurations. Figure 8 shows how each addition clears boxes that the preceding components leave unresolved. The program may try an inexpensive specialised check before a more general one; the order affects only which record a box receives.

4.3 Domain component: skipping boxes outside the reduced domain

The simplest component uses the inequalities defining D in Equation (3.11). If one of them is strictly violated throughout B , then $B \cap D = \emptyset$. We call this the *Domain* component. The justification is the representative-coverage result: the configurations discarded here already have representatives in D . They need not be poke configurations.

Affine inequalities. The viewing-triangle inequality and the twelve relative-rotation inequalities are affine. Write one as $p(c) \leq 0$, subtracting its right side from its left side. To minimise p on B , choose the lower endpoint in every coordinate with a positive coefficient and the upper endpoint for a negative coefficient; a zero coefficient permits either choice. This gives the exact minimum. If it is strictly positive, the entire box violates the inequality. Equality is insufficient because the inequalities defining D are non-strict.

Fold inequalities. The fold inequalities require more care. A *corner* of a box is obtained by choosing an endpoint of each parameter interval. We group these choices according to what they control. The allowed pairs (s, t) fill a rectangle in the (s, t) -plane, which we call the *viewing rectangle*. Choosing an endpoint for each of s and t gives its four corners, each specifying a viewing direction. Independently, the allowed vectors $\mathbf{r} = (r_1, r_2, r_3)$ fill a three-dimensional *rotation box*, whose eight corners choose an endpoint for each of r_1, r_2, r_3 . Each viewing corner can be combined with any rotation corner, giving the $4 \times 8 = 32$ corner configurations of B .

The square of an expression can be large at both ends of an interval while the expression changes sign and vanishes inside. For this reason, evaluating only the squared fold inequalities at the corners would not justify eliminating the box. We also check the sign before squaring.

Proposition 4.4 (Whole-box fold violation). *Fix $g \in \mathcal{Q}_+$. The inequality*

$$\ell_g(c)^2 > 1 + s^2 + t^2$$

holds throughout B if and only if it holds at every corner of B and the values of ℓ_g at all corners have one common nonzero sign.

Proof. Suppose first that the corner conditions hold, and let $\sigma \in \{-1, 1\}$ be the common sign. At every corner,

$$\sigma \ell_g(c) - \|\mathbf{u}\| > 0, \quad \mathbf{u} = (s, t, 1).$$

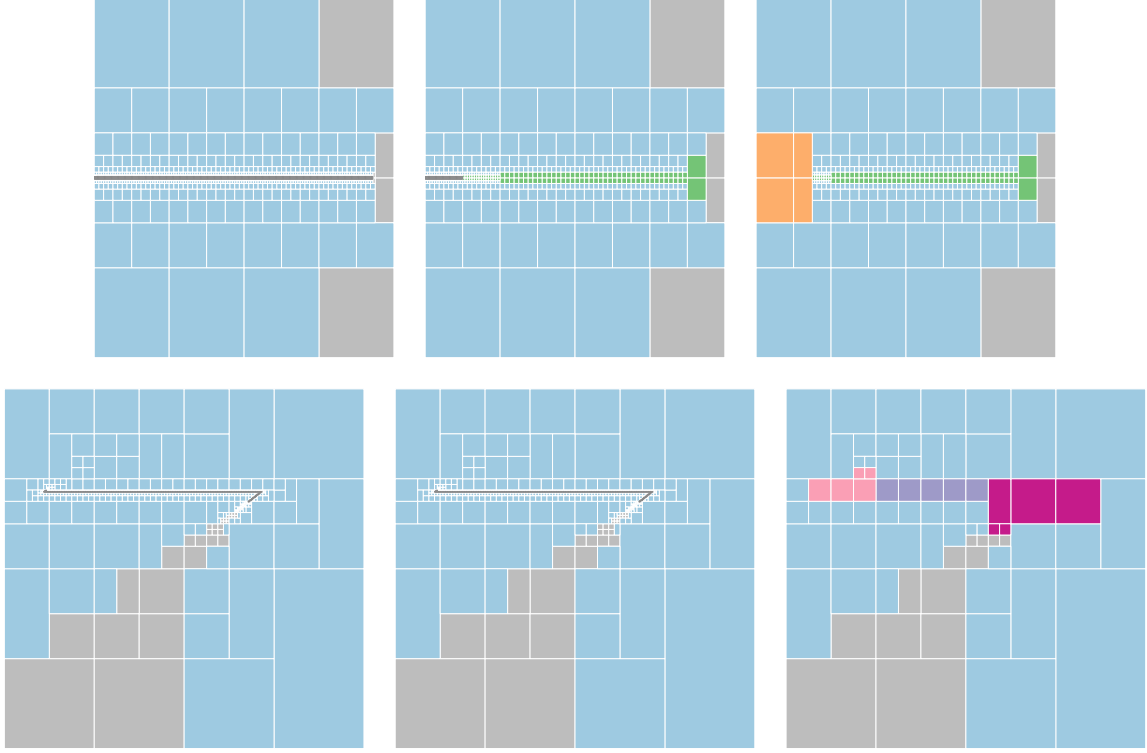


Figure 8: The collection built up component by component on two planes of configurations, each halved alternately until a component eliminates a piece or the piece is 16 (top) or 18 (bottom) halvings deep; pieces still unresolved at that depth are black. Top: the plane $t = 0$, $r_1 = r_2 = 0$ through the square configuration, with $s \in [0, 2/3]$ horizontally and $r_3 \in [-2/5, 2/5]$ vertically. Bottom: the plane Π_1 of Section 9, with $s \in [3/16, 5/8]$ horizontally and $\theta \in [-5/16, 1/8]$ vertically. Left: the Domain (grey) and Global (light blue) components leave the aligned configurations $r_3 = 0$ and the two arcs unresolved. Middle: the Local component (green) eliminates the aligned configurations except near the square configuration $s = 0$, and nothing on the arcs. Right: the whole collection, which leaves nothing; the Exotic covers (square light orange, arc lavender, endpoint light pink, crossing magenta) are tried before the Local component, so the square cover also takes pieces that the Local component would eliminate.

Fix $\mathbf{r}$ at one of the eight rotation corners. The expression ℓ_g is affine in (s, t) , as follows from the quaternion product in Equation (3.7). The norm of $(s, t, 1)$ is convex, so the displayed difference is concave in (s, t) . Every pair (s, t) in the viewing rectangle is a convex combination of its four corners. Concavity therefore makes the difference at least the corresponding combination of four strictly positive values. It is strictly positive throughout that rectangle.

This holds for each rotation corner. Now fix arbitrary $(s, t) \in [l_1, h_1] \times [l_2, h_2]$. The same difference is affine in $\mathbf{r}$, so interpolation between the eight rotation corners proves strict positivity throughout the rotation box. Thus $\sigma \ell_g(c) > \|\mathbf{u}\| > 0$ everywhere in B , and squaring proves the required inequality.

Conversely, if the inequality holds throughout B , then $|\ell_g(c)| > \|\mathbf{u}\| \geq 1$. The continuous function ℓ_g cannot change sign on the connected box without vanishing. Its corner values consequently have the same nonzero sign and satisfy the strict inequalities. The proof also applies to zero-width intervals, where some of the corner choices coincide. $\square$

Checking a box in the implementation. The implementation writes every inequality defining D in the form $q(c) \leq 0$ for a polynomial q : the affine inequalities as above, and each fold inequality as $q = \ell_g^2 - (1 + s^2 + t^2)$. To eliminate B , it certifies $q > 0$ throughout B by applying the Bernstein sign test of Section 5.2 to $-q$. For an affine inequality the coefficients are exactly the corner values, so this is the endpoint rule above. For a fold inequality it is a sufficient condition, and Proposition 4.4 describes exactly when a box violates the inequality throughout. The component tries only the inequalities whose value at the midpoint of the box, computed in floating point, is positive. An inequality that holds at the midpoint cannot be violated throughout the box, and a rounding error in this choice can only leave a box unresolved; the elimination itself rests on the exact test.

All the corner values and coefficients lie in $\mathbb{Q}(\sqrt{5})$, since the box endpoints are rational and the coefficients lie in that field. Their exact signs can be determined using integers; Section 5.1 gives these operations in the next section. A certificate record produced by the Domain component identifies the domain inequality violated throughout the box, so its checker can reconstruct and repeat the whole-box test.

If none of the tests succeeds, the box may still be disjoint from D : different parts of it could violate different inequalities. Further subdivision can separate those parts. In particular, every point of $B_0 \setminus D$ strictly violates at least one defining inequality. Continuity preserves that violation in a closed neighbourhood, and the exact test accepts every sufficiently small box contained in it, by the convergence of Bernstein coefficients proved in Lemma 5.2. Thus the Domain component handles sufficiently small boxes around every point outside D . Configurations on the retained boundary still require the geometric components. Before developing those components, we explain how to carry out these calculations exactly and certify the sign of more general polynomials on a box.

5 Reduction to exact arithmetic

The Domain component reduces its whole-box test to finitely many sign comparisons in $\mathbb{Q}(\sqrt{5})$. We explain how to perform these comparisons using integers, then develop a sign test for the polynomial inequalities needed by the geometric components.

For a general polynomial, negative values at the midpoint, or even at every corner, do not establish negativity throughout a box. The Domain component's corner criterion relies on the particular affine and concave expressions in its proof. To handle more general expressions, we write a polynomial on a box as a weighted average of finitely many numbers, its Bernstein coefficients, which can be computed exactly. When all of them are negative, so is the polynomial throughout the box, including its boundary. Every elimination in this article that is not a direct endpoint comparison ends in this single test.

5.1 Exact representation of field elements with integers

The standard arithmetic of $\mathbb{Q}(\sqrt{5})$ gives exact field operations and sign and equality tests using only integer calculations. As introduced in Section 2.3, we represent $a + b\sqrt{5}$ by the rational pair (a, b) , storing each rational number as a ratio of arbitrary-size integers with positive denominator. The operations are as follows:

- *Addition and subtraction* act componentwise on the pairs.

- *Multiplication* uses

$$(a + b\sqrt{5})(a' + b'\sqrt{5}) = (aa' + 5bb') + (ab' + ba')\sqrt{5}.$$

- *Division* uses the reciprocal of a nonzero field element:

$$\frac{1}{a + b\sqrt{5}} = \frac{a - b\sqrt{5}}{a^2 - 5b^2}.$$

The denominator cannot vanish unless $a = b = 0$, since $\sqrt{5}$ is irrational.

- *Equality and sign comparisons* are exact. An element is zero exactly when $a = b = 0$. If either coefficient is zero, or both have the same sign, the sign of $a + b\sqrt{5}$ is immediate. If their signs differ, compare the rational squares a^2 and $5b^2$:

$$\text{sign}(a + b\sqrt{5}) = \text{sign}(a) \text{sign}(a^2 - 5b^2).$$

The preliminary sign distinction is essential before using this squared comparison. Comparing two field elements reduces to testing their difference.

These operations supply the exact comparisons required by the tests of the Domain component in [Section 4.3](#).

5.2 Bernstein coefficients

To eliminate a box, a component must show that some polynomial is strictly negative at every point of the box, not only at finitely many sampled configurations. We use a classical tool for this [[Garloff, 1986](#), Section 3]. Consider first a polynomial p of degree at most n in one variable x on an interval $[l, h]$ of width $w = h - l \geq 0$. Substituting $x = l + wy$ gives a polynomial in $y \in [0, 1]$, which can be written uniquely as

$$p(l + wy) = \sum_{k=0}^n b_k \binom{n}{k} y^k (1 - y)^{n-k}. \quad (5.1)$$

The numbers b_k are the *Bernstein coefficients* of p on $[l, h]$. The $n + 1$ polynomials $\binom{n}{k} y^k (1 - y)^{n-k}$ are nonnegative on $[0, 1]$, and by the binomial theorem they add up to $(y + (1 - y))^n = 1$. Every value of p on the interval is therefore a weighted average of its Bernstein coefficients, and if all of them are negative, so is p .

A worked example. The polynomial $p(x) = -1 + 3x - 3x^2$ is negative on $[0, 1]$: its largest value there is $p(1/2) = -1/4$. Its Bernstein coefficients on $[0, 1]$ are $(-1, 1/2, -1)$, so the test does not yet succeed. After halving, the coefficients are $(-1, -1/4, -1/4)$ on $[0, 1/2]$, where $p(y/2) = -1 + \frac{3}{2}y - \frac{3}{4}y^2$, and $(-1/4, -1/4, -1)$ on $[1/2, 1]$. Both halves pass. This is typical: halving brings the coefficients closer to the values of the polynomial, as we make precise below.

Several variables. Let p be a polynomial in $x_1, \dots, x_k$ with coefficients in $\mathbb{Q}(\sqrt{5})$, of degree n_j in x_j , and let $B = \prod_j [l_j, h_j]$ be a closed box with rational endpoints and widths $w_j = h_j - l_j \geq 0$. With $x_j = l_j + w_j y_j$, the products

$$B_{\mathbf{i}}(\mathbf{y}) = \prod_{j=1}^k \binom{n_j}{i_j} y_j^{i_j} (1 - y_j)^{n_j - i_j}, \quad 0 \leq i_j \leq n_j,$$

form a basis of the polynomials of degree at most n_j in each y_j . The unique coefficients $b_{\mathbf{i}}$ with

$$p(l_1 + w_1 y_1, \dots, l_k + w_k y_k) = \sum_{\mathbf{i}} b_{\mathbf{i}} B_{\mathbf{i}}(\mathbf{y})$$

are the *tensor Bernstein coefficients* of p on B ; there are $\prod_j (n_j + 1)$ of them.

Lemma 5.1 (Bernstein sign test). *If every tensor Bernstein coefficient of p on B is strictly negative, then $p(\mathbf{x}) < 0$ for every $\mathbf{x} \in B$.*

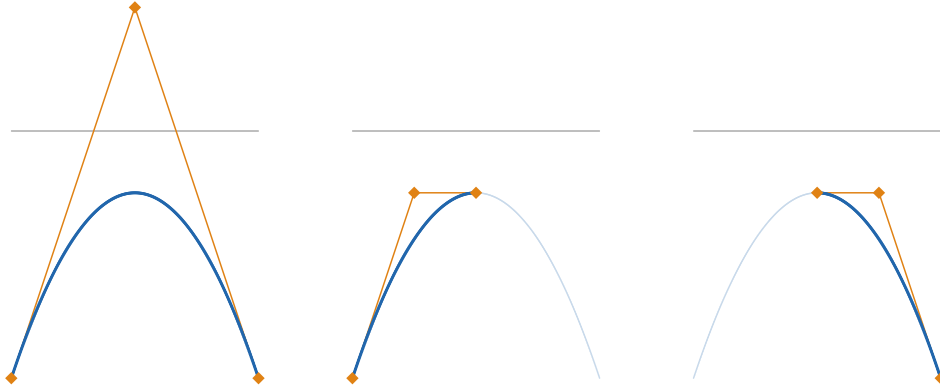


Figure 9: The polynomial $p(x) = -1 + 3x - 3x^2$ (blue), its Bernstein coefficients (orange diamonds, at $x = l + k(h-l)/2$) and their control polygon; grey is the axis $p = 0$. Left: on $[0, 1]$ the middle coefficient is positive although p is negative throughout, and the sign test refuses. Middle and right: on the two halves, shown with the whole graph for reference, every coefficient is negative and the test succeeds. On each interval the graph lies between the smallest and the largest coefficient.

Proof. Each B_i is nonnegative on $[0, 1]^k$, and their sum is $\prod_j (y_j + (1 - y_j))^{n_j} = 1$. A point $\mathbf{x} \in B$ corresponds to $y_j = (x_j - l_j)/w_j$, or to any $y_j \in [0, 1]$ when $w_j = 0$. So $p(\mathbf{x})$ is a convex combination of the coefficients and at most their maximum, which is negative. This covers every real point of the closed box, including its faces and corners and points with irrational coordinates. $\square$

The coefficient with every $i_j \in \{0, n_j\}$ is the value of p at the corresponding corner of B . The test therefore refuses a polynomial that is nonnegative at some corner, and more generally anywhere in B . For a polynomial of degree at most one in each variable, the coefficients are exactly its corner values, so the test coincides with the endpoint rule of the Domain component in Section 4.3. A refusal never proves anything: it only leaves the box unresolved.

Exact computation. The coefficients are computed one variable at a time. Fix j , write $n = n_j$, and expand along $y = y_j$ as $p = \sum_k a_k y^k$, with coefficients that are polynomials in the other variables. The identity

$$y^k = \sum_{r=k}^n \frac{\binom{r}{k}}{\binom{n}{k}} \binom{n}{r} y^r (1-y)^{n-r}$$

converts powers of y into the Bernstein basis, giving

$$b_r = \sum_{k=0}^r \frac{\binom{r}{k}}{\binom{n}{k}} a_k. \quad (5.2)$$

Substitution, expansion and this change of basis use only exact field operations in $\mathbb{Q}(\sqrt{5})$. In practice we first multiply by a common denominator, so that every step is an operation on integers of $\mathbb{Z}[\sqrt{5}]$ followed by a known positive scale, and read the sign of each coefficient with the exact test of Section 5.1.

Convergence under subdivision. Refinement helps wherever a polynomial is strictly negative, because the coefficients approach the values of the polynomial as the box shrinks.

Lemma 5.2 (Convergence of Bernstein coefficients). *Let p be a polynomial and K a bounded box. There is a constant C such that for every box $B \subseteq K$ whose widths are all at most $w \leq 1$, every tensor Bernstein coefficient of p on B differs from the value of p at the lower corner of B by at most Cw . In particular, if $p < 0$ throughout a closed box B_* , every box $B \subseteq B_*$ with sufficiently small widths passes the test of Lemma 5.1.*

Proof. In one variable, $a_0 = p(l)$ and $a_k = p^{(k)}(l)w^k/k!$ by Taylor's formula. In Equation (5.2), every weight lies in $[0, 1]$ and the weight of a_0 is one, so

$$|b_r - p(l)| \leq \sum_{k \geq 1} |a_k| \leq w \sum_{k \geq 1} \frac{|p^{(k)}(l)|}{k!}.$$

The derivatives are bounded on K . Converting one variable at a time applies this bound to polynomials whose coefficients are themselves bounded on K , and the errors add, giving the constant C . For the second claim, $p \leq -\gamma$ on B_* for some $\gamma > 0$ by compactness; every coefficient on a subbox of widths at most $w < \gamma/C$ is then at most $-\gamma + Cw < 0$. $\square$

On a search box, every polynomial we test has degree at most two in each of the five parameters, so a test inspects at most $3^5 = 243$ coefficients. The same test applies after a polynomial change of variables, which is how the later components use it, through the zoom lemma of Section 7.3.

6 Global component

A projected plug vertex protruding beyond the hole's shadow prevents a fit. The *Global* component certifies that such protrusion persists throughout a box. This geometric idea also underlies the global theorem of Steininger and Yurkevich [2026, Section 4, Theorem 17].

6.1 Eliminating *poke* configurations

Fix a poke configuration. At least one projected plug vertex lies outside the hole's shadow: otherwise convexity would put the entire plug shadow inside it. We use the orthogonal screen for the separation picture. There is a direction along which that vertex extends further than every point of the hole's shadow. To see the comparison, project the two shadows onto a line parallel to this direction. The selected plug vertex lies beyond the interval occupied by the hole shadow.

For a nonzero vector $\mathbf{n}$ in the orthogonal screen plane, the largest value of $\mathbf{n} \cdot \mathbf{h}$ over points $\mathbf{h}$ of the hole shadow is its *support value* in that direction. The line where this value is attained is a *supporting line*: the whole shadow lies on one side of it. We call $\mathbf{n}$ a *support direction*. Since $\mathbf{n} \cdot \mathbf{u} = 0$, taking a dot product with $\mathbf{n}$ gives the same value before and after projection. Thus a selected plug vertex $\mathbf{v}_p \in V$ is separated from the hole shadow if

$$\mathbf{n} \cdot R_P(\mathbf{r})\mathbf{v}_p > \max_{\mathbf{v} \in V} \mathbf{n} \cdot \mathbf{v}. \quad (6.1)$$

Due to convexity and the linearity of the dot product, the maximum over vertices is also the maximum over their convex hull.

Stability of the margin. A sufficiently small change of configuration preserves a strict separation. Before deriving the inequalities used to check a whole box, we quantify this stability in terms of the margin introduced in Section 2.4.

Lemma 6.1 (Lipschitz bound for the margin). *For configurations $c, c' \in B_0$,*

$$|M_{\mathcal{R}}(c) - M_{\mathcal{R}}(c')| \leq 16\|c - c'\|, \quad (6.2)$$

where the distance is Euclidean distance between the five parameters, as in Section 4.2.

Proof. First, $\mathcal{R}$ contains the ball of radius two about zero and lies in the ball of radius five. For the inner bound, average the three cyclic permutations of $(1, 1, \varphi^3)$, applying the same coordinate sign choices to each. This places all eight corners of the cube with half-side $(4 + \sqrt{5})/3 > 2$ inside $\mathcal{R}$. For the outer bound, all vertices whose coordinates are given in Equation (2.1) have distance $\sqrt{11 + 4\sqrt{5}} < 5$ from the origin. Orthogonal shadows therefore give $M_{\mathcal{R}}(c) \leq 5/2$ for every configuration. The same bound holds in rational screen coordinates: applying the same invertible linear map to both shadows preserves scaled containment and hence the margin, by Lemma 2.3.

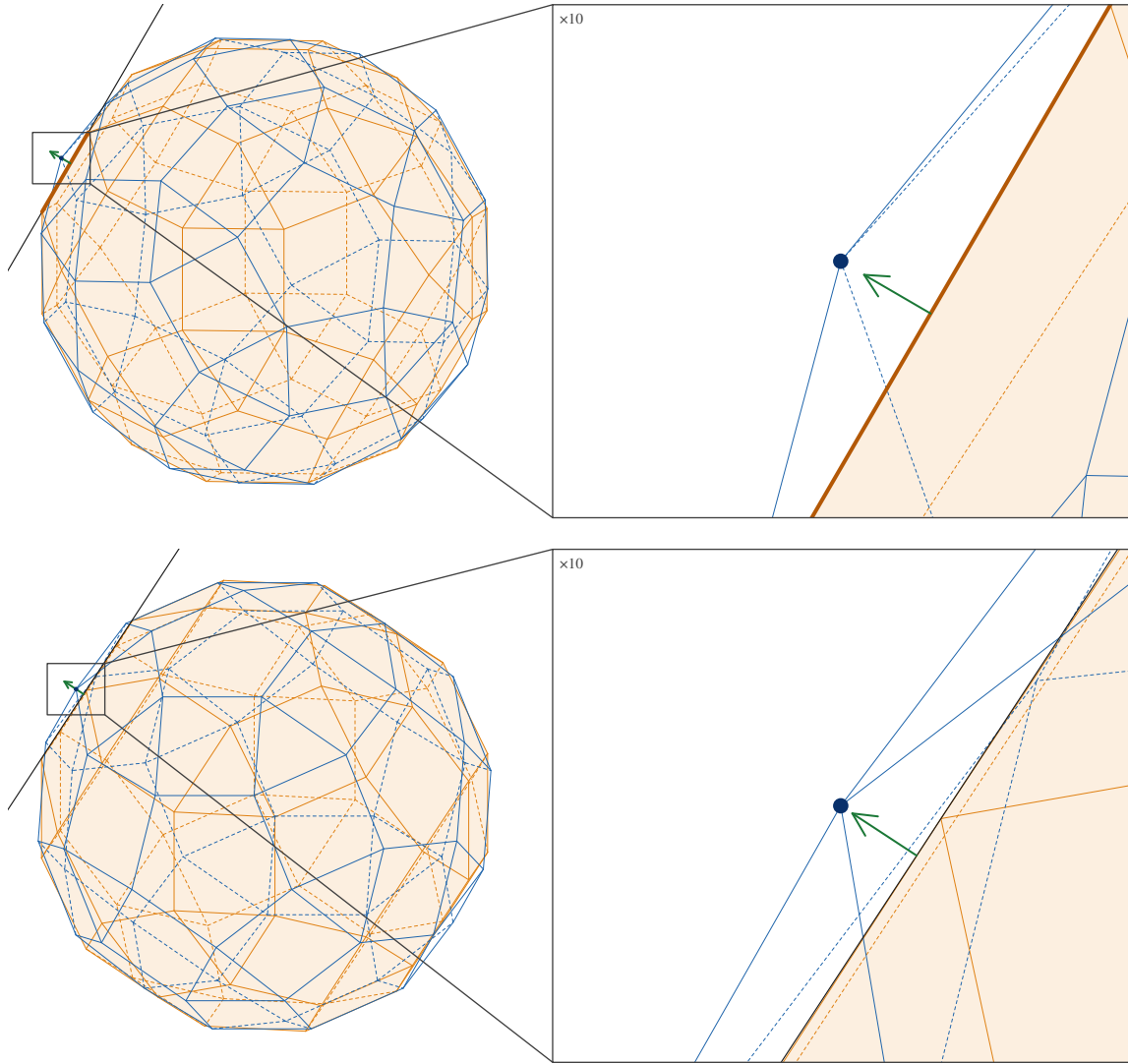


Figure 10: Two poke configurations eliminated by the Global component, at the midpoints of two search boxes of the final certificate: $[7/48, 7/80, (1/20, -1/20, 1/4)]$ and $[3/16, 23/80, (-1/40, -13/40, -1/8)]$. Left: the projected edges of the hole (orange) and of the plug (blue), dashed on each body's far side, with the hole's shadow lightly tinted. Right: the marked windows, enlarged ten times. Black: the supporting line of the hole's shadow for the direction of the hole edge named in the box's record; thick brown: the part of the hole's silhouette on that line (on the right, a single vertex attains it); green: the outward normal, drawn from the foot of the perpendicular from the record's plug vertex (dark blue dot), which lies strictly beyond the line. The component certifies this separation at every configuration of the box.

Rotation and shadow displacement. We use the Euclidean operator norm for matrices, $\|A\| = \max_{\|\mathbf{v}\|=1} \|A\mathbf{v}\|$. Put $\delta_{s,t} = \|(s, t) - (s', t')\|$ and $\delta_{\mathbf{r}} = \|\mathbf{r} - \mathbf{r}'\|$. To bound the motion of the shadow vertices, we use

$$\begin{aligned} \|R_P(\mathbf{r}) - R_P(\mathbf{r}')\| &\leq 2\delta_{\mathbf{r}}, \\ \|L_{s,t} - L_{s',t'}\| &= \delta_{s,t}, \\ \|L_{s,t}\| &\leq \frac{4}{3}. \end{aligned} \tag{6.3}$$

These estimates follow from the rotation and screen formulas [Equations \(2.3\) and \(2.4\)](#) and the ranges defining B_0 . For the rotation estimate, write $[\mathbf{r}]_{\times} \mathbf{v} = \mathbf{r} \times \mathbf{v}$. The vector triple-product identity gives $[\mathbf{r}]_{\times}^2 = \mathbf{r}\mathbf{r}^T - \|\mathbf{r}\|^2 I$, and $[\mathbf{r}]_{\times} \mathbf{r} = \mathbf{0}$. Multiplying by $I - [\mathbf{r}]_{\times}$ therefore verifies the inverse formula

$$(I - [\mathbf{r}]_{\times})^{-1} = \frac{I + [\mathbf{r}]_{\times} + \mathbf{r}\mathbf{r}^T}{1 + \|\mathbf{r}\|^2}.$$

The rotation formula [Equation \(2.3\)](#) can therefore be written as

$$R_P(\mathbf{r}) = 2(I - [\mathbf{r}]_{\times})^{-1} - I.$$

Also, orthogonality of a cross product to its factors gives

$$\|(I - [\mathbf{r}]_{\times})\mathbf{v}\|^2 = \|\mathbf{v}\|^2 + \|[\mathbf{r}]_{\times} \mathbf{v}\|^2,$$

so this inverse has operator norm at most one. Apply the inverse-difference identity $A^{-1} - B^{-1} = A^{-1}(B - A)B^{-1}$ to $A = I - [\mathbf{r}]_{\times}$ and $B = I - [\mathbf{r}']_{\times}$. Since $\|\mathbf{a} \times \mathbf{v}\| \leq \|\mathbf{a}\|\|\mathbf{v}\|$, we obtain

$$\|R_P(\mathbf{r}) - R_P(\mathbf{r}')\| \leq 2\|[\mathbf{r}]_{\times} - [\mathbf{r}']_{\times}\| \leq 2\|\mathbf{r} - \mathbf{r}'\| = 2\delta_{\mathbf{r}}.$$

By [Equation \(2.4\)](#), the difference $L_{s,t} - L_{s',t'}$ sends $\mathbf{v}$ to $-(s - s')v_3, -(t - t')v_3$, so its operator norm is $\delta_{s,t}$. Moreover,

$$L_{s,t} L_{s,t}^T = \begin{pmatrix} 1 + s^2 & st \\ st & 1 + t^2 \end{pmatrix}$$

has eigenvalues 1 and $1 + s^2 + t^2$. The ranges $0 \leq s \leq 2/3$ and $0 \leq t \leq 2/5$ in [Equation \(3.10\)](#) therefore give

$$\|L_{s,t}\| = \sqrt{1 + s^2 + t^2} \leq \sqrt{1 + \frac{4}{9} + \frac{4}{25}} = \frac{19}{15} < \frac{4}{3}.$$

Using the vertex bound above, set $\delta_H = 5\delta_{s,t}$ and $\delta_P = 5\delta_{s,t} + (40/3)\delta_{\mathbf{r}}$. Corresponding hole vertices satisfy

$$\|L_{s,t}\mathbf{v} - L_{s',t'}\mathbf{v}\| \leq 5\delta_{s,t} = \delta_H.$$

For plug vertices, write the difference as

$$(L_{s,t} - L_{s',t'})R_P(\mathbf{r})\mathbf{v} + L_{s',t'}(R_P(\mathbf{r}) - R_P(\mathbf{r}'))\mathbf{v}.$$

Rotations preserve length, so the estimates just proved bound its norm by

$$5\delta_{s,t} + \frac{4}{3} \cdot 2\delta_{\mathbf{r}} \cdot 5 = 5\delta_{s,t} + \frac{40}{3}\delta_{\mathbf{r}} = \delta_P.$$

Taking convex combinations of corresponding vertices with the same coefficients gives the same bounds for points of the filled shadows, in both directions. Each rational-screen shadow contains the radius-two disk, since the spatial body contains the radius-two ball and $L_{s,t}(v_1, v_2, 0) = (v_1, v_2)$.

From displacement to margin. Let $m = M_{\mathcal{R}}(c)$ be the margin at c . Each point $\mathbf{p}' \in P_{c'}$ of the plug shadow at c' is within δ_P of some $\mathbf{p} \in P_c$. Since $P_c \subseteq mH_c$, write $\mathbf{p} = m\mathbf{h}$ for a point $\mathbf{h} \in H_c$ of the hole shadow at c . Choose $\mathbf{h}' \in H_{c'}$ within δ_H of $\mathbf{h}$. Then

$$\|\mathbf{p}' - m\mathbf{h}'\| \leq m\delta_H + \delta_P.$$

The radius-two disk and convexity of $H_{c'}$ imply

$$P_{c'} \subseteq \left(m + \frac{m\delta_H + \delta_P}{2}\right) H_{c'}.$$

Indeed, dividing $\mathbf{p}' = m\mathbf{h}' + (\mathbf{p}' - m\mathbf{h}')$ by the scale factor $m + (m\delta_H + \delta_P)/2$ gives a convex combination of $\mathbf{h}'$ and a point in the radius-two disk. Apply [Lemma 2.3](#), use $m \leq 5/2$, and interchange c, c' to obtain

$$|M_{\mathcal{R}}(c) - M_{\mathcal{R}}(c')| \leq \frac{(5/2)\delta_H + \delta_P}{2} \leq 16\|c - c'\|. \quad (6.4)$$

For the exact derivation of the last inequality, substitute the displacement bounds above:

$$\frac{(5/2)\delta_H + \delta_P}{2} = \frac{35}{4}\delta_{s,t} + \frac{20}{3}\delta_{\mathbf{r}}.$$

Since the parameter distance is Euclidean, $\|c - c'\|^2 = \delta_{s,t}^2 + \delta_{\mathbf{r}}^2$. Both $\delta_{s,t}$ and $\delta_{\mathbf{r}}$ are therefore at most $\|c - c'\|$, and

$$\frac{35}{4}\delta_{s,t} + \frac{20}{3}\delta_{\mathbf{r}} \leq \left(\frac{35}{4} + \frac{20}{3}\right)\|c - c'\| = \frac{185}{12}\|c - c'\| \leq 16\|c - c'\|.$$

□

Consequently, if $M_{\mathcal{R}}(c) = 1 + \varepsilon$ with $\varepsilon > 0$, then every $c' \in B_0$ with $\|c' - c\| \leq \varepsilon/32$ satisfies $M_{\mathcal{R}}(c') \geq 1 + \varepsilon/2 > 1$. This gives an explicit neighbourhood consisting entirely of poke configurations, including when c lies on a boundary of B_0 . The constant is a convenient bound derived from the geometry. The program checks specific support inequalities. We must still prove that those checks succeed on sufficiently small boxes; we do so in [Section 6.3](#).

Whole-box support inequalities. To choose a support direction that remains perpendicular to the changing viewing vector, take two distinct hole vertices $\mathbf{v}_a, \mathbf{v}_b$ and a sign $\sigma \in \{-1, 1\}$, and set

$$\mathbf{n}(s, t) = \sigma \mathbf{u} \times (\mathbf{v}_b - \mathbf{v}_a), \quad \mathbf{u} = (s, t, 1). \quad (6.5)$$

This vector is perpendicular to both the viewing direction and the segment joining the projected vertices. At a fixed configuration, choosing a pair along a silhouette edge gives a normal to its supporting line. Throughout a box, however, the pair need not remain on the silhouette or even be an edge of the polyhedron: the comparison with every hole vertex is what justifies the result. If the pair projects to a single point, $\mathbf{n} = \mathbf{0}$ and no strict separation can be certified with it.

Write $\hat{R}_P(\mathbf{r}) = dR_P(\mathbf{r})$ for the polynomial numerator of the rotation, where $d = 1 + \|\mathbf{r}\|^2 > 0$ is the denominator of the rotation formula. For each hole vertex $\mathbf{v} \in V$, define the *gap polynomial*

$$p_{\mathbf{v}}(c) = \mathbf{n}(s, t) \cdot (\hat{R}_P(\mathbf{r})\mathbf{v}_p - d\mathbf{v}). \quad (6.6)$$

Its total degree is at most three, and its coefficients lie in $\mathbb{Q}(\sqrt{5})$. If

$$p_{\mathbf{v}}(c) > 0 \quad \text{for every } c \in B \text{ and every } \mathbf{v} \in V, \quad (6.7)$$

then division by $d > 0$ proves [Equation \(6.1\)](#) at every configuration in the box. No hole vertex is singled out: the comparison with all sixty of them is what justifies the result, so a change inside the box in which vertices form the silhouette does not by itself invalidate the test. The certificate record stores the plug vertex and the ordered hole-vertex pair, whose order fixes the sign; these data determine all the polynomials to be checked.

Choosing vertices and a support direction. Enumerating the vertex pairs from [Equation \(2.1\)](#) gives exactly 120 pairs at squared distance four. Since the RID has 120 edges, each of length two in the normalisation of [Section 2.1](#), this list contains exactly its edges. Comparing their difference vectors up to sign gives fifteen distinct unoriented edge vectors. One orientation of each suffices. Reversing the sign of $\mathbf{n}$ and replacing the plug vertex $\mathbf{v}_p$ by its antipode $-\mathbf{v}_p$ only permutes the sixty gap polynomials, since

$$-\mathbf{n} \cdot (\hat{R}_P(\mathbf{r})\mathbf{v}_p - d\mathbf{v}) = \mathbf{n} \cdot (\hat{R}_P(\mathbf{r})(-\mathbf{v}_p) - d(-\mathbf{v}))$$

and the vertex set is centrally symmetric. The component therefore considers $15 \times 60 = 900$ candidates: one oriented edge from each class of parallel edges, with each plug vertex.

Only one candidate is tried on each box. It is chosen in floating point, as the candidate whose projected plug vertex lies farthest beyond the hole shadow's extent in the direction $\mathbf{n}$ at the worst of the box's 32 corners, measured as a distance in the shadow plane. If no candidate lies beyond at every

corner, none is tried. This choice only decides which exact test to run: a rounding error can make the component try a poorer candidate and leave the box unresolved, but never eliminate a box. Choosing at the worst corner rather than at the midpoint matters in practice, as the midpoint favours candidates that fail near the edges of the box.

Figure 10 shows two such certified separations, enlarged.

We apply the Bernstein sign test of Section 5.2 to each polynomial $-p_{\mathbf{v}}$. Testing the differences between plug and hole values directly preserves cancellations that would be lost by bounding plug and hole values separately. Clearing the rotation denominator preserves the inequalities' direction because $d = 1 + \|\mathbf{r}\|^2 \geq 1$ for every rotation parameter $\mathbf{r}$. Each gap polynomial has degree at most one in s and t and two in each r_i , so each test inspects at most $2 \cdot 2 \cdot 3^3 = 108$ coefficients.

Proposition 6.2 (Soundness of the *Global* component). *Suppose that, for a plug vertex and a signed edge direction, every tensor Bernstein coefficient on B of each of the sixty polynomials $-p_{\mathbf{v}}$ in Equation (6.6) is strictly negative. Then every configuration in B is a poke configuration, and the box can be eliminated.*

Proof. Lemma 5.1 gives $p_{\mathbf{v}} > 0$ throughout B for every hole vertex $\mathbf{v}$. Dividing by the positive d proves that the chosen projected plug vertex lies beyond every projected hole vertex in direction $\mathbf{n}(s, t)$. It therefore lies outside their convex hull, so $P_c \not\subseteq H_c$ and $M_{\mathcal{R}}(c) > 1$. $\square$

6.2 Robustly detecting potential Rupertness

The same support comparisons can identify an unexpected fit during the search. If $c \in B_0$ and $M_{\mathcal{R}}(c) = 1 - \varepsilon$ with $\varepsilon > 0$, then Lemma 6.1 gives

$$M_{\mathcal{R}}(c') \leq 1 - \frac{\varepsilon}{2} < 1 \quad \text{whenever } c' \in B_0 \text{ and } \|c' - c\| \leq \frac{\varepsilon}{32}.$$

This quantifies the openness of the set of fit configurations proved in Section 2.4. Sufficiently deep search boxes containing c lie entirely in this fitting neighbourhood by Lemma 4.1, so any point chosen inside such a box is a fit configuration. We also need an exact test that recognises it.

Exact detection at a configuration. The candidate normals derived from RID edges in Section 6.1 supply that test. Every point outside a convex polygon violates the supporting inequality of at least one of its edges, because the polygon is the intersection of the closed half-planes defined by these inequalities. An edge of the hole shadow is the projection of a supporting face of $\mathcal{R}$, obtained by maximizing the corresponding spatial dot product. Here that face can be a RID edge or one of its polygonal faces. That face contains a RID edge whose projection is a nonzero segment parallel to the shadow edge: otherwise all of its edges would project to points, and so would the face. Hence one of the signed directions in Equation (6.5), using a RID edge, strictly separates a protruding projected plug vertex.

At a fixed configuration, discard any candidate for which $\mathbf{n} = \mathbf{0}$. A necessary and sufficient condition for fit is

$$\max_{\mathbf{v} \in V} \mathbf{n} \cdot \widehat{R}_P(\mathbf{r})\mathbf{v} < d \max_{\mathbf{v} \in V} \mathbf{n} \cdot \mathbf{v} \quad \text{for every remaining candidate } \mathbf{n}. \quad (6.8)$$

Necessity follows because the compact plug shadow lies in the interior of the hole shadow, giving a strict support inequality in every nonzero direction. Conversely, the candidate family includes a normal to every edge of the hole shadow. Since $d > 0$, these strict inequalities put the entire plug shadow on the inner side of every edge's supporting line, and hence in the interior of the hole shadow.

Call the left side minus the right side of Equation (6.8) the *score* of the candidate; fit requires every nonzero candidate's score to be negative. At a configuration with rational parameters, all quantities in Equation (6.8) lie in $\mathbb{Q}(\sqrt{5})$. The finite maxima, zero tests and strict comparisons can therefore be computed exactly using Section 5.1. The test finishes even when vertices have coincident projections or several vertices have the same support value; equality between the two sides of Equation (6.8) fails the fit condition. In particular, checking only that no candidate has a positive score would also accept aligned configurations, whose scores are all zero.

Detection during subdivision. We can add this test to the procedure in Section 4.1: before subdividing an unresolved box, test its midpoint and stop if it is a fit configuration. The midpoint is rational because every interval endpoint is rational. Any other point in the box with rational parameters could be used instead. The sampled point need not belong to D : it still describes a valid physical configuration, and the containment test applies throughout B_0 .

Proposition 6.3 (Finite detection of fit). *Suppose every elimination component is sound and each component call finishes. If the increasing-depth search uses cyclic midpoint subdivision and the exact point test above, without a stopping limit, then if a fit exists, one is detected after finitely many processed boxes.*

Proof. If a fit exists, Proposition 3.1 supplies a fit representative $c_* \in D$. No sound component can eliminate a box containing c_* . Whenever such a box is subdivided, at least one of its closed children still contains c_* , even if the point lies on their shared boundary.

Write $M_{\mathcal{R}}(c_*) = 1 - \varepsilon$, where $\varepsilon > 0$. By Equation (4.2), at a sufficiently large finite depth every box containing c_* has diameter less than $\varepsilon/32$. Every point in such a box is then a fit configuration by the Lipschitz bound above, including its sampled rational point, which passes Equation (6.8). There are finitely many boxes up to that depth, and each component call and point test finishes, so increasing-depth processing reaches this box and detects a fit unless it has already detected one earlier. $\square$

The sufficient depth depends on the fitting margin, so this argument gives no uniform practical runtime estimate. On a Nopert input, completion still requires the elimination components to resolve all boxes. For bodies without central symmetry, the corresponding search would also need the translation parameters discussed in Section 2.2 and effective exact comparisons for their vertex coordinates.

The current certificate generator does not run this companion test and has a finite depth limit; the proposition describes the procedure with the test added and continued without such a limit.

6.3 Range and insufficiency for *touch* configurations

For whole-box elimination, finding a separating direction at one configuration is only the first step: its inequalities must be certified on the entire box. We describe the range of the component that may try every one of its 900 candidates. The implementation tries only the candidate proposed by its floating-point ranking; on boxes drawn uniformly from the final search, this never lost a box that another candidate eliminates. The final proof does not rely on either statement: it rests on the finished certificate, checked as described in Section 11.

Proposition 6.4 (Range of the *Global* component). *The range of the Global component is exactly*

$$\{c \in B_0 : \text{poke}(c)\}.$$

Proof. Fix a poke configuration c_0 . The edge argument in Section 6.2 gives a plug vertex and signed edge direction for which every gap polynomial is strictly positive at c_0 . By continuity, these sixty polynomials remain strictly positive on a closed neighbourhood U of c_0 in B_0 , which we may take to be a box. By Lemma 5.2, applied to each of them on U , every box contained in U whose widths are small enough passes the test of Proposition 6.2 with this candidate. Shrink U around c_0 so that all of its widths are that small. Then every search box contained in U is accepted, and U contains search boxes at arbitrarily large depths by Lemma 4.1. This is precisely the range condition in Definition 4.2.

Conversely, by Proposition 6.2, any accepted box contains only poke configurations. If c is a fit or touch configuration, no box containing it can be accepted. By Lemma 4.1, every neighbourhood of c contains a search box containing c , including when c lies on a subdivision boundary. Thus c cannot belong to this component's range. $\square$

Together with the compactness reasoning in Section 4.2, this shows that sufficiently fine subdivision handles every compact subset of the set of poke configurations. It does not give a uniform useful depth as such a subset approaches touch configurations. The Lipschitz bound quantifies how far the margin remains above one, while the proof above additionally establishes that the whole-box checks succeed locally.

Limitations at touch configurations. At a touch configuration, every projected plug vertex is either inside the hole shadow or on its silhouette. There is no protrusion, so the component's strict support inequalities cannot hold throughout a box containing that configuration. Further subdivision makes these boxes smaller without removing the touch configuration from every child. Aligned configurations already provide a whole family of such points.

The next component is designed to eliminate neighbourhoods of aligned configurations, including the touch configurations themselves, by studying how the plug's vertices move under sufficiently small relative rotations.

Figure 11 shows this on two slices.

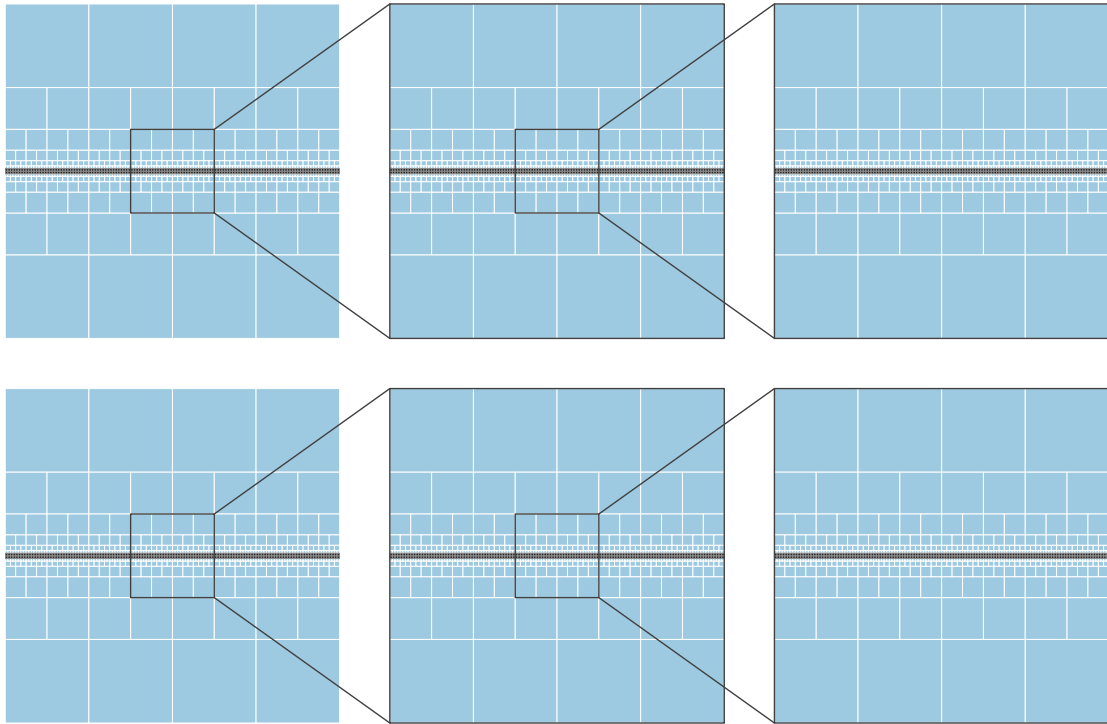


Figure 11: Subdivision with the Domain and Global components alone, near aligned configurations with ordinary viewing directions. Top: the plane $t = 1/10$, $r_1 = r_2 = 0$ around $s = 1/4$, $r_3 = 0$ (s horizontally, r_3 vertically). Bottom: the plane $s = 1/5$, $r_2 = r_3 = 0$ around $t = 1/8$, $r_1 = 0$ (t horizontally, r_1 vertically). From left to right, windows of half-width $1/8$, $1/32$ and $1/128$, each halved 14 times; light blue pieces are eliminated by the Global component, black pieces are unresolved. At every magnification a band of unresolved pieces remains along the aligned configurations: refinement alone never reaches a touch configuration.

7 Local component

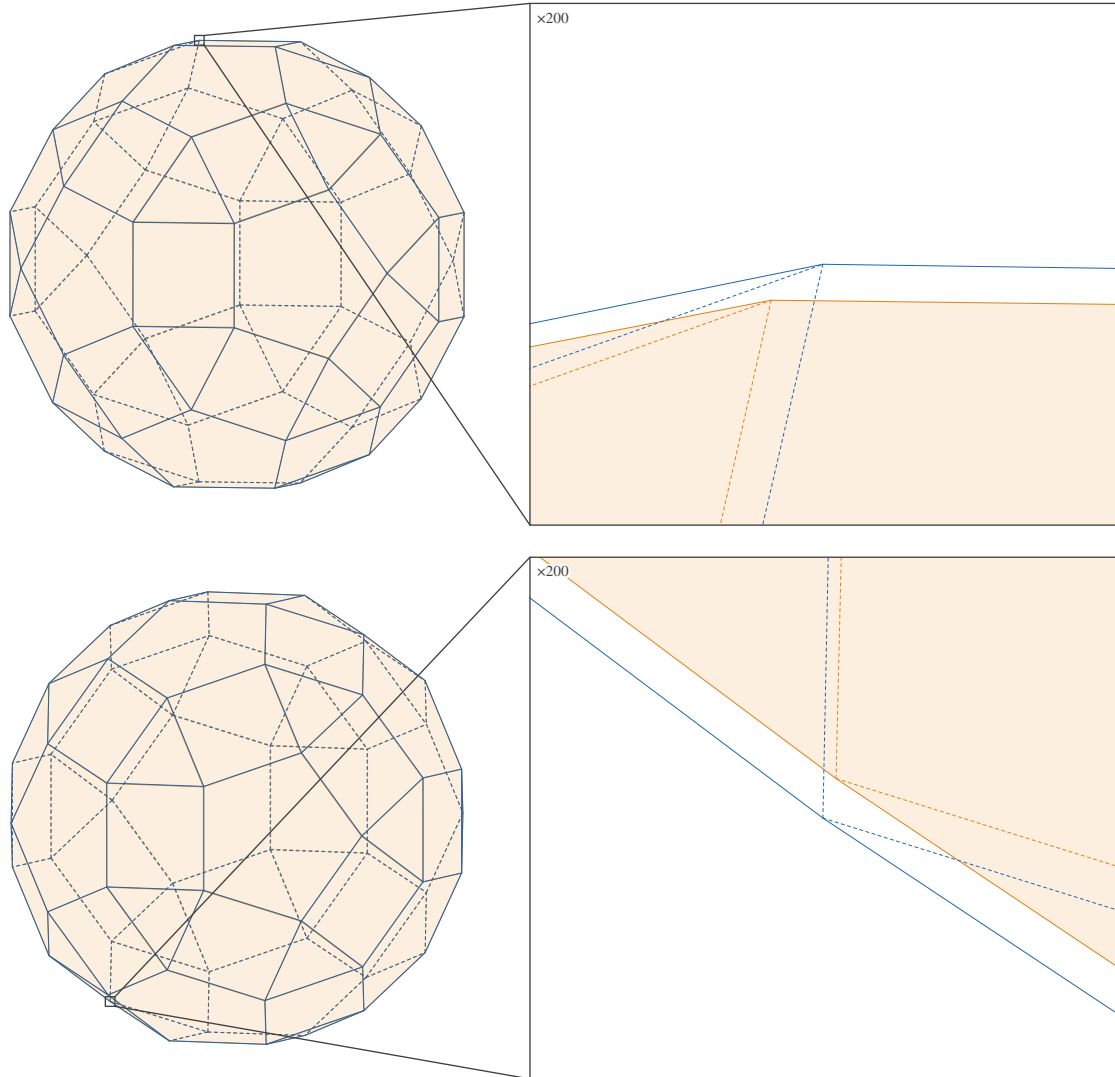


Figure 12: Two configurations near aligned ones, in the regions of Local covers: $[1/4, 1/20, (1/1000, 1/2000, -1/1000)]$ and $[2/5, 1/10, (-1/1000, 1/1000, 1/2000)]$. The **hole** (orange) and the **plug** (blue) as in Figure 10. The two shadows almost coincide; the enlargements, 200 times, show a projected plug vertex protruding beyond the hole's shadow by about 0.004, the outward motion that the face zooms certify for every small rotation.

At an aligned configuration, the plug and hole coincide. A small relative rotation can nevertheless make a projected plug vertex protrude beyond the hole's shadow. The *Local* component uses this outward motion to eliminate whole neighbourhoods containing aligned configurations, extending the Global component, which cannot eliminate any box containing an aligned configuration, as established in Section 6.3.

7.1 Geometry of *aligned* configurations

Use the representative $\mathbf{r} = \mathbf{0}$ for alignment, so the plug initially has the hole's orientation. Fix the viewing parameters (s, t) and rotate the plug about a fixed axis through the origin. A vertex off the axis moves on a circle; its orthogonal projection traces an ellipse, possibly reduced to a line segment. A vertex on the axis stays fixed. The rational screen map preserves these descriptions, since it is an invertible linear change of coordinates on the projection plane.

Choose a vertex whose initial projection lies on a supporting line of the hole shadow. If its projection moves beyond that line, away from the hole shadow, the plug cannot fit. We seek a positive interval of rotation throughout which this protrusion persists, starting immediately after zero rotation. An outward initial velocity can establish the direction of motion, but the curvature of the trajectory also matters when determining how far the plug can rotate before the vertex returns to the supporting line.

For a five-dimensional neighbourhood, this conclusion must hold uniformly as both (s, t) and the rotation axis vary. Different axes may require different vertices and supporting lines. We therefore use a finite family of such choices, each valid throughout a specified region of viewing parameters and rotation directions. Together they must account for every direction of rotation and give a common positive bound on its magnitude. When these bounds hold, they certify that the neighbourhood contains no fit configurations: zero-rotation configurations remain touch configurations, while every nonzero rotation within the bound yields a poke configuration.

7.2 Eliminating *aligned* configurations

For a nonzero rotation vector $\mathbf{r} = (r_1, r_2, r_3)$, separate its magnitude and direction by writing

$$\delta = \|\mathbf{r}\|_\infty := \max\{|r_1|, |r_2|, |r_3|\} > 0, \quad \hat{\mathbf{r}} = \frac{\mathbf{r}}{\delta}, \quad \mathbf{r} = \delta \hat{\mathbf{r}}. \quad (7.1)$$

The normalised vector lies on the boundary of the cube $[-1, 1]^3$. We denote its six closed faces by

$$F_i^\sigma = \{\hat{\mathbf{r}} \in [-1, 1]^3 : \hat{r}_i = \sigma\}, \quad i \in \{1, 2, 3\}, \quad \sigma \in \{-1, 1\}.$$

On F_i^σ , coordinate i is fixed at σ , and the other two belong to $[-1, 1]$. When several coordinates have maximum absolute value, the vector belongs to several faces, so the six faces together contain every direction. The two free face coordinates and δ describe the three rotational degrees of freedom.

We use the maximum norm because its unit sphere is made of six squares: the rotation directions on one face form a box, which is where the Bernstein sign test works, and a bound on δ bounds all three rotation coordinates directly. Here $\hat{\mathbf{r}}$ need not have Euclidean length one, and δ is not the rotation angle. By Equation (2.3), the unit axis and angle are respectively

$$\frac{\hat{\mathbf{r}}}{\|\hat{\mathbf{r}}\|}, \quad \theta = 2 \arctan(\delta \|\hat{\mathbf{r}}\|).$$

Supporting lines throughout a viewing region. Fix a closed axis-parallel rectangle $W \subseteq \mathbb{R}^2$ of viewing parameters (s, t) . Select a hole vertex $\mathbf{v} \in V$ and a normal $\mathbf{n}(s, t)$ perpendicular to $\mathbf{u} = (s, t, 1)$. For every $(s, t) \in W$, the projection of $\mathbf{v}$ must lie on a supporting line of the hole shadow with outward normal $\mathbf{n}(s, t)$:

$$\mathbf{n}(s, t) \cdot (\mathbf{w} - \mathbf{v}) \leq 0 \quad \text{for all } \mathbf{w} \in V. \quad (7.2)$$

As in the Global component, equalities are allowed. Since $\mathbf{n} \perp \mathbf{u}$, the same inequalities compare the orthogonal projections onto the screen. As the plug rotates, the projection of the vertex starting at $\mathbf{v}$ protrudes beyond the hole shadow as soon as $\mathbf{n} \cdot (R_P(\mathbf{r})\mathbf{v} - \mathbf{v}) > 0$.

The normal must stay perpendicular to $\mathbf{u}$ as (s, t) varies. We use either the moving direction Equation (6.5) of the Global component, or the direction

$$\mathbf{n}(s, t) = (n_1, n_2, -sn_1 - tn_2), \quad n_1, n_2 \in \mathbb{Q}(\sqrt{5}), \quad (n_1, n_2) \neq (0, 0), \quad (7.3)$$

which corresponds to the fixed planar support direction (n_1, n_2) in the screen coordinates $L_{s,t}$ of Equation (2.4). For either form, each of the sixty inequalities in Equation (7.2) is affine in (s, t) , so checking them at the four corners of W establishes support throughout W .

Displacement under rotation. To determine how long this protrusion persists, we need both its initial direction of motion and the bending of its circular trajectory. For a vertex off the rotation axis, $\hat{\mathbf{r}} \times \mathbf{v}$ points along the initial tangent, while $\hat{\mathbf{r}} \times (\hat{\mathbf{r}} \times \mathbf{v})$ points towards the axis. Taking their components along the outward normal gives the two displacement polynomials

$$p_1 = \mathbf{n} \cdot (\hat{\mathbf{r}} \times \mathbf{v}), \quad p_2 = \mathbf{n} \cdot (\hat{\mathbf{r}} \times (\hat{\mathbf{r}} \times \mathbf{v})). \quad (7.4)$$

Their variables are s, t and the two free coordinates of $\hat{\mathbf{r}}$ on F_i^σ . A positive p_1 indicates initial outward motion, whereas a negative p_2 opposes it as the rotation grows. Subtracting $\mathbf{v}$ from Equation (2.3), substituting $\mathbf{r} = \delta\hat{\mathbf{r}}$, and using

$$\hat{\mathbf{r}} \times (\hat{\mathbf{r}} \times \mathbf{v}) = \hat{\mathbf{r}}(\hat{\mathbf{r}} \cdot \mathbf{v}) - \|\hat{\mathbf{r}}\|^2 \mathbf{v}$$

gives the exact identity

$$\mathbf{n} \cdot (R_P(\delta\hat{\mathbf{r}})\mathbf{v} - \mathbf{v}) = \frac{2\delta}{1 + \delta^2\|\hat{\mathbf{r}}\|^2} (p_1 + \delta p_2). \quad (7.5)$$

Dividing out the size of the rotation. Both sides of Equation (7.5) vanish at $\delta = 0$, where the configuration is aligned. This is why no box containing an aligned configuration can be eliminated by testing the displacement directly: its sign is zero there, and the Bernstein sign test must refuse it. The factor δ , however, is explicit. In terms of the gap polynomial Equation (6.6) of the Global component, with the plug vertex $\mathbf{v}_p = \mathbf{v}$ and the rotation denominator $d = 1 + \delta^2\|\hat{\mathbf{r}}\|^2$, the identity reads

$$p_{\mathbf{v}} = \mathbf{n} \cdot (\hat{R}_P(\delta\hat{\mathbf{r}})\mathbf{v} - d\mathbf{v}) = 2\delta (p_1 + \delta p_2), \quad (7.6)$$

an identity of polynomials in s, t, δ and the two free coordinates of $\hat{\mathbf{r}}$. If the quotient $2(p_1 + \delta p_2)$ is positive throughout a closed region of these five variables with $0 \leq \delta \leq \delta_{\max}$, then the vertex protrudes at every positive δ in the region, and the configurations with $\delta = 0$ are aligned touch configurations, which are not fits. The Bernstein sign test can certify the quotient directly, and it can succeed even at $\delta = 0$, where the quotient is $2p_1$: it only needs outward initial motion.

This observation is the idea behind every remaining component of the proof. Near a set of configurations that are already known not to be fits, we change coordinates so that some coordinates measure the distance from that set, divide a necessary inequality exactly by a power of those distances, and certify the quotient with Bernstein coefficients.

7.3 The zoom lemma

The rest of the proof uses the same step with other sets in place of the aligned configurations and other distances in place of δ . We therefore state it once, in a form that covers every use.

Several of the coordinate changes below scale the viewing direction, so we allow the viewing vector to be any positive multiple of $(s, t, 1)$: a vector $\mathbf{u}$ with $u_3 > 0$ and a rotation vector $\mathbf{r}$ describe the configuration $[u_1/u_3, u_2/u_3, \mathbf{r}]$.

Definition 7.1 (No-fit sets and witnesses). A *no-fit set* is a set of configurations containing no fit. A *witness* is a polynomial p in $\mathbf{u}$ and $\mathbf{r}$, homogeneous in $\mathbf{u}$, together with finitely many linear forms $\lambda_1, \dots, \lambda_m$ in $\mathbf{u}$, possibly none, its *validity conditions*, such that $p \leq 0$ at every fit in D whose viewing vector satisfies $\lambda_j(\mathbf{u}) \geq 0$ for all j .

By homogeneity, the signs of p and of each λ_j do not depend on the positive multiple chosen for $\mathbf{u}$. The aligned configurations form a no-fit set: by Equation (2.3), $\mathbf{r} = \mathbf{0}$ gives $P_c = H_c$, and a compact shadow never lies in its own interior. Two kinds of witnesses suffice for the whole proof.

- A gap polynomial Equation (6.6), with $\mathbf{n}$ written as a linear function of $\mathbf{u}$ (for Equation (7.3), as $(u_3 n_1, u_3 n_2, -u_1 n_1 - u_2 n_2)$), whose validity conditions are the support inequalities Equation (7.2) of its hole vertex $\mathbf{v}$. Where they hold, the projection of $\mathbf{v}$ lies on a supporting line of the hole shadow with outward normal $\mathbf{n}$; a fit puts the projected plug vertex in the interior of the hole shadow, strictly on the inner side of that line, so the gap polynomial is negative there.
- A *domain polynomial*: the polynomial q of an inequality $q \leq 0$ defining D , written homogeneously in $\mathbf{u}$ (the fold inequality, for example, as $\ell_g^2 - \|\mathbf{u}\|^2$), without validity conditions. It is nonpositive on all of D .

Definition 7.2 (Zooms). A *zoom* is a polynomial map $\mathcal{Z}$ from a box Y of *zoom coordinates* $\mathbf{y} = (y_1, \dots, y_5)$ to viewing and rotation vectors, $\mathcal{Z}(\mathbf{y}) = (\mathbf{u}(\mathbf{y}), \mathbf{r}(\mathbf{y}))$, together with a set of designated *distance coordinates*, such that

1. $u_3(\mathbf{y}) > 0$ on Y , and each component of $\mathbf{u}(\mathbf{y})$ has degree at most one in each zoom coordinate;
2. each distance coordinate is nonnegative on Y , and wherever one of them vanishes, $\mathcal{Z}(\mathbf{y})$ lies in a fixed no-fit set, the zoom's no-fit set.

A monomial $\mathbf{y}^{\mathbf{a}} = \prod_j y_j^{a_j}$ in the distance coordinates is a *zoom factor*, and a closed box $C \subseteq Y$ is a *zoom cell*.

The name describes what a zoom does near its no-fit set. A distance coordinate measures how far a configuration is from that set, and the other coordinates describe the direction of departure in units of that distance. A thin neighbourhood of the no-fit set is thereby spread out over a whole box, much as a camera zooms in on a detail, and once the zoom factor is divided out, the behaviour of a witness arbitrarily close to the no-fit set becomes the behaviour of a polynomial on a box, where Bernstein coefficients can see it. The degree condition in the first requirement is what allows validity conditions to be checked at the corners of a cell.

Lemma 7.3 (Zoom lemma). *Let $\mathcal{Z}$ be a zoom, C a zoom cell and p a witness. Suppose that*

1. $p(\mathcal{Z}(\mathbf{y})) = \mathbf{y}^{\mathbf{a}} N(\mathbf{y})$ exactly, as polynomials, for a zoom factor $\mathbf{y}^{\mathbf{a}}$;
2. every validity condition of p holds at the viewing vectors $\mathbf{u}(\mathbf{y})$ of the corners $\mathbf{y}$ of C ;
3. every tensor Bernstein coefficient of N on C is strictly positive.

Then no configuration $\mathcal{Z}(\mathbf{y})$ with $\mathbf{y} \in C$ is a fit lying in D . More precisely, each such configuration lies in the zoom's no-fit set or satisfies $p > 0$ and the validity conditions of p ; for a gap polynomial, the latter makes it a poke configuration.

Proof. By Lemma 5.1 applied to $-N$, we have $N > 0$ throughout C . Let $\mathbf{y} \in C$. If a distance coordinate with $a_j > 0$ vanishes at $\mathbf{y}$, then $\mathcal{Z}(\mathbf{y})$ lies in the no-fit set. Otherwise $\mathbf{y}^{\mathbf{a}} > 0$, so $p(\mathcal{Z}(\mathbf{y})) > 0$, and by homogeneity $p > 0$ at the configuration itself. Each validity condition $\lambda_j(\mathbf{u}(\mathbf{y}))$ is linear in $\mathbf{u}$, and $\mathbf{u}(\mathbf{y})$ has degree at most one in each zoom coordinate, so the condition is affine in each coordinate separately; varying one coordinate at a time, its minimum over C is attained at a corner. The validity conditions therefore hold throughout C , and since p is a witness, $p > 0$ excludes a fit in D . For a gap polynomial, $p > 0$ places the projected plug vertex strictly beyond a supporting line of the hole shadow. $\square$

The lemma separates what is proved by hand from what is checked by the computer. By hand: that a set is a no-fit set, that a polynomial is a witness, and that the zooms reach every configuration of the region concerned. By exact computation, cell by cell: the division, the validity conditions at the corners and the signs of the Bernstein coefficients. The division must be exact; a single term that is not a multiple of the zoom factor means the cell is refused.

The tests of the preceding components are the simplest case, with nothing divided out. With the identity map on a search box as the zoom, no distance coordinates and the zoom factor one, the lemma for a domain polynomial is exactly the test of the Domain component. The test of the Global component is the same idea for gap polynomials, except that instead of checking the validity conditions of one hole vertex at the corners, it requires the gap polynomials of all sixty hole vertices to be positive, so that whichever hole vertex supports the shadow, its gap polynomial is positive.

Tube zooms. The zooms of this paper are of two kinds. Both use coordinates adapted to a no-fit set: after an invertible affine change of variables, the *base* coordinates $\mathbf{z}_b$ run along the no-fit set and the *offset* coordinates $\mathbf{z}_o$ measure the displacement away from it, so that the configurations with $\mathbf{z}_o = \mathbf{0}$ lie in the no-fit set. As for the rotation vector above, the offsets are normalised on the faces of a box. Let $\Phi \subseteq [-1, 1]^m$ be a box whose every side is $[-1, 0]$, $[0, 1]$ or $[-1, 1]$; a side $[0, 1]$ or $[-1, 0]$ records that an offset coordinate has a fixed sign. The *cone* of Φ is the set of vectors whose coordinates have the signs that Φ allows: nonnegative where its side is $[0, 1]$, nonpositive where it is $[-1, 0]$. For $\mathbf{z} \neq \mathbf{0}$ in the cone, $\mathbf{z}/\|\mathbf{z}\|_\infty$ lies on a *face* of Φ , a side where one coordinate equals ± 1 .

Let the base coordinates range over a box Σ_b , and let $\delta_{\max} > 0$. For each face F of Φ , the *tube zoom* $\mathcal{T}_F$ has as zoom coordinates the base coordinates $\mathbf{z}_b \in \Sigma_b$, the distance coordinate $\delta \in [0, \delta_{\max}]$ and the free coordinates of $\hat{\mathbf{f}} \in F$, and it sets $\mathbf{z}_o = \delta \hat{\mathbf{f}}$. The *covered set* of the family is

$$\{(\mathbf{z}_b, \mathbf{z}_o) : \mathbf{z}_b \in \Sigma_b, \mathbf{z}_o \in \delta_{\max} \Phi\}. \quad (7.7)$$

Lemma 7.4 (Tube coverage). *Every configuration in the covered set of a family of tube zooms lies in the no-fit set or is the image of a point in the domain of one of them.*

Proof. Let $\delta = \|\mathbf{z}_o\|_\infty \leq \delta_{\max}$. If $\delta = 0$, the configuration lies in the no-fit set. Otherwise $\hat{\mathbf{f}} = \mathbf{z}_o/\delta$ lies on a face F of Φ , and $(\mathbf{z}_b, \delta, \hat{\mathbf{f}})$ is a point of the domain of $\mathcal{T}_F$. $\square$

Tube zooms divide out one distance, the size of the offset. They are used around the aligned configurations below and along two families of non-aligned touch configurations in [Section 9](#). At a few special points, one distance is not enough, and [Section 8](#) introduces *point zooms*, which divide out two: the distance along the no-fit set from the special point, and the offset measured relative to it. Where two families of touch configurations cross, [Section 10](#) adds a *hand-over* from zooms around one family to zooms around the other.

Zoom covers. The cells of a family of zooms are found by subdivision. The domain of each zoom is bisected in one coordinate at a time, keeping both children, until every terminal cell carries a witness and a zoom factor satisfying the hypotheses of [Lemma 7.3](#). The union of the children is their parent, so by induction the terminal cells cover the domain. A family of zooms with such subdivisions and witnesses is a *zoom cover*; its covered set is the covered set of its family. The cells and witnesses are found by the program's generator (`rid generate`), but every cell is checked exactly before use, so how they were found does not matter.

7.4 Face zooms and the *Local* component

The *face zooms* are the tube zooms around the aligned configurations. Their base coordinates are the viewing parameters $(s, t) \in W$, their offset is the rotation vector $\mathbf{r}$ with $\Phi = [-1, 1]^3$, whose six faces are the F_i^σ , and

$$\mathbf{u} = (s, t, 1), \quad \mathbf{r} = \delta \hat{\mathbf{r}}, \quad \hat{\mathbf{r}} \in F_i^\sigma, \quad 0 \leq \delta \leq \delta_{\max}.$$

Where the distance coordinate δ vanishes, $\mathbf{r} = \mathbf{0}$ and the configuration is aligned. By [Equation \(7.6\)](#), a gap polynomial whose plug vertex is its hole vertex is divisible by the zoom factor δ . The same holds for any plug vertex $\mathbf{v}_p$ whose unrotated projection lies on the same supporting line throughout the cell, for example the other end of a hole edge along it, since the gap then vanishes at $\delta = 0$; the exact division checks this. The covered set of the six face zooms is the region

$$W \times [-\delta_{\max}, \delta_{\max}]^3. \quad (7.8)$$

Proposition 7.5 (Soundness of the *Local* component). *Suppose a zoom cover of face zooms with covered set [Equation \(7.8\)](#) has been checked, with a gap polynomial and a zoom factor δ or 1 in every cell. Then every configuration in that region is a touch configuration when $\mathbf{r} = \mathbf{0}$ and a poke configuration otherwise. Any search box contained in the region can therefore be eliminated.*

Proof. By [Lemma 7.4](#), a configuration of the region with $\mathbf{r} \neq \mathbf{0}$ is the image of a point of some face zoom's domain, and a terminal cell contains this point. By [Lemma 7.3](#), the cell's witness is positive there, so a projected plug vertex lies strictly beyond a supporting line of the hole shadow and the configuration is a poke configuration. Zero rotation yields an aligned and therefore touch configuration. $\square$

Checking a search box. The *Local* component has a finite list of viewing rectangles W with checked zoom covers and bounds $\delta_{\max}$. It accepts a search box that lies in one of the regions [Equation \(7.8\)](#): the product of its two viewing intervals must lie in W , and each of its three rotation intervals in $[-\delta_{\max}, \delta_{\max}]$. Its certificate record names the region, so that these endpoint comparisons can be repeated. The covers are checked once, before the search; the test for a box is only this inclusion. [Figure 13](#) shows the effect of adding the *Local* component.

7.5 Range and exotic viewing directions

The *Local* component has 30 zoom covers of this kind, with bounds $\delta_{\max}$ between $1/200$ and $1/20$ and 833 terminal cells in all. Their viewing rectangles come from 30 rectangles with disjoint interiors, each enlarged by $1/1024$ on every side (and cut to the viewing rectangle $[0, 2/3] \times [0, 2/5]$ of B_0), so that neighbouring rectangles overlap and a small search box on the boundary between two of them lies inside one. For example, one of the regions is

$$\left[\frac{1}{6} - \frac{1}{1024}, \frac{1}{3} + \frac{1}{1024} \right] \times \left[0, \frac{1}{10} + \frac{1}{1024} \right] \times \left[-\frac{1}{50}, \frac{1}{50} \right]^3.$$

Since $\|\mathbf{r}\|_\infty \leq \|\mathbf{r}\| = \tan(\theta/2)$, such a region contains the rotations about every axis through angles $0 \leq \theta \leq 2 \arctan \delta_{\max}$.

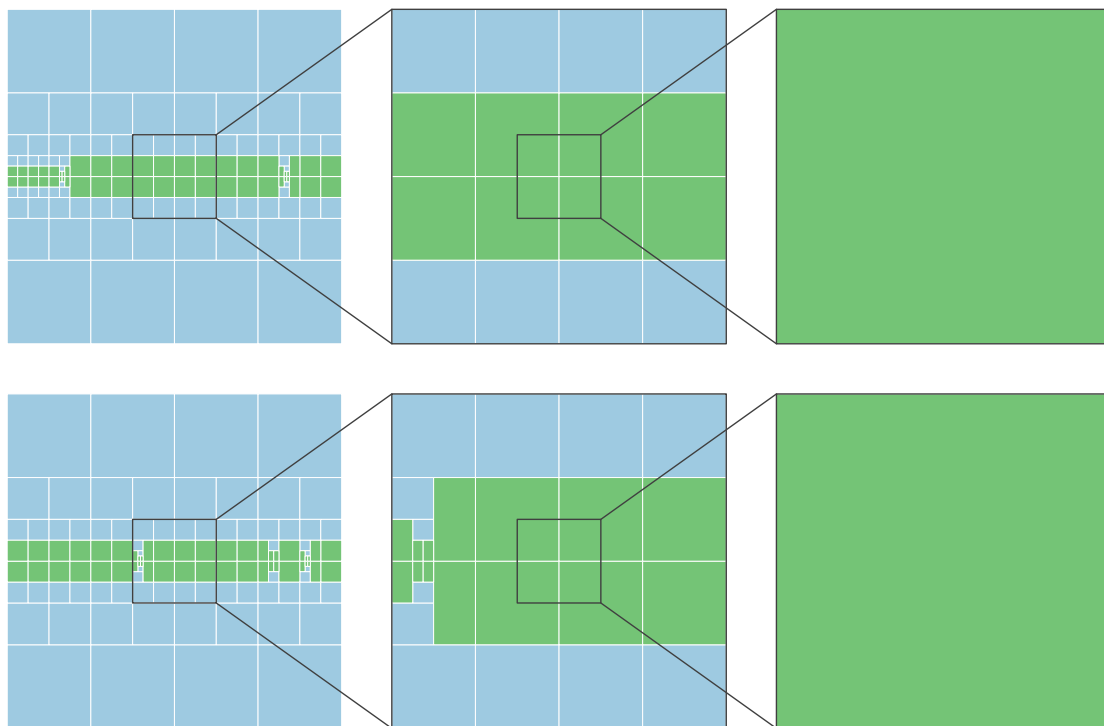


Figure 13: The slices of Figure 11 with the Local component added (green). The unresolved band is gone: near the aligned configurations every piece lies in the region of a Local cover, and in the smallest window the whole window does at once.

The relative interior in B_0 of each region lies in the Local component's range (Definition 4.2): every configuration there has a neighbourhood inside the region, and every search box within that neighbourhood passes the inclusion test. In Section 8.3 we show that these regions, together with the covers of the next section, contain a neighbourhood of every aligned configuration of D .

Exotic viewing directions. The rectangles leave out two viewing directions, and no face zoom cover can reach them. At each of them, for every vertex and every supporting line valid throughout a neighbourhood of that viewing direction, the quotient $2p_1$ in Equation (7.6) vanishes at $\delta = 0$ for some rotation directions, so no face zoom cell containing them can pass the test, however small. Both are directions from which faces of the RID are seen edge-on: the direction $(s, t) = (0, 0)$, along which the four square faces with normals $\pm \mathbf{e}_1$ and $\pm \mathbf{e}_2$ are seen edge-on, and a direction on the boundary of the viewing triangle from which a pentagonal face is seen edge-on. Near such a direction, a small change of the viewing direction changes which vertices of the edge-on faces belong to the silhouette, and supports that work on one side fail on the other. We located the two directions by subdivision: with the Global and Local components alone, the unresolved boxes concentrate around them (Figure 14). Section 8 divides out a second distance there.

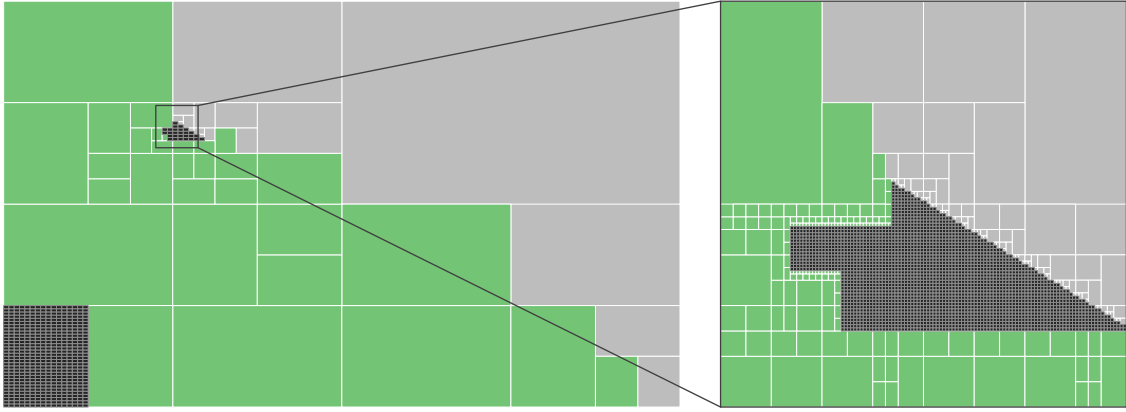


Figure 14: The plane $\mathbf{r} = \mathbf{0}$ of aligned configurations, with $s \in [0, 2/3]$ horizontally and $t \in [0, 2/5]$ vertically, halved 14 times with the Domain (grey), Global and Local (green) components; right, the window of half-width $1/48$ around the point where the edge-on line $s = (\varphi - 1)t$ meets the long side of the viewing triangle. Every configuration of this plane is a touch configuration, so the Global component eliminates none of it: beyond the viewing triangle the Domain component removes it, and inside only the Local covers eliminate pieces. They reach every viewing direction of the triangle except those near the corner $(0, 0)$ and near that point, where the pieces stay unresolved (black) at any depth.

8 Square and pentagon configurations

The two viewing directions left out by the Local component belong to two aligned configurations. We call the aligned configuration with $(s, t) = (0, 0)$ the *square* configuration: its viewing line passes through opposite square faces. The second, the *pentagon* configuration, has

$$(s, t) = \left(\frac{-5 + 3\sqrt{5}}{10}, \frac{5 - \sqrt{5}}{10} \right), \quad (8.1)$$

where the boundary $\varphi s + \varphi^2 t = 1$ of the viewing triangle meets the line $s = (-1 + \varphi)t$, along which a pentagonal face with normal $(1, 1 - \varphi, 0)$ is seen edge-on.

Three things make these configurations harder than the other aligned ones, and the rest of the section deals with them in turn.

1. At both, for every vertex and supporting line valid near the special viewing direction, the initial outward motion vanishes for some directions of rotation, and dividing out the size of the rotation

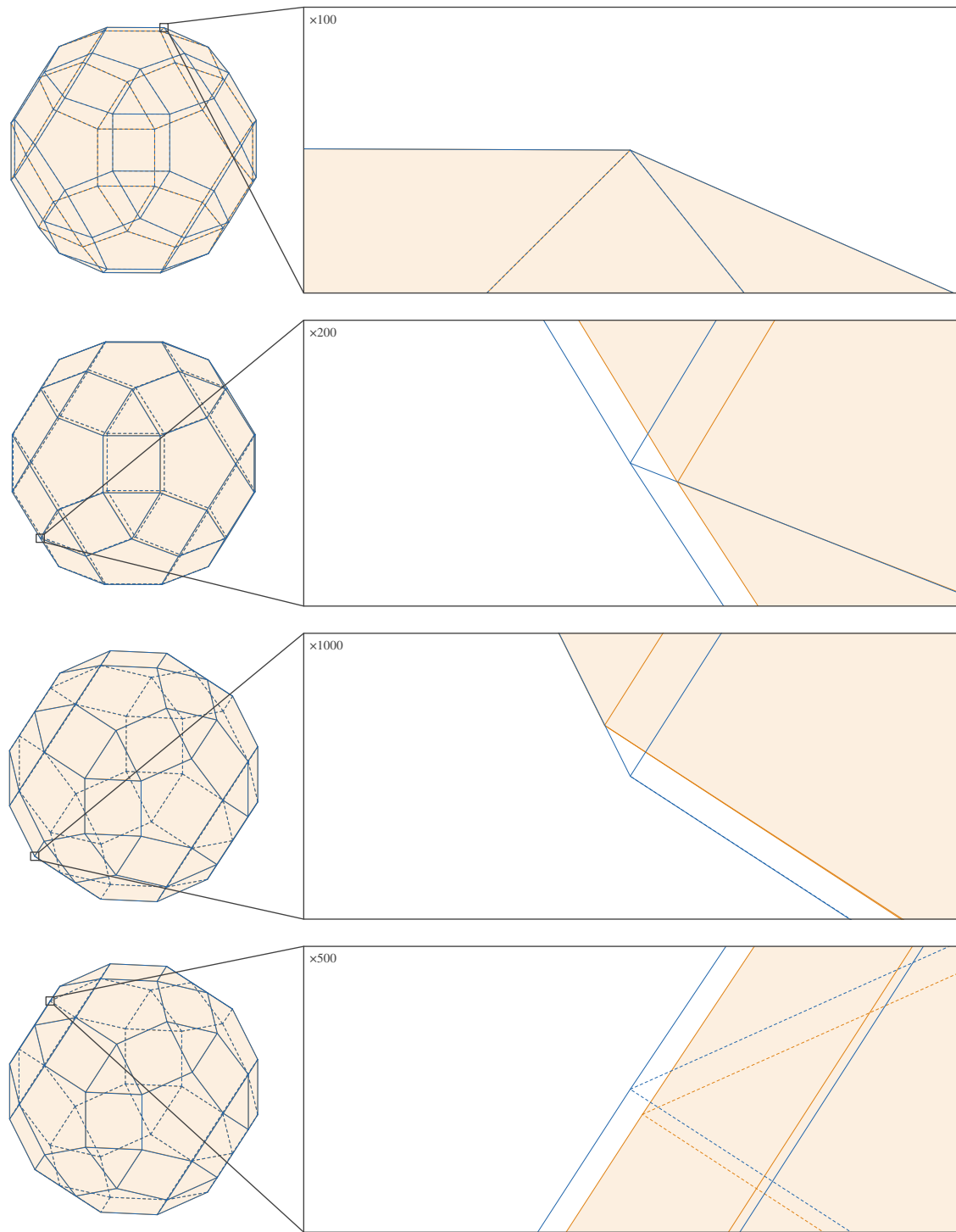


Figure 15: Near the square configuration (top two rows) and the pentagon configuration (bottom two); the **hole** (orange) and the **plug** (blue) as in Figure 10. First: the configuration $[1/16, 1/16, (-1/16, 1/16, 0)]$ of the half-turn family, whose plug shadow and projected edges coincide with the hole's, so that a single wireframe is visible, enlarged 100 times at a vertex of the outline; it lies outside D . Second: the poke configuration $[1/64, 1/128, (1/1000, -1/1000, -1/1000)]$, with a plug vertex protruding by about 0.005 (enlarged 200 times). Third and fourth: the viewing parameters $(\eta, \xi) = (1/200, -1/400)$ and $(-1/200, -1/400)$ of Section 8.2, on the two sides of the line along which a pentagonal face is seen edge-on, with the rotation $\mathbf{r} = (1/1000, 0, 0)$; a different plug vertex protrudes on each side (enlarged 1000 and 500 times).

is not enough. *Point zooms*, below, divide out a second distance: the distance of the viewing parameters from those of the special configuration.

2. At the square configuration, a family of touch configurations with nonzero rotation reaches the aligned configuration. The fold inequalities of Equation (3.11) remove it (Section 8.1).
3. At the pentagon configuration, the vertices of the edge-on face that form the silhouette differ on the two sides of the edge-on line, and so do the useful supports (Section 8.2).

Both configurations are handled by one further component, the *Exotic* component. Like the Local component, it eliminates a search box by checking that the box lies in the covered set of a zoom cover that was checked once, before the search. It has two covers here, one around each configuration, and six more in Sections 9 and 10.

Why one distance is not enough. A model shows the difficulty. Let $x \geq 0$ measure how far the viewing parameters are from those of the special configuration, and let $y \geq 0$ be the size of the rotation. Suppose a witness behaves like

$$p = y(x + y).$$

A face zoom divides by y and leaves $x + y$, which is positive except at $x = y = 0$, where it vanishes. No zoom cell containing the special configuration can then pass the test of Lemma 7.3, however small. The Local component meets exactly this behaviour at the square and pentagon configurations: in the directions of rotation described in Section 7.5, the initial outward motion of every projected vertex, measured against any support valid near the special configuration, is proportional to x .

Point zooms. The remedy is to measure the rotation relative to x , using two families of zooms around the special configuration (Figure 16).

- In *family A*, the rotation is at most ρ_0 times the distance of the viewing parameters: that distance is μ , and the rotation has size $\rho\mu$ with $0 \leq \rho \leq \rho_0$. Both μ and ρ are distance coordinates. In the model, $p = \mu^2\rho(1 + \rho)$.
- In *family B*, the rotation is larger: its size is δ and the distance of the viewing parameters is $b\delta$ with $0 \leq b \leq 1/\rho_0$. The distance coordinate is δ . In the model, $p = \delta^2(b + 1)$.

After division by the zoom factors $\mu^2\rho$ and δ^2 , both quotients are at least one throughout their zoom boxes, including the special configuration itself, which has become the whole face $\mu = 0$ or $\delta = 0$. The ratio $\rho_0 > 0$ is a parameter; any positive value works in the model.

In general, we use base and offset coordinates as for the tube zooms of Section 7.3, now around a point: the special configuration is $\mathbf{z}_b = \mathbf{z}_o = \mathbf{0}$, and the configurations with $\mathbf{z}_o = \mathbf{0}$ lie in the no-fit set. Both parts are normalised on the faces of boxes: the offsets on the faces of a box $\Phi \subseteq [-1, 1]^m$, and the base directions on the faces of a box $\Sigma \subseteq [-1, 1]^{5-m}$, each with sides $[-1, 0]$, $[0, 1]$ or $[-1, 1]$.

Given a base radius $\mu_{\max}$, an offset radius $\delta_{\max}$ and the ratio ρ_0 , the *point zooms* are

- $\mathcal{A}_{G,F}$, for each face G of Σ and F of Φ : $\mathbf{z}_b = \mu\hat{\mathbf{b}}$ and $\mathbf{z}_o = \rho\mu\hat{\mathbf{f}}$, with $\hat{\mathbf{b}} \in G$, $\hat{\mathbf{f}} \in F$, $0 \leq \mu \leq \mu_{\max}$ and $0 \leq \rho \leq \rho_0$;
- $\mathcal{B}_F$, for each face F of Φ : $\mathbf{z}_b = \delta\hat{\mathbf{b}}$ and $\mathbf{z}_o = \delta\hat{\mathbf{f}}$, with $\hat{\mathbf{f}} \in F$, $0 \leq \delta \leq \delta_{\max}$ and $\hat{\mathbf{b}}$ in the cone of Σ with $\|\hat{\mathbf{b}}\|_{\infty} \leq 1/\rho_0$.

The free coordinates of $\hat{\mathbf{b}}$ and $\hat{\mathbf{f}}$ on their faces are zoom coordinates, and every coordinate of these maps is affine in each zoom coordinate separately. Where a distance coordinate vanishes, $\mathbf{z}_o = \mathbf{0}$. The *covered set* of the family is

$$\{(\mathbf{z}_b, \mathbf{z}_o) : \mathbf{z}_b \in \mu_{\max}\Sigma, \mathbf{z}_o \in \delta_{\max}\Phi\}. \quad (8.2)$$

Lemma 8.1 (Point coverage). *Every configuration in the covered set Equation (8.2) lies in the no-fit set or is the image of a point in the domain of one of the zooms $\mathcal{A}_{G,F}$ or $\mathcal{B}_F$.*

Proof. Let $\mu = \|\mathbf{z}_b\|_{\infty} \leq \mu_{\max}$ and $\delta = \|\mathbf{z}_o\|_{\infty} \leq \delta_{\max}$. If $\delta = 0$, the configuration lies in the no-fit set. Otherwise $\hat{\mathbf{f}} = \mathbf{z}_o/\delta$ lies on a face F of Φ . If $0 < \delta \leq \rho_0\mu$, then $\hat{\mathbf{b}} = \mathbf{z}_b/\mu$ lies on a face G of Σ , and $\rho = \delta/\mu \in (0, \rho_0]$ gives a point of the domain of $\mathcal{A}_{G,F}$. If $\delta > \rho_0\mu$, then $\hat{\mathbf{b}} = \mathbf{z}_b/\delta$ lies in the cone with $\|\hat{\mathbf{b}}\|_{\infty} = \mu/\delta < 1/\rho_0$, a point of the domain of $\mathcal{B}_F$. $\square$

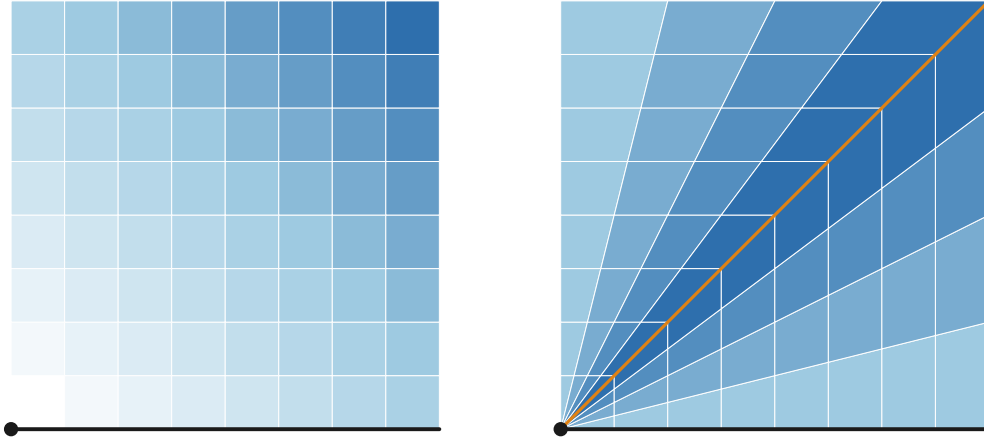


Figure 16: The model $p = y(x + y)$: the distance $x \geq 0$ of the viewing parameters horizontally and the size $y \geq 0$ of the rotation vertically, both in $[0, 1]$; the no-fit set $y = 0$ is the black line and the special configuration the black dot. Each cell is shaded by the smallest value of its quotient, from white for 0 to dark blue for 2. Left: the cells of a face zoom, which divides by y ; the quotient $x + y$ fades to 0 at the special configuration, so the cell there fails however small it is. Right: the point zooms with $\rho_0 = 1$. Below the diagonal (orange), family A, with the segments of constant distance μ and the rays of constant ratio ρ ; above it, family B, with the segments of constant rotation size δ and the rays of constant ratio b . The quotients $1 + \rho$ and $1 + b$ are at least 1 on every cell, including the cells at the special configuration.

The proof bounds each face G of Σ separately, so the base radius may also depend on the face; we use this in [Section 10.1](#). A search box is accepted by a cover when all its configurations lie in the covered set. For the covers of this section, every base and offset coordinate is an affine combination of s , t and $\mathbf{r}$, so its exact range over a box is attained at corners, and the test consists of exact endpoint comparisons.

8.1 Eliminating the *square* configuration

Around the square configuration, the base coordinates are (s, t) and the offset coordinates are $\mathbf{r}$, with $\Sigma = [0, 1]^2$, $\Phi = [-1, 1]^3$, $\mu_{\max} = \delta_{\max} = 1/8$ and $\rho_0 = 1$. The distance of the viewing parameters is $\mu = \max\{s, t\}$, and the normalised viewing parameters $(\hat{s}, \hat{t}) = (s, t)/\mu$ lie on one of the two sides of the unit square where a coordinate equals one. The rotation in family A is $\rho\mu\hat{\mathbf{r}}$ with $\hat{\mathbf{r}}$ on a face F_i^σ of the cube. This gives $2 \times 6 = 12$ zooms in family A and six in family B.

For example, one zoom of family A is

$$[s, t, \mathbf{r}] = [\mu, \mu\hat{t}, \mu\rho(-1, y_1, y_2)],$$

with zoom coordinates $\mu, \rho, \hat{t}, y_1, y_2$. On the zoom cell $\mu \in [0, 1/8]$, $\rho \in [1/8, 1/4]$, $\hat{t} \in [0, 1/4]$ and $(y_1, y_2) \in [-1, 1]^2$, the cell's witness is a gap polynomial with a fixed screen direction [Equation \(7.3\)](#). Pulled back along the zoom, it has 216 Bernstein coefficients, ranging from 0 to about 0.315, and 54 of them are zero, so the test must refuse it. After exact division by the zoom factor $\mu\rho$, its 108 coefficients range from about 2.03 to 11.8, and the cell is proved.

The half-turn family. By the quaternion identity [Equation \(3.9\)](#), the configurations

$$c = [s, t, (-t, s, 0)]$$

have the plug rotation $R_P = J_{\mathbf{u}}Z$. The half-turn Z is a symmetry of $\mathcal{R}$, while the half-turn $J_{\mathbf{u}}$ about the viewing line reverses every projected vector, and central symmetry makes the reversed shadow identical to the original. So $P_c = H_c$, as in [Equation \(3.3\)](#): these are touch configurations with nonzero rotation, and as $(s, t) \rightarrow (0, 0)$ they approach the square configuration. Their rotation size equals the distance of their viewing parameters, $\rho = 1$, on the boundary between the two families, so they meet the zoom

domains, and no gap polynomial is positive on them. Cells meeting them use instead the fold inequality of Equation (3.8) for the half-turn Z , $g = k$, which is violated on the whole family except at $s = t = 0$. The zoom lemma then shows that every configuration of such a cell outside the no-fit set violates the fold, and so lies outside D . This is the only place in this section where the fold is needed.

The cover has 877 terminal cells: 782 with gap polynomials and 95 with the fold inequality.

Proposition 8.2 (The square neighbourhood). *Every configuration $c \in D$ with $(s, t) \in [0, 1/8]^2$ and $\|\mathbf{r}\|_\infty \leq 1/8$ is a touch configuration if $\mathbf{r} = \mathbf{0}$ and a poke configuration otherwise.*

Proof. By Lemma 8.1, c is aligned, and therefore a touch configuration, or the image of a point in a zoom domain, which lies in a checked cell. If that cell's witness is the fold inequality and c is not aligned, Lemma 7.3 shows that c violates the fold, contradicting $c \in D$. Otherwise the witness is a gap polynomial, and the lemma shows that c is aligned or a poke configuration. $\square$

Remark 8.3 (Why finitely many cells suffice). Soundness needs only the checked cells, but it is worth seeing why a finite set of cells exists at all. Divide each witness by its zoom factor and let $\mu \rightarrow 0$ in family A. What remains depends only on the normalised viewing parameters $(\hat{s}, \hat{t})$ and on the rotation measured in units of μ , $\mathbf{z} = \mathbf{r}/\mu$. Write $\boldsymbol{\tau} = (z_2, -z_1)$ for the direction in which the plug's third axis tilts, and z_3 for its spin about the viewing line. The four square faces with normals $\pm \mathbf{e}_1$ and $\pm \mathbf{e}_2$ are seen edge-on, and we write $\mathbf{a} = (\varphi^2, 2 + \varphi, 0)$ for a vertex of the silhouette. Expanding each witness to lowest order in μ gives the following limits.

Witnesses	What protrudes	Limit after division
$L_x^\pm, L_y^\pm$	a corner of an edge-on square with third coordinate -1	$-4(\tau_1 \pm z_3), -4(\tau_2 \mp z_3)$
$U_x^\pm, U_y^\pm$	a corner of the same square with third coordinate $+1$	$4(\tau_1 \mp z_3) - 4\hat{s}, 4(\tau_2 \pm z_3) - 4\hat{t}$
O_1, O_2	the vertex $\mathbf{a}$, past one of two edges through it	$4z_3, -4z_3$
radial	the vertex $\mathbf{a}$, in the direction $\mathbf{m}_0$ of its projection	$2l(l_0 - l)$ when $z_3 = 0$
fold	nothing: the fold inequality for $g = k$	$2(\hat{s}\tau_1 + \hat{t}\tau_2) - \hat{s}^2 - \hat{t}^2$

Here $\mathbf{m}_0 = (\varphi^2, 2 + \varphi)$, $l = \mathbf{m}_0 \cdot \boldsymbol{\tau}$ and $l_0 = \mathbf{m}_0 \cdot (\hat{s}, \hat{t})$; each L witness compares a corner with third coordinate -1 with the supporting line of the square's edge through it, and each U witness compares the opposite corner, with third coordinate $+1$, with that same line (hence the terms $-4\hat{s}$ and $-4\hat{t}$). Some witness has a positive limit at every point:

1. if the plug spins, $z_3 \neq 0$, then O_1 or O_2 is positive;
2. without spin, if some $\tau_i < 0$, then an L witness is positive;
3. without spin, if $\boldsymbol{\tau} \geq \mathbf{0}$ exceeds $(\hat{s}, \hat{t})$ in some coordinate, then U_x^+ or U_y^+ is positive;
4. without spin, if $\mathbf{0} \leq \boldsymbol{\tau} \leq (\hat{s}, \hat{t})$ with $\boldsymbol{\tau}$ equal to neither end, then $l > 0$ and $l_0 - l > 0$, because both entries of $\mathbf{m}_0$ are positive, and the radial witness is positive;
5. $\boldsymbol{\tau} = (\hat{s}, \hat{t})$ without spin is the half-turn family, where the fold limit is $\hat{s}^2 + \hat{t}^2 > 0$;
6. the limits $\mathbf{z} \rightarrow \mathbf{0}$, the face $\rho = 0$ of family A, and $\mathbf{z} \rightarrow \infty$, which is family B, depend only on the direction of the rotation, and the same witnesses cover every direction.

The limit set is compact and the Bernstein coefficients converge, by Lemma 5.2. So some finite set of cells, each with one of these witnesses, covers every zoom domain near $\mu = 0$ or $\delta = 0$; away from those faces, the reasoning of the Local component applies. The program that found the cover tries witnesses of these kinds, and the case analysis explains why it could succeed.

8.2 Eliminating the *pentagon* configuration

At the pentagon configuration, a pentagonal face is seen edge-on along a line of viewing parameters, and, for supports valid near the configuration, a small rotation about one axis in the shadow plane moves no silhouette vertex outward at first order. On the two sides of the edge-on line, different vertices of that face form the silhouette, so the useful supports change across it (Figures 15 and 17). Moreover, the configuration lies on the boundary of the viewing triangle, so a search box around it always reaches outside D .

The edge-on line and the boundary of the triangle suggest coordinates that describe this geometry directly. Write

$$\bar{\varphi} = 1 - \varphi = -\frac{1}{\varphi} = \frac{1 - \sqrt{5}}{2}$$

for the *algebraic conjugate* of φ , the other root of $x^2 - x - 1 = 0$, and define

$$\eta = s + \bar{\varphi}t, \quad \xi = s + \varphi t + \bar{\varphi}. \quad (8.3)$$

The pentagon configuration has $\eta = \xi = 0$. The line $\eta = 0$ is $s = -\bar{\varphi}t$, along which the pentagonal face is seen edge-on, and

$$\varphi\xi = \varphi s + \varphi^2 t - 1$$

shows that the viewing-triangle inequality of D is exactly $\xi \leq 0$. The inverse change of coordinates is

$$s = \frac{-5 + 3\sqrt{5}}{10} + \frac{\varphi\eta - \bar{\varphi}\xi}{\sqrt{5}}, \quad t = \frac{5 - \sqrt{5}}{10} + \frac{\xi - \eta}{\sqrt{5}}, \quad (8.4)$$

so a rectangle in (η, ξ) describes a parallelogram in (s, t) .

Around the pentagon configuration, the base coordinates are (η, ξ) and the offset coordinates are $\mathbf{r}$. Since the domain requires $\xi \leq 0$, the base box is $\Sigma = [-1, 1] \times [-1, 0]$, with the three faces $\eta = \pm 1$ and $\xi = -1$, and $\Phi = [-1, 1]^3$. The radii are $\mu_{\max} = 1/32$ and $\delta_{\max} = 1/16$, and the ratio is $\rho_0 = 2$. This gives $3 \times 6 = 18$ zooms in family A and six in family B. The cover has 6,173 terminal cells, all with gap polynomials; each carries its own witness, and the cells on the two sides of the edge-on line use supports from the two sides.

Every configuration with $\xi > 0$ lies outside D and needs no witness, so we drop the upper bound on ξ from the covered set: a search box is accepted when all its configurations satisfy $|\eta| \leq 1/32$, $\xi \geq -1/32$ and $\|\mathbf{r}\|_{\infty} \leq 1/16$. This is what allows boxes of the cyclic bisection, whose sides never meet the irrational boundary line exactly, to be eliminated around the pentagon configuration.

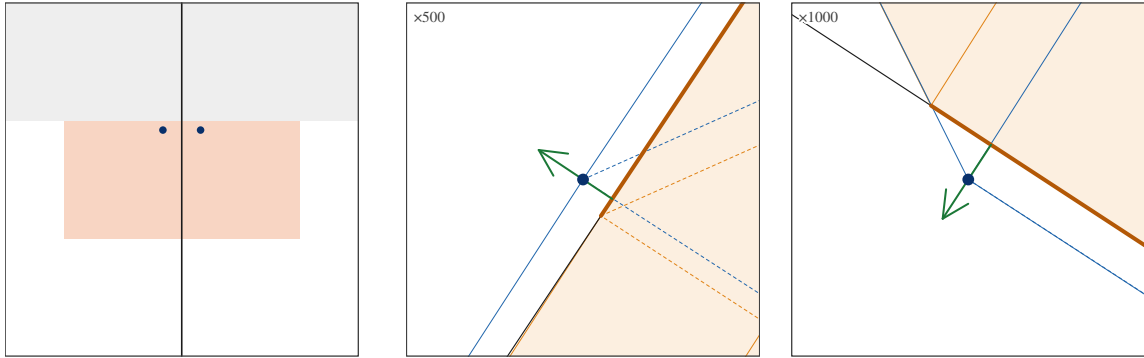


Figure 17: The two sides of the edge-on line near the pentagon configuration. Left: the pentagon cover's viewing parameters in the coordinates (η, ξ) , with $\eta \in [-3/64, 3/64]$ horizontally and $\xi \in [-1/16, 1/32]$ vertically: the covered rectangle $|\eta| \leq 1/32$, $-1/32 \leq \xi \leq 0$ (red-orange), the views beyond the viewing triangle, $\xi > 0$ (grey), the edge-on line $\eta = 0$ (black) and two examples (dots) at $(\eta, \xi) = (-1/200, -1/400)$ and $(1/200, -1/400)$, both with the rotation $\mathbf{r} = (1/1000, 0, 0)$. Middle and right: their shadows near the protruding plug vertex, enlarged 500 and 1000 times, with the hole (orange) and the plug (blue) as in Figure 10 and a supporting line of the hole's shadow that this vertex (dark blue dot) lies beyond. On the two sides, different plug vertices protrude beyond different hole edges.

Proposition 8.4 (The pentagon neighbourhood). *Every configuration $c \in D$ with $|\eta| \leq 1/32$, $-1/32 \leq \xi \leq 0$ and $\|\mathbf{r}\|_{\infty} \leq 1/16$ is a touch configuration if $\mathbf{r} = \mathbf{0}$ and a poke configuration otherwise.*

Proof. As for Proposition 8.2, using Lemma 8.1 and Lemma 7.3; every cell's witness is a gap polynomial. $\square$

8.3 A neighbourhood of all aligned configurations

The square and pentagon covers, together with the 30 covers of the Local component, give a uniform neighbourhood of all aligned representatives. Recall that a configuration lies in the range of the collection when the collection accepts every search box in some neighbourhood of it (Definition 4.2). Write Δ for the viewing triangle Equation (3.2), $W_0 = [0, 2/3] \times [0, 2/5]$ for the viewing rectangle of B_0 , and $\delta_0 = 1/200$ for the smallest rotation bound among the 32 covers of aligned configurations. Each of these covers has a set of viewing parameters: a rectangle, or, for the pentagon cover, the half-strip $|\eta| \leq 1/32$, $\xi \geq -1/32$ of its acceptance test (the upper bound on ξ is dropped because configurations with $\xi > 0$ lie outside D). Each cover contains every configuration whose viewing parameters lie in its set and whose rotation satisfies $\|\mathbf{r}\|_\infty \leq \delta_0$. Shrink each set by $\beta = 1/4096$ in the maximum norm, except along the sides of W_0 (the half-strip has no upper side to shrink). An exact computation, bisecting W_0 and comparing endpoints in $\mathbb{Q}(\sqrt{5})$, shows that the shrunk sets still cover Δ ; without the margin of $1/1024$ around each Local rectangle, which makes neighbouring rectangles overlap, this would fail.

Proposition 8.5 (A neighbourhood of all aligned representatives). *For every $(s_0, t_0) \in \Delta$, some cover of aligned configurations contains every configuration $[s, t, \mathbf{r}] \in B_0$ with $\max\{|s - s_0|, |t - t_0|\} \leq \beta$ and $\|\mathbf{r}\|_\infty \leq \delta_0$. Every such configuration in D is a touch configuration when $\mathbf{r} = \mathbf{0}$ and a poke configuration otherwise. In particular, the union*

$$\bigcup_{(s_0, t_0) \in \Delta} \{[s, t, \mathbf{r}] \in B_0 : \max\{|s - s_0|, |t - t_0|\} < \beta, \|\mathbf{r}\|_\infty < \delta_0\} \quad (8.5)$$

lies in the collection's range.

Proof. The pair (s_0, t_0) lies in the shrunk set of some cover, so the maximum-norm ball of radius β around it, cut to W_0 , lies in that cover's set of viewing parameters. The geometric claim follows from Propositions 7.5, 8.2 and 8.4.

For the range, let c lie in the union Equation (8.5). It has a neighbourhood in B_0 inside one of the closed sets just described, and hence inside the covered set of one cover. Every search box contained in that neighbourhood lies in the covered set, so the cover's exact inclusion test accepts it. $\square$

Because of the overlaps, the conclusion does not depend on where the boundaries of the covers lie relative to the halving lines of the search, and every aligned representative lies in the range, even on a boundary of the viewing triangle.

Remaining touch configurations. With the Global and Local components and these two covers, the unresolved boxes no longer concentrate around aligned configurations. They concentrate instead along two segments of configurations in which the plug and the hole have different orientations but their shadows still touch (Figure 18). We describe them and the covers along them in Section 9.

9 Arc configurations

The boxes left unresolved by the preceding components cluster around configurations with $t = 0$, a range of values of s , and a relative rotation close to a turn through 36° about a five-fold axis of the RID, followed by a small turn about the second coordinate axis. For $t = 0$, that axis lies in the shadow plane, and a turn about it does not change how far the plug reaches in its direction. The 36° turn happens to leave that reach equal to the hole's, so whatever the second turn, the plug's shadow reaches exactly as far as the hole's in that direction, and none of these configurations is a fit. This gives two planes of configurations, each a no-fit set for a reason as simple as for the aligned configurations. Around the touch configurations in them, the Exotic component uses tube zooms that divide out the distance from such a plane, much as the Local component divides out the size of the rotation.

9.1 Geometry of *non-aligned touch* configurations

The arc planes. Let

$$a_1 = \frac{-5 + 3\sqrt{5}}{10}, \quad a_3 = \frac{-5 + \sqrt{5}}{10}, \quad \mathbf{r}_A = (a_1, 0, a_3),$$

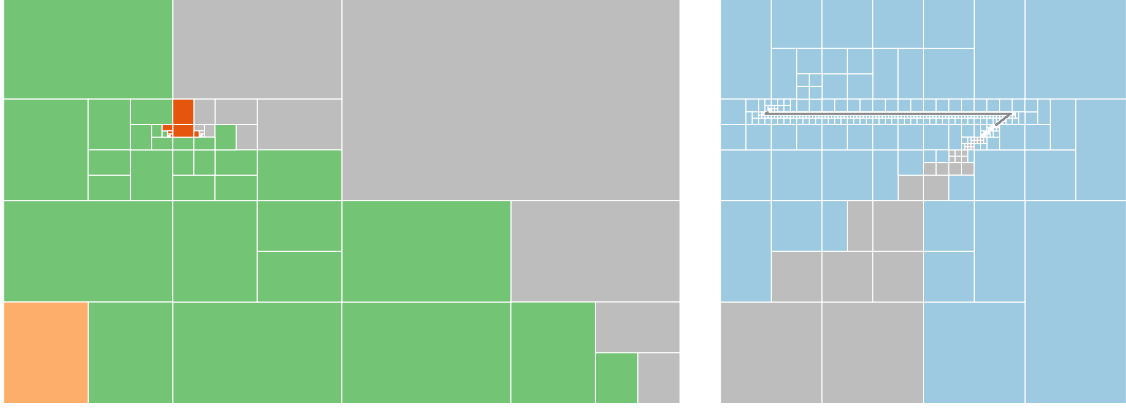


Figure 18: With the square and pentagon covers added. Left: the plane $\mathbf{r} = \mathbf{0}$ of Figure 14, now eliminated entirely, near the corner by the square cover (light orange) and near the edge-on point by the pentagon cover (red-orange). Right: a plane of non-aligned configurations with $t = 0$, with $s \in [3/16, 5/8]$ horizontally and a rotation parameter $\theta \in [-5/16, 1/8]$ vertically (it is the plane Π_1 of Section 9). Unresolved pieces (black) remain along two segments that meet at $s = 1/2$.

and, for $\sigma \in \{-1, 1\}$, $\mathbf{w}_\sigma = (\sigma a_3, 1, -\sigma a_1)$. The two *arc planes* are the families of configurations

$$\Pi_\sigma = \{[s, 0, \sigma \mathbf{r}_A + \theta \mathbf{w}_\sigma] : s, \theta \in \mathbb{R}\}, \quad \sigma \in \{-1, 1\}, \quad (9.1)$$

two-dimensional sets of configurations with $t = 0$. Since $\|\mathbf{r}_A\| = \tan 18^\circ$, the rotation $R_P(\mathbf{r}_A)$ is a turn through 36° about the axis of $\mathbf{r}_A$, which is a five-fold axis of the RID; its square is a symmetry of $\mathcal{R}$, but it is not one itself. With $\mathbf{e}_2 = (0, 1, 0)$, the quaternion product gives

$$(1, \theta \mathbf{e}_2)(1, \sigma \mathbf{r}_A) = (1 - \theta \sigma \mathbf{e}_2 \cdot \mathbf{r}_A, \theta \mathbf{e}_2 + \sigma \mathbf{r}_A + \theta \sigma \mathbf{e}_2 \times \mathbf{r}_A) = (1, \sigma \mathbf{r}_A + \theta \mathbf{w}_\sigma),$$

because the second coordinate of $\mathbf{r}_A$ is zero. Hence $R_P(\sigma \mathbf{r}_A + \theta \mathbf{w}_\sigma) = R_P(\theta \mathbf{e}_2)R_P(\sigma \mathbf{r}_A)$: the plug is first turned through 36° and then through the angle $2 \arctan \theta$ about the second coordinate axis.

Lemma 9.1 (The arc planes contain no fit). *No configuration in Π_1 or Π_{-1} is a fit.*

Proof. On Π_σ we have $t = 0$, so $\mathbf{e}_2$ is perpendicular to $\mathbf{u} = (s, 0, 1)$: it is a direction in the shadow plane, and $\mathbf{e}_2 \cdot \mathbf{x}$ is unchanged by projection. A turn about the second coordinate axis does not change $\mathbf{e}_2 \cdot \mathbf{x}$ either. So the plug's support value in the direction $\mathbf{e}_2$ is

$$\max_{\mathbf{v} \in V} \mathbf{e}_2 \cdot R_P(\sigma \mathbf{r}_A) \mathbf{v} = 2 + \sqrt{5} = \max_{\mathbf{v} \in V} \mathbf{e}_2 \cdot \mathbf{v},$$

the hole's support value; the middle equality is an exact comparison over the sixty vertices of Equation (2.1). A fit places the compact plug shadow in the interior of the hole shadow, which requires a strictly smaller support value in every direction of the shadow plane. $\square$

The two planes describe the same plug positions, since $R_P(\mathbf{r}_A)^2$ is a symmetry of the plug, but they are different subsets of B_0 . Both lie in the hyperplanes on which the following inequalities hold with equality, so their configurations in D lie on the boundary of D : on Π_σ , the relative-rotation comparison

$$\sigma(\varphi^{-1} r_1 - r_3) \leq 2 - \varphi \quad (9.2)$$

of Equation (3.11) (the one with $i = 3$, $\epsilon = -\sigma$ and $\delta = \sigma$) holds with equality, because $\varphi^{-1} a_1 - a_3 = 2 - \varphi$ and $a_1 + \varphi^{-1} a_3 = 0$. Along the plane, two representatives of the same configuration tie, and the comparison decides between them; we call Equation (9.2) the *tie inequality* of Π_σ . Every construction below comes in two copies, one for each sign σ ; the map $(r_1, r_2, r_3) \mapsto (-r_1, r_2, -r_3)$ exchanges the two planes, keeping s and θ , together with their tie inequalities.

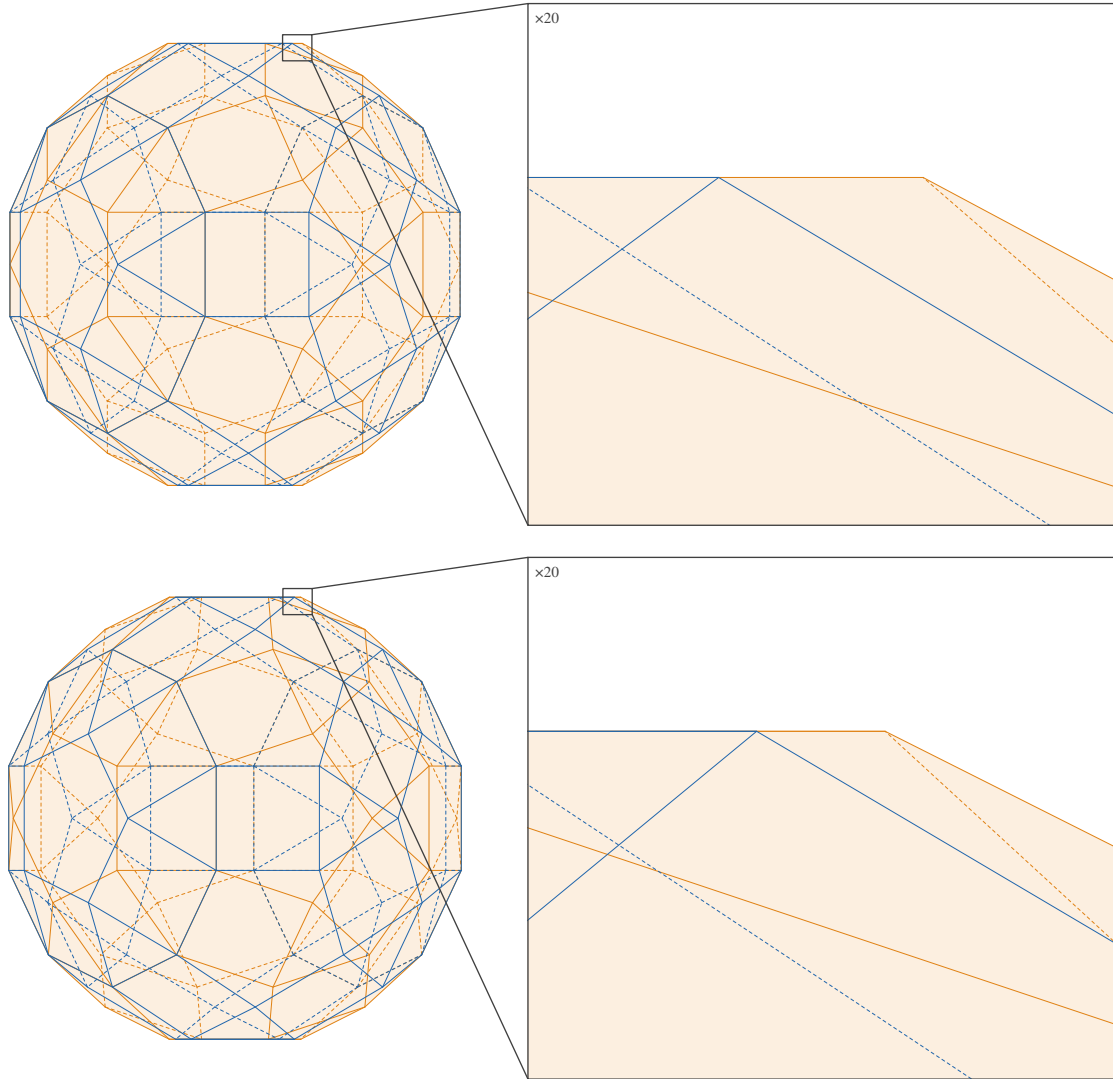


Figure 19: Two touch configurations on the first arc of the plane Π_1 , at $e = 1/10$ and $e = 3/20$ ($t = 0$, $\theta = 0$); the **hole** (orange) and the **plug** (blue) as in Figure 10. Both shadows reach exactly as far in the upward screen direction, which is $\mathbf{e}_2$, and share their top edge. The enlargements, 20 times at the right ends of the two top edges, show the plug's top edge ending inside the hole's.

Arc coordinates. Near Π_σ we use the coordinates

$$e = \frac{1-2s}{2+s}, \quad v = \frac{5t}{2+s}, \quad \theta = r_2, \quad \zeta_1 = r_1 - \sigma(a_1 + a_3\theta), \quad \zeta_3 = r_3 - \sigma(a_3 - a_1\theta), \quad (9.3)$$

defined for $s > -2$. Then

$$\mathbf{u} = (1-2e, v, 2+e) = \frac{5}{2+s}(s, t, 1), \quad \mathbf{r} = \sigma \mathbf{r}_A + \theta \mathbf{w}_\sigma + (\zeta_1, 0, \zeta_3),$$

so $\mathbf{u}$ is a positive multiple of $(s, t, 1)$ with components affine in e and v , as zooms require, and every witness is again a polynomial in the new coordinates. The plane Π_σ is $v = \zeta_1 = \zeta_3 = 0$. The coordinate e is chosen so that both families of touch configurations below are straight lines in the plane, crossing at $e = 0$; this matters for the shear in [Section 10.2](#).

The arcs. Inside each plane lie two segments of touch configurations ([Figures 19](#) and [22](#)):

- the *first arc*, $\theta = 0$ with $0 \leq e \leq e_* = \sqrt{5} - 2$, that is $s \in [\sqrt{5} - 2, 1/2]$;
- the *second arc*, $\theta = -e$ with $0 \leq e \leq e_*$.

Along the first arc, the two shadows share their top edge in the direction $\mathbf{e}_2$, the plug's top edge ends inside the hole's, and the rest of the plug's shadow lies inside the hole's. At the *endpoint* $e = e_*$, the ends of the two top edges coincide; beyond it, the plug's top edge is the longer one, and its corner pokes out. The two arcs meet at the *crossing*, $e = 0$, where $s = 1/2$. Elsewhere in the plane, the configurations are poke configurations, and the Global component handles them ([Figure 20](#)). These properties explain where zooms are needed; the proof itself uses only [Lemma 9.1](#).

The second arc lies outside D except at the crossing. On it, the polynomial of one of the fold inequalities of [Equation \(3.11\)](#), written homogeneously in the viewing vector $(1-2e, v, 2+e)$, equals $5e^2 + 5e^4$, which is positive for $e > 0$. Sufficiently small boxes around its configurations with $e > 0$ are therefore eliminated by the Domain component, and no cover follows the second arc; near the crossing it is handled in [Section 10](#).

9.2 Eliminating arc configurations

Along the first arc, a small displacement off the plane Π_σ moves some projected plug vertex beyond the hole shadow at a rate proportional to the displacement. This is the situation of the aligned configurations, and the tube zooms of [Section 7.3](#) apply.

The arc covers. Along the first arc of Π_σ , the base coordinate is e and the offset coordinates are $(v/2, \theta, \zeta_1, \zeta_3)$, weighting v by one half. The offset box is $\Phi = [0, 1] \times [-1, 1]^3$, since $t \geq 0$ forces $v \geq 0$. This gives seven tube zooms, with $e \in [3/64, 3/16]$ and $\delta_{\max} = 1/32$, so the covered set consists of the configurations with

$$\frac{3}{64} \leq e \leq \frac{3}{16}, \quad 0 \leq v \leq \frac{1}{16}, \quad |\theta|, |\zeta_1|, |\zeta_3| \leq \frac{1}{32}.$$

The no-fit set of the zooms is the plane Π_σ : where $\delta = 0$, the offsets vanish, so the configuration lies on the first arc. The view $\mathbf{u} = (1-2e, v, 2+e)$ has $u_3 = 2+e > 0$ and components affine in each zoom coordinate, as [Definition 7.2](#) requires. The weight of v , the radii and the range of e are tuning choices: the proof needs only that every cell passes its check and that the covers overlap as described in [Section 10.3](#).

Most cells carry gap polynomials with the zoom factor δ . Part of the covered set lies on the far side of the tie inequality, outside D ; there the tie inequality itself serves as the witness. The cover for $\sigma = 1$ has 343 terminal cells, 32 of them with the tie inequality, and the cover for $\sigma = -1$ has 288, with 72.

Proposition 9.2 (The arc neighbourhoods). *For each $\sigma \in \{-1, 1\}$, no configuration of D in the covered set above is a fit.*

Proof. By [Lemma 7.4](#), such a configuration lies in Π_σ , which contains no fit by [Lemma 9.1](#), or in a checked cell. There [Lemma 7.3](#) applies to the cell's witness: a gap polynomial excludes a fit, and the tie inequality shows that the configuration lies outside D . $\square$

A search box lies in the covered set exactly when the ranges of the five arc coordinates over the box lie in the stated intervals. The coordinates e and v are monotone in s and in t separately, and θ, ζ_1 and ζ_3 are affine in $\mathbf{r}$, so each range is attained at corners of the box and the inclusion test consists of exact comparisons.

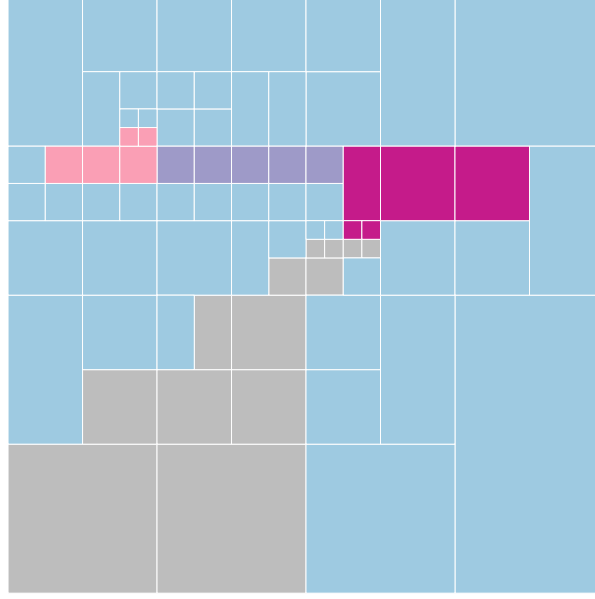


Figure 20: A slice of the final collection through the plane Π_1 : horizontally $s \in [3/16, 5/8]$, vertically $\theta \in [-5/16, 1/8]$ (with $\mathbf{r}_A$ rounded to rationals within 10^{-14}), halved alternately until one component eliminates each piece, which is coloured by that component: **light blue** Global, **grey** Domain, **lavender** the arc cover, **light pink** the endpoint cover (near $s = \sqrt{5} - 2$) and **magenta** the crossing cover (near $s = 1/2$). The first arc is the segment $\theta = 0$ between these two values. The Global component handles the configurations of the plane away from the arcs, and the Domain component the pieces outside D , including those along the second arc away from the crossing.

Where the arc covers stop. The arc covers stop short of both $e = 0$ and $e = e_*$. The rate at which a displacement off the plane moves some projected vertex outward shrinks linearly towards both ends of the first arc, like e at the crossing and like $e_* - e$ at the endpoint. At those two points the quotient after dividing by δ vanishes in some directions, exactly as at the square and pentagon configurations, and the unresolved boxes concentrate there (Figure 21). We treat the endpoint and the crossing with point zooms in Section 10.

10 *Endpoint and crossing configurations*

At the two ends of the first arc, the rate of outward motion off the arc plane shrinks to zero, as it did at the square and pentagon configurations. The remedy is the same: point zooms that divide out two distances, the distance along the arc from the special configuration and the displacement off the arc measured relative to it. The crossing needs one further idea, because the second arc passes through it.

Both covers use the arc coordinates Equation (9.3) of one plane Π_σ and come in two copies, one for each sign σ . Their base coordinate is e measured from the special configuration, and their offset coordinates are those of the arc covers, with v weighted.

10.1 Eliminating the *endpoint* configuration

At the endpoint $e = e_*$, the first arc ends: beyond it, the configurations of the plane poke. Around it we use a point zoom cover as in Section 8, with base coordinate $e - e_*$ and $\Sigma = [-1, 1]$, offset coordinates $(v/2, \theta, \zeta_1, \zeta_3)$ with $\Phi = [0, 1] \times [-1, 1]^3$, and ratio $\rho_0 = 1$. The base radius is $1/16$ on the face $e < e_*$ and $1/32$ on the face $e > e_*$, and the offset radius is $\delta_{\max} = 1/32$. The two base faces and seven offset faces give 14 zooms in family A and seven in family B. The covered set consists of the configurations with

$$-\frac{1}{16} \leq e - e_* \leq \frac{1}{32}, \quad 0 \leq v \leq \frac{1}{16}, \quad |\theta|, |\zeta_1|, |\zeta_3| \leq \frac{1}{32}.$$

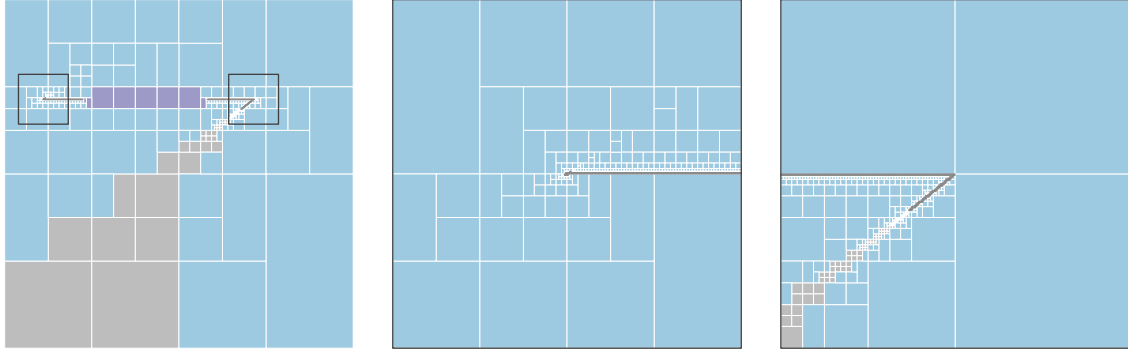


Figure 21: The plane Π_1 as in Figure 20, halved 18 times with the arc covers added (lavender) but not yet the endpoint and crossing covers. Unresolved pieces (black) remain only near the two ends of the first arc. Middle: the window of half-width $1/32$ around the endpoint, $s = \sqrt{5} - 2$, $\theta = 0$; right: the same around the crossing, $s = 1/2$, $\theta = 0$, where the second arc leaves the first. Away from the crossing, the second arc lies outside D (grey).

The no-fit set of the zooms is the plane Π_σ , by Lemma 9.1. As at the square configuration, some directions of displacement move no projected vertex outward at first order at the endpoint itself, and the cells there use two kinds of witnesses: one whose limit after division contains the base coordinate, and a radial one.

Proposition 10.1 (The endpoint neighbourhoods). *For each $\sigma \in \{-1, 1\}$, no configuration of D in the covered set above is a fit.*

Proof. As for Proposition 9.2, with Lemma 8.1 in place of Lemma 7.4. □

10.2 Eliminating the *crossing* configuration

At the crossing $e = 0$, the second arc $\theta = -e$ meets the first. The point zoom cover around it has base coordinate e with $\Sigma = [-1, 1]$, offset coordinates $(4v/5, \theta, \zeta_1, \zeta_3)$ with $\Phi = [0, 1] \times [-1, 1]^3$, radii $\mu_{\max} = \delta_{\max} = 1/16$ and ratio $\rho_0 = 5/4$: 14 zooms in family A and seven in family B, with the covered set

$$|e| \leq \frac{1}{16}, \quad 0 \leq v \leq \frac{5}{64}, \quad |\theta|, |\zeta_1|, |\zeta_3| \leq \frac{1}{16}.$$

The distance coordinates of these zooms vanish only on the segment $\theta = 0$, $|e| \leq 1/16$ of Π_σ , which contains the first arc near the crossing; the no-fit set of the zooms is Π_σ , by Lemma 9.1. The second arc, however, is also a family of touch configurations, and it runs through the domain of one zoom. In the family A zoom on the base face $e = \mu > 0$ and the offset face $\theta = -1$, the second arc is the set where $\rho = 1$ and the other offset coordinates vanish. No gap polynomial is positive at a touch configuration. The fold inequality is violated on the second arc for $e > 0$ (Section 9), but at the crossing its polynomial has first-order terms in v , ζ_1 and ζ_3 : in this zoom it is divisible only by μ , and its quotient at $\mu = 0$ is ρ times a linear form in the off-plane offsets, which vanishes on the second arc. So no cell containing the end $\mu = 0$ of the second arc can pass the test. Whatever the ratio, the second arc meets some zoom domain; with $\rho_0 = 5/4 > 1$ it lies inside family A only, away from the boundary $\rho = \rho_0$ between the two families, and only cells of family A need the following treatment.

Handing cells over. The cells of family A along the second arc are covered by other zooms, built in the same way around the second arc instead of the first. The *sheared* offset $\theta' = \theta + e$ measures the displacement from the second arc. The sheared zooms $\mathcal{A}'_{G,F}$, for the base face G with $e > 0$ and every offset face F , are the zooms of family A with θ' in place of θ and the ratio $\rho' \in [0, \rho'_0]$, $\rho'_0 = 1/4$. Their distance coordinates vanish only on the second arc, which lies in the no-fit plane Π_σ . A cell of family A whose configurations, measured from the second arc, stay within the domains of the sheared zooms needs no witness of its own: it is *handed over*, and the checked cells of the sheared zooms take its place.

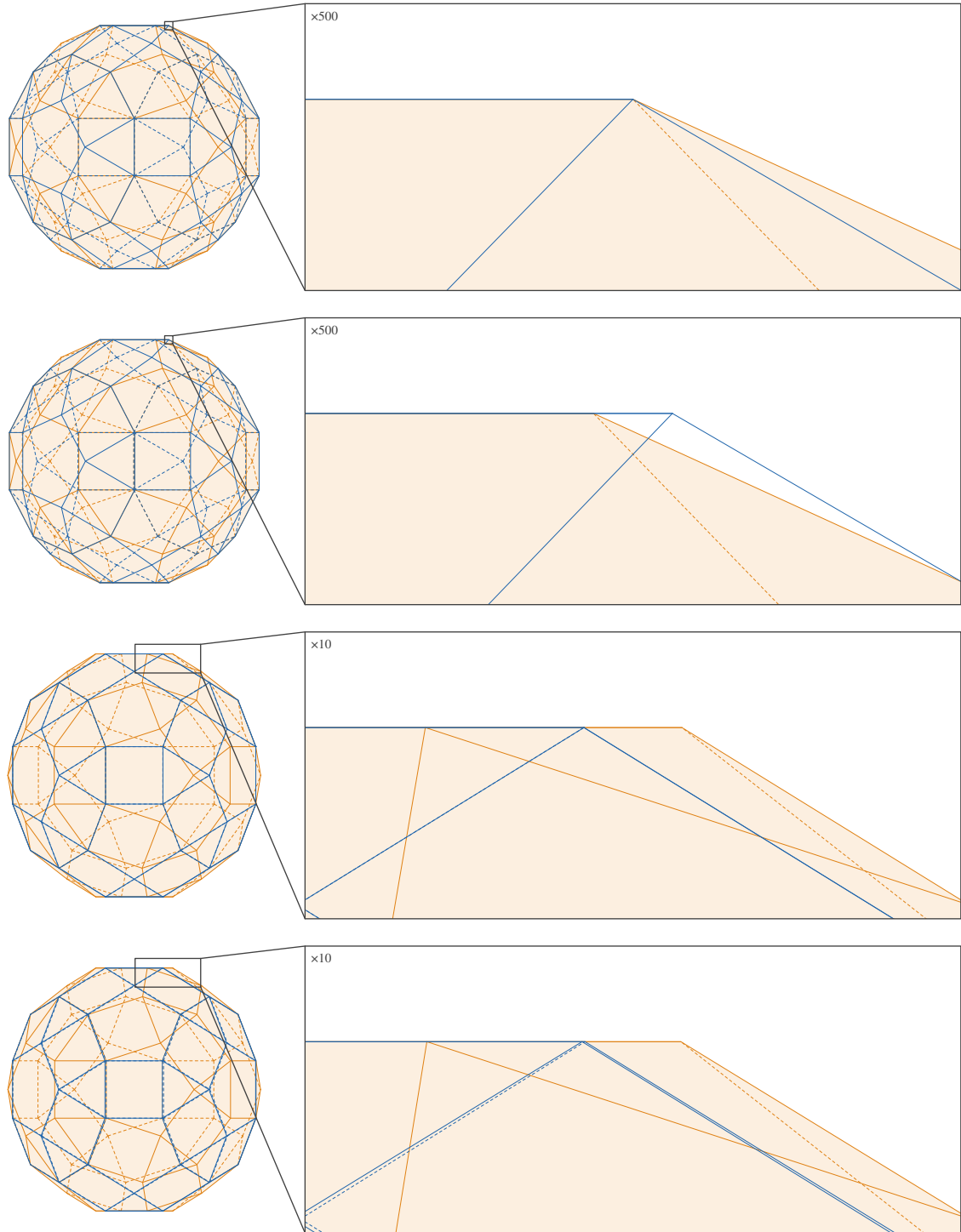


Figure 22: Configurations in the plane Π_1 , enlarged at the right ends of the two top edges as in Figure 19, 500 times for the first two and 10 times for the last two; the **hole** (orange) and the **plug** (blue) as in Figure 10. First: the endpoint $e = e_*$, where the ends of the two top edges coincide. Second: $e = e_* + 1/256$ on the continuation of the first arc, where the plug's top edge is longer than the hole's and its corner protrudes, a poke configuration. Third: the crossing $e = 0$. Fourth: the configuration $e = 1/200$, $\theta = -1/200$ of the second arc, which lies outside D .

To state this precisely, let $\mathcal{A}_{G,F}$ be a zoom of family A on the base face G with $e > 0$, so that $e = \mu$, and write

$$\mathbf{o}(\mathbf{y}) = \rho \hat{\mathbf{f}} + (0, 1, 0, 0)$$

for the offset measured from the second arc in units of μ : the shear adds $e = \mu$ to the second offset coordinate θ .

Lemma 10.2 (Hand-over). *Let C be a cell of a zoom $\mathcal{A}_{G,F}$ of family A, on the base face G with $e > 0$, such that $\mathbf{o}(\mathbf{y}) \in \rho'_0 \Phi$ for every $\mathbf{y} \in C$. Then every configuration in the image of C lies in the no-fit set or is the image of a point in the domain of one of the sheared zooms $\mathcal{A}'_{G,F'}$.*

Proof. At $\mathbf{y} \in C$, the base coordinate is μ and the sheared offset is $\mu \mathbf{o}(\mathbf{y})$. If $\mathbf{o}(\mathbf{y}) = \mathbf{0}$, the configuration lies on the second arc, in the no-fit set. Otherwise $\rho' = \|\mathbf{o}(\mathbf{y})\|_\infty \in (0, \rho'_0]$, and $\hat{\mathbf{f}}' = \mathbf{o}(\mathbf{y})/\rho'$ lies on a face F' of Φ , giving a point of the domain of $\mathcal{A}'_{G,F'}$ with the same configuration. $\square$

The hypothesis is checked exactly for each handed-over cell: each coordinate of $\mathbf{o}$ is the product of two zoom coordinates, plus a constant for the second, so its exact range over the cell follows from interval products. Cells are handed over only from family A to the sheared family, never back. The cover for $\sigma = 1$ hands 34 cells over, and the cover for $\sigma = -1$ hands over 14 (Figure 23).

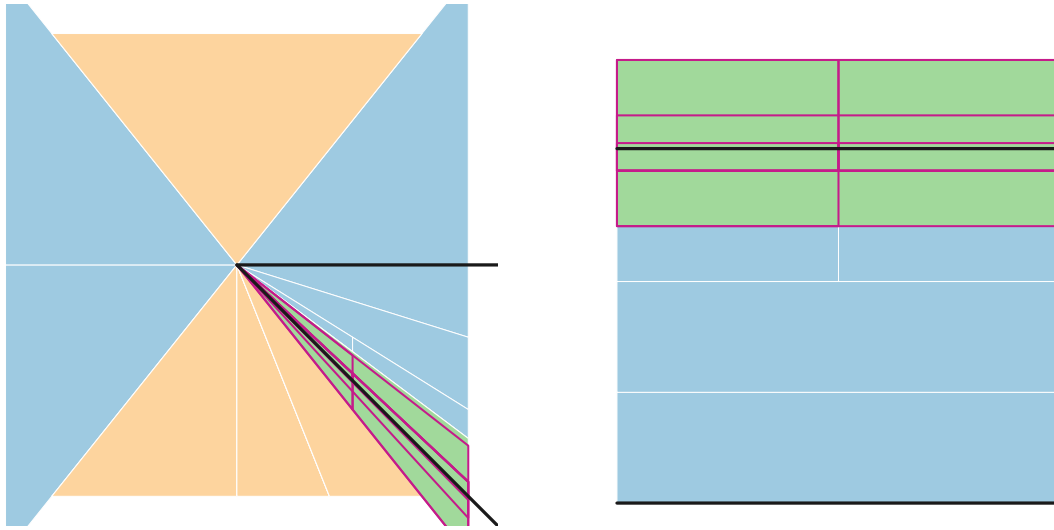


Figure 23: The crossing cover of Π_1 where it meets the plane itself ($v = \zeta_1 = \zeta_3 = 0$). Left: in the coordinates (e, θ) , both in $[-9/128, 9/128]$, the cells of family A (light blue; $|\theta| \leq \frac{5}{4}|e|$, cut by rays of constant ratio), of family B (light orange; $|e| \leq \frac{4}{5}|\theta|$) and of the sheared zooms around the second arc (green); the cells of family A handed over to the sheared zooms are outlined in magenta, and the first arc $\theta = 0$ and the second arc $\theta = -e$ are black. Right: the family A zoom on the base face $e = \mu > 0$ and the offset face $\theta = -1$, in its own coordinates ($20\mu \in [0, 5/4]$ horizontally, $\rho \in [0, 5/4]$ vertically), in which the first arc is $\rho = 0$ and the second arc $\rho = 1$ (black): its cells in the plane (light blue) and, along $\rho = 1$, the cells it hands over (green, outlined in magenta). Most of the refinement of this cover happens in v , ζ_1 and ζ_3 , so few cells meet the plane.

Proposition 10.3 (The crossing neighbourhoods). *For each $\sigma \in \{-1, 1\}$, no configuration of D in the covered set above is a fit.*

Proof. By Lemma 8.1, a configuration of the covered set lies in the no-fit set Π_σ , in a checked cell with a witness, or in a handed-over cell. The first two cases are as for Proposition 9.2. In the third, Lemma 10.2 places the configuration on the second arc or in the domain of a sheared zoom, whose cells are checked by Lemma 7.3. $\square$

10.3 The arcs together

The covers along the first arc overlap: the arc cover and the crossing cover share $3/64 \leq e \leq 1/16$, and the arc cover and the endpoint cover share $e_* - 1/16 \leq e \leq 3/16$. Every configuration of the first arc with $0 \leq e \leq e_*$ therefore lies in the interior, relative to B_0 , of the covered set of one cover, and so in the collection's range. The second arc lies outside D for $e > 0$ and meets the crossing cover at $e = 0$. Together with [Proposition 8.5](#), the collection's range contains a neighbourhood of every touch configuration of D that the subdivision revealed. That the collection eliminates every box is confirmed by the search itself, in [Section 11](#).

11 Generating and verifying the certificate

The components of the preceding sections are combined in a single search over B_0 . Its result is a certificate: one record for every eliminated box. The certificate is checked separately from the search that produced it, and this check, together with the mathematics of the preceding sections, proves [Theorem 1.1](#).

11.1 Complete coverage and the main theorem

The collection. The final collection has four components: the Domain component of [Section 4.3](#), the Global component of [Section 6](#), the Local component of [Section 7](#), and the Exotic component with the eight covers of [Sections 8 to 10](#). [Table 2](#) lists the 38 zoom covers. The Local and Exotic components work in the same way, by inclusion in a checked cover; we keep them apart because the Local covers form one uniform family, indexed by viewing rectangles, while each Exotic cover belongs to a particular configuration. The components are tried in the order Domain, Exotic, Local, Global, since the first three tests are fast and the Global component's test is the most expensive.

Cover	No-fit set	Zooms	Section	Zoom cells	Records
Local 0, ..., 29	aligned	tube (face zooms)	7.4	833	21,792
square	aligned	point	8.1	877	56
pentagon	aligned	point	8.2	6,173	315
arc $\pm$	$\Pi_{\pm 1}$	tube	9.2	343, 288	135, 68
endpoint $\pm$	$\Pi_{\pm 1}$	point	10.1	305, 246	55, 23
crossing $\pm$	$\Pi_{\pm 1}$	point, hand-over	10.2	1,431, 687	86, 58

Table 2: The zoom covers of the Local component (first row) and of the Exotic component, with the number of their terminal cells that carry witnesses and the number of certificate records naming them. The pairs are for $\sigma = 1$ and $\sigma = -1$. The parameters of every cover are listed in [Appendix B](#).

Records. A search box is named by its *path*: the sequence of choices, 0 for the lower half and 1 for the upper half, that leads to it from B_0 under the cyclic bisection of [Section 4.1](#). A certificate record consists of the path, the component that eliminated the box, and the data its test needs: the violated inequality of D for the Domain component, the hole edge and plug vertex for the Global component, and the zoom cover containing the box for the Local and Exotic components. For example, the record with path 11 names the viewing-triangle inequality: the box $[1/3, 2/3] \times [1/5, 2/5] \times [-2/5, 2/5]^3$ lies beyond the boundary $\varphi s + \varphi^2 t = 1$.

Checking a certificate. A certificate is accepted when two checks succeed.

1. *Coverage.* The paths must be the leaves of a finite binary tree rooted at B_0 : no path is a prefix of another, and every sequence of choices meets one of them after finitely many steps. The checker verifies this by reading the paths in order of increasing depth and, within a depth, in lexicographic order, while maintaining the list of boxes not yet accounted for; the list must be empty at the end.
2. *Records.* Each record's test is repeated exactly on its box: the Bernstein sign test of the named inequality, the Bernstein sign tests of the sixty gap polynomials, or the exact inclusion in the named cover's covered set. The covers themselves are rebuilt and all their cells checked, as described in [Appendix B](#), before any record is read.

Proposition 11.1 (Complete coverage). *If both checks succeed, the boxes of the records cover B_0 , and every box satisfies Equation (4.1).*

Proof. The children of a box cover it, so by induction on the tree, its leaves cover the root B_0 . Each record passed its component's test, and the components are sound: the Domain component by Section 4.3, the Global component by Proposition 6.2, the Local component by Proposition 7.5, and the Exotic component by Propositions 8.2, 8.4, 9.2, 10.1 and 10.3. $\square$

Proof of Theorem 1.1. The final certificate passes both checks. If the RID were Rupert, a fit would exist and, by Proposition 3.1, a fit would lie in $D \subseteq B_0$. By Proposition 11.1, it would lie in an eliminated box, contradicting Equation (4.1). $\square$

What the proof rests on. Unwinding the propositions cited above, the proof uses:

- the reduction to the domain D : Section 2 and Proposition 3.1, with the symmetry data of Appendix A;
- the Bernstein sign test (Lemma 5.1) and the exact arithmetic of Section 5;
- the Domain and Global tests (Section 4.3 and Proposition 6.2);
- the zoom lemma (Lemma 7.3) with its two kinds of witnesses, gap polynomials and domain polynomials;
- the two no-fit sets: the aligned configurations (Section 7.3) and the arc planes (Lemma 9.1);
- the coverage lemmas for tube zooms, point zooms and the hand-over (Lemmas 7.4, 8.1 and 10.2), which give the neighbourhood propositions of Sections 7 to 10;
- a correct implementation of the two checks above, for which the independent checker described below provides a second implementation, and of the cover checks of Appendix B, which that checker takes as given and which the program's tests re-verify with independent reimplementations.

The list of touch configurations, the analyses of why finitely many cells suffice, the floating-point proposals and the program that found the cells are not needed.

11.2 Implementation, execution, and reproducibility

The search and the checker form one program, written in Rust, which combines the performance needed for a search over hundreds of thousands of boxes with a type system that keeps exact numbers, polynomials, boxes and records strictly apart. Arbitrarily large integers and rationals come from the standard `num` libraries; numbers of $\mathbb{Q}(\sqrt{5})$, polynomials in five variables and their Bernstein coefficients are implemented in the program itself, as described in Section 5. Floating point is used only to choose which witness to try; every acceptance is decided exactly. The search processes boxes in a fixed order, so its certificate does not depend on the number of threads, and it can be interrupted and resumed from the certificate alone.

The final search, on 16 threads of a workstation, evaluated 385,391 boxes and wrote 192,696 records in about a minute; checking them takes about half a minute. Most records are Global records (157,458); the Local, Domain and Exotic components account for 21,792, 12,650 and 796. Figure 24 shows how the components share a slice of the configurations.

Testing and validation. The program is tested extensively, with unit, integration and adversarial tests of every part, including independent reimplementations of the exact arithmetic, the zooms and the covers. Two experiments validate the collection: every configuration of a list of 683 configurations chosen near the touch configurations of the preceding sections has a neighbourhood that the collection eliminates, and each component and cover is the only one to eliminate the boxes around some configuration, so none of them is redundant. Finally, an independent checker written in Python from the definitions alone re-verified every record of the final certificate and its coverage of B_0 , taking the cells of the zoom covers as checked by the program.

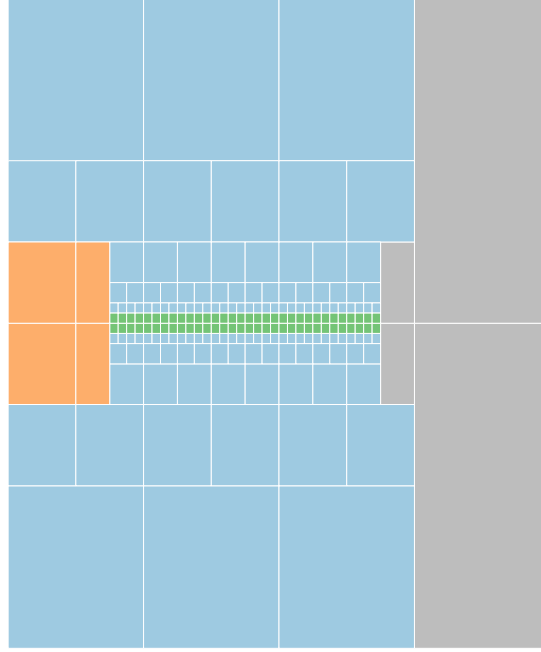


Figure 24: A slice of the final collection: $t = 1/10$ and $r_1 = r_2 = 0$, with $s \in [0, 2/3]$ horizontally and $r_3 \in [-2/5, 2/5]$ vertically, halved alternately until one component eliminates each piece, and coloured by that component: light blue Global, green Local, light orange the square cover, grey Domain (beyond the boundary of the viewing triangle). The Local component takes over in a thin band around $r_3 = 0$, where the configurations are nearly aligned. The search itself halves five-dimensional boxes.

Reproducibility. The program, its zoom cover data, the final certificate and the independent checker are available in the repository <https://github.com/bence-hervay/nopert-rid>. From its source directory, building the program and running its check on the certificate repeats every test of Section 11.1; running the search reproduces the certificate byte for byte. The SHA-256 hash of the final certificate is `df1f8dbab68b561f22fa9daeca48153d71bde7b6fba3345d9f6b7b8077df4b5f`.

12 Remarks and further directions

12.1 Relation to previous work

Most of the tools used here are established. Subdividing boxes of parameters until a sufficient condition holds on each is the classical branch and bound method. Exact arithmetic in a quadratic field, support separation of projected vertices, and the centring argument for centrally symmetric bodies are standard in this context; separation by a supporting line also underlies the global theorem of [Steininger and Yurkevich \[2026, Section 4, Theorem 17\]](#). Bernstein coefficients as bounds for the range of a multivariate polynomial on a box go back at least to [Garloff \[1986\]](#).

Their combination, however, is new in this context. The parametrisation turns every comparison into a polynomial inequality with coefficients in $\mathbb{Q}(\sqrt{5})$, so that no trigonometric function has to be bounded and every step can be checked exactly. Bernstein coefficients then serve as the single sign test of the whole proof, from the domain inequalities to the gap polynomials, and the zoom lemma extends that same test to the neighbourhoods of touch configurations. Together they make the entire argument a finite list of exact checks, which a certificate records and an independent program can repeat.

What the RID required, and what we supply, is a way to eliminate the neighbourhoods of touch configurations, where no strict inequality can hold and ordinary subdivision cannot finish. [Steininger and Yurkevich \[2026\]](#) proved the Noperthedron Nopert with a global and a local theorem, and identified an exotic viewing direction of the RID where their local theorem does not apply; they indicated that its neighbourhood could be treated with polynomial inequalities, and reported further difficult regions [[Steininger and Yurkevich, 2026, Section 9.1](#)]. Here the aligned configurations, the square and pentagon

configurations and two non-aligned families of touch configurations are all treated by one statement, [Lemma 7.3](#). The idea is to recognise sets of configurations that contain no fit for a simple reason, change coordinates so that some coordinates measure the distance from such a set, and divide a necessary inequality exactly by a power of those distances before testing its sign. This is reminiscent of blowing up a singular point in algebraic geometry, although we use nothing from that theory. The specific ingredients for the RID are the symmetry-reduced domain with its fold inequalities, the observation that the non-aligned touch families lie in planes where the plug reaches exactly as far as the hole in one screen direction, and the explicit zoom covers of [Appendix B. Theorem 1.1](#) settles the conjecture of [Steininger and Yurkevich \[2023\]](#) that the RID is not Rupert.

12.2 Extensions and possible automation

What transfers. The argument uses few properties of the RID beyond its exact vertex coordinates, its central symmetry and its symmetry group. For another centrally symmetric polyhedron with vertex coordinates in a real number field, the Global component applies unchanged, the Local component applies in the same form with new covers, and the Domain component needs that polyhedron’s own reduced domain. The Exotic component’s covers are specific to the RID, but their construction is not: each consists of a no-fit set, an affine change to base and offset coordinates, and the boxes and radii of a tube or point zoom. Much of the work is already automatic. The zoom cells and their witnesses were found by a program, the parameters of each cover amount to a few numbers, and exact arithmetic decides the equalities on which the no-fit sets rest. The step that required insight was finding the touch configurations and the no-fit sets containing them. A natural next step is to apply the method to other polyhedra conjectured to be Noperth.

Towards a decision procedure. We expect these techniques to lead to a practical algorithm that decides Rupert’s property for centrally symmetric polyhedra, once the special cases can be enumerated algorithmically. The missing ingredient is a finite procedure that finds, for a given polyhedron, every family of touch configurations together with a no-fit set containing it, and the points on these families where a second distance must be divided out. The aligned configurations are always touch configurations, and the exotic viewing directions of the Local component might be enumerated: if the component fails only where faces are seen edge-on, the candidates are finitely many intersections of great circles on the viewing sphere. Non-aligned touch families, such as the arcs of [Section 9](#), need a separate analysis, and we do not know whether they always lie in no-fit sets as simple as the arc planes. Given such an enumeration, the remaining steps are mechanical: placing zoom covers around the families, searching for cells and witnesses, and running the subdivision with the resulting collection, which would also detect a fit if one existed ([Section 6.2](#)). We expect this to be practical for polyhedra whose vertex coordinates are algebraic of low degree, whose symmetry group is large enough to make the reduced configuration space small, and which are regular in other respects, as the RID is; how far it extends is a question for further work.

The checks performed on the certificate are few and elementary, and together with the lemmas of [Sections 2, 3 and 5 to 10](#) they would be natural candidates for formalisation in a proof assistant.

12.3 History of this work

The author has worked on this problem since 2024, as a self-motivated project pursued in free time, over roughly 1,500 hours. Already in 2024, the author developed, independently, the two methods on which the proof for the Noperthedron by [Steininger and Yurkevich \[2026\]](#) rests: the global separation theorem, which became the Global component, and a version of their local theorem around aligned configurations, which became the Local component. These first versions used an angular parametrisation with transcendental functions, less practical than the one used here, together with interval arithmetic. The author also knew at that stage about the exotic configurations of the RID, where neither method applies, and suspected that a polyhedron could be constructed for which the two components alone would suffice. Nevertheless, the author chose to pursue the case of the RID itself, whose exotic configurations these methods proved hard to reach. In the meantime, [Steininger and Yurkevich \[2026\]](#) constructed the Noperthedron, the first polyhedron proved to be Noperth.

The next approach treated each exotic case with its own collection of polynomial inequalities, designed around the geometric features of that case. It was sufficient, but considerably more complicated: the proof was far more involved, and generating the full certificate required thousands of times as much

computation as the proof presented here. The zoom lemma of [Section 7.3](#) then unified all these cases, simplifying both the theoretical foundations of the proof and its implementation. It also contains the earlier components: the Local component is its simplest instance with a distance divided out, and the Domain and Global tests are the same idea with nothing divided out.

To complete the work without further delay, AI tools were used to produce a well-tested implementation of the final pipeline, to simplify the author's research pipelines, which had grown to more than 100,000 lines of code over the years, and to help write this article. The ideas, the mathematical arguments and the design of the pipeline are the author's.

A Symmetry data and checks

This appendix gives the finite symmetry data and calculations used in [Section 3](#). The symmetry actions, representative selection and resulting domain inequalities are stated and explained there. All finite comparisons below use exact arithmetic in $\mathbb{Q}(\sqrt{5})$.

A.1 Quaternion families and the full symmetry group

The set $\mathcal{Q}$ defined in [Section 3.1](#) consists of the unit quaternions in these three families:

$$\begin{array}{ll} \text{all coordinate permutations of } (\pm 1, 0, 0, 0), & 8, \\ \frac{1}{2}(\pm 1, \pm 1, \pm 1, \pm 1), & 16, \\ \text{even coordinate permutations of } \frac{1}{2}(0, \pm 1, \pm \varphi, \pm(-1 + \varphi)), & 96. \end{array}$$

Signs are chosen independently. There are 120 quaternions altogether: the unit icosians, which form the binary icosahedral group [\[Conway and Smith, 2003\]](#). The two quaternions in each pair $g, -g$ represent the same rotation. Composing this rotation with central inversion gives an orientation-reversing symmetry. The 60 quaternion pairs therefore give 60 rotations in $\mathcal{G}$ and 60 orientation-reversing symmetries in $-\mathcal{G}$.

The finite calculations behind this description use the exact vertex coordinates from [Equation \(2.1\)](#). They verify that every listed quaternion has norm one, that the represented rotations are distinct up to quaternion sign and closed under composition, and that each takes every vertex to another vertex. All these checks use exact arithmetic in $\mathbb{Q}(\sqrt{5})$.

Completeness of the symmetry list. To see that there are no further body symmetries, fix an ordered triangle formed by three vertices at pairwise distance two. Exact enumeration gives 20 unordered triangles, with each vertex belonging to exactly one, and verifies that the three position vectors of each triangle are linearly independent. A body symmetry must take the chosen triangle to one of them. There are 60 choices for its first image vertex and two for its second; the third is then fixed. The three independent images determine the linear map, so there are at most $60 \cdot 2 = 120$ orthogonal body symmetries. The 60 rotations and their products with central inversion attain this bound. Of the 60 orientation-reversing symmetries, 15 are reflections; the whole group is the full icosahedral group, isomorphic to $A_5 \times \{\pm 1\}$.

A.2 Reflection normals and the viewing cone

The fifteen normal lines come from the quaternions in $\mathcal{Q}$ whose scalar part is zero. Orient their unit normals $\mathbf{n}$ toward $\mathbf{h} = (1, 2, 9)$ as in [Section 3.2](#). The intersection of their closed halfspaces $\mathbf{n} \cdot \mathbf{u} \geq 0$ is the cone in [Equation \(3.1\)](#). This identification needs only two finite checks on the listed normals. For the inclusion of the intersection in the cone, note that the three planes bounding the cone are among the fifteen, with inward normals proportional to $(1, 0, 0)$, $(0, 1, 0)$ and $(-\varphi, -\varphi^2, 1)$. For the reverse inclusion, every oriented normal has nonnegative dot product with each of the three vectors

$$(0, 0, 1), \quad (1, 0, \varphi), \quad (0, 1, \varphi^2).$$

These vectors generate the whole cone: a vector satisfying [Equation \(3.1\)](#) can be written as

$$\mathbf{u} = u_1(1, 0, \varphi) + u_2(0, 1, \varphi^2) + (u_3 - \varphi u_1 - \varphi^2 u_2)(0, 0, 1),$$

with all coefficients nonnegative, so every vector of the cone lies in all fifteen halfspaces. The normal orientations and these dot products are checked exactly in $\mathbb{Q}(\sqrt{5})$.

A.3 Selected quaternions for the rotation comparisons

The representative $(1, \mathbf{r})$ has the largest scalar part among its right multiples by elements of $\mathcal{Q}$, which gives the inequality

$$\text{scalar}((1, \mathbf{r})g) \leq 1$$

for every $g \in \mathcal{Q}$. The domain D uses only twelve of them, those in Equation (3.6); the others hold as well, but D needs only to contain a representative of every configuration. Use cyclic subscripts, with $r_4 = r_1$. For each cyclic pair $i, i+1$ and signs $\epsilon, \delta \in \{-1, 1\}$, $\mathcal{Q}$ contains a quaternion with scalar part $\varphi/2$ and vector coordinates $-\epsilon/2$ and $-(-1 + \varphi)\delta/2$ in those two positions, with the remaining coordinate zero. Substituting these twelve quaternions gives precisely Equation (3.6).

B Zoom covers

The Local and Exotic components use 38 zoom covers: 30 around the aligned configurations and eight around the square and pentagon configurations and the first arc. Each is specified by a small set of parameters, fixed in the program's source, from which its zooms are built: the coordinate system (the configuration parameters, or the arc coordinates Equation (9.3) of one plane), an invertible affine map to base and offset coordinates, the boxes Σ and Φ , the radii and, for a point zoom cover, the ratio ρ_0 . The two tables below list these parameters; together with the coordinates given in the text, they determine every cover. The terminal cells and their witnesses are stored in one data file per cover. Before any search, every cover is rebuilt from its parameters and checked: the cells of each zoom must tile its domain exactly, every cell must satisfy the hypotheses of Lemma 7.3, and every handed-over cell must satisfy the hypothesis of Lemma 10.2. The covered sets are then those of Lemmas 7.4 and 8.1, except that the pentagon cover drops the upper bound $\xi \leq 0$, beyond which no configuration lies in D (Section 8.2). The checks take a few seconds; the data files were produced by the program's generator (`rid generate`), which is not part of the proof: every cell is checked exactly before use.

Each Local cover consists of the six face zooms of Section 7.4 over one viewing rectangle W : its base coordinates are $(s, t) \in W$, its offset is $\mathbf{r}$ with $\Phi = [-1, 1]^3$, and its offset radius is $\delta_{\max}$. The rectangles of Table 3 have disjoint interiors; each cover's rectangle W is the row enlarged by $1/1024$ on every side and cut to $W_0 = [0, 2/3] \times [0, 2/5]$, so that neighbouring covers overlap. The rows do not tile W_0 : the enlarged rows leave gaps around the square and pentagon configurations, which those covers fill, and the parts of W_0 beyond the viewing triangle need no cover, since the Domain component eliminates them. Figure 25 shows the rows with the square and pentagon covers. That the covers of aligned configurations together reach every viewing direction of the triangle is the computation of Section 8.3.

Cover	s from	s to	t from	t to	$\delta_{\max}$	Cells
0	1/12	1/6	0	1/10	1/96	24
1	0	1/6	1/10	1/5	1/80	45
2	1/6	1/3	0	1/10	1/50	26
3	1/6	1/4	1/10	1/5	1/50	27
4	1/4	1/3	1/10	3/20	1/50	16
5	1/4	1/3	3/20	1/5	1/50	20
6	0	1/12	1/5	3/10	1/50	18
7	1/12	1/8	1/5	9/40	1/50	27
8	1/12	1/8	9/40	1/4	1/50	15
9	1/8	1/6	1/5	1/4	1/200	34
10	1/12	1/8	1/4	3/10	1/50	15
11	1/8	7/48	1/4	11/40	1/50	12
12	7/48	1/6	1/4	21/80	1/200	49
13	7/48	5/32	21/80	11/40	1/50	12
14	5/32	31/192	21/80	43/160	1/200	45
15	1/8	1/6	11/40	3/10	1/50	15
16	0	1/6	3/10	2/5	1/50	39
17	1/6	5/24	1/5	9/40	1/50	22
18	1/6	3/16	9/40	1/4	1/200	19
19	3/16	5/24	9/40	1/4	1/50	67
20	5/24	1/4	1/5	9/40	1/50	17
21	5/24	1/4	9/40	1/4	1/200	25
22	1/6	3/16	1/4	21/80	1/200	31
23	3/16	5/24	1/4	21/80	1/200	35
24	5/24	11/48	1/4	11/40	1/200	17
25	1/4	1/3	1/5	1/4	1/50	29
26	1/3	1/2	0	1/10	1/50	19
27	1/3	1/2	1/10	1/5	1/50	17
28	1/2	7/12	0	1/10	1/50	11
29	7/12	5/8	0	1/20	1/20	85

Table 3: The 30 covers of the Local component, each consisting of the six face zooms over a viewing rectangle. The table gives the rectangles before enlargement; each cover's rectangle is the row enlarged by $1/1024$ on every side and cut to $[0, 2/3] \times [0, 2/5]$. All cells use gap polynomials, 833 in all.

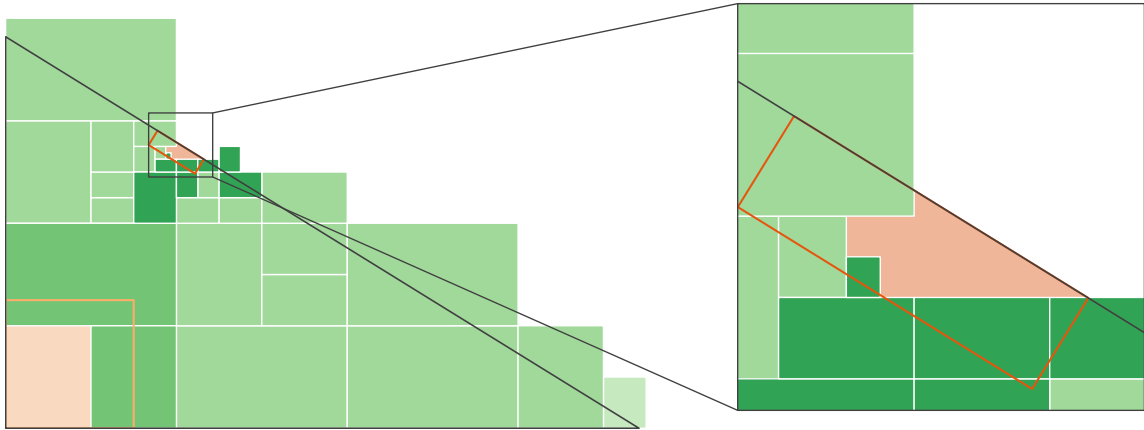


Figure 25: The viewing parameters of the covers around aligned configurations, $s \in [0, 2/3]$ horizontally and $t \in [0, 2/5]$ vertically, with the viewing triangle outlined: the 30 rows of Table 3 before enlargement, shaded by their radius (lightest $1/20$, then $1/50$, $1/80$ and $1/96$, darkest $1/200$); the square cover's views $[0, 1/8]^2$ (light orange); and the pentagon cover's parallelogram $|\eta| \leq 1/32, -1/32 \leq \xi \leq 0$ (red-orange). The square and pentagon covers, each a single region, are lightly tinted beneath the rows, which overlap them, and outlined on top. Right: the neighbourhood of the pentagon's viewing parameters, enlarged, where the rows approach the edge-on line in steps.

Cover	Kind	Base; offset	Parameters	Zooms	Cells	Domain	Handed
square	point	$(s, t), \Sigma = [0, 1]^2; \mathbf{r}$	$\mu_{\max} = \delta_{\max} = \frac{1}{8}, \rho_0 = 1$	18	877	95	–
pentagon	point	$(\eta, \xi), \Sigma = [-1, 1] \times [-1, 0]; \mathbf{r}$	$\mu_{\max} = \frac{1}{32}, \delta_{\max} = \frac{1}{16}, \rho_0 = 2$	24	6,173	0	–
arc $\pm$	tube	$e \in [\frac{3}{64}, \frac{3}{16}]; (\frac{v}{2}, \theta, \zeta_1, \zeta_3)$	$\delta_{\max} = \frac{1}{32}$	7	343, 288	32, 72	–
endpoint $\pm$	point	$e - e_*; (\frac{v}{2}, \theta, \zeta_1, \zeta_3)$	$\mu_{\max} = \frac{1}{16}, \frac{1}{32}, \delta_{\max} = \frac{1}{32}, \rho_0 = 1$	21	305, 246	10, 50	–
crossing $\pm$	point	$e; (\frac{4v}{5}, \theta, \zeta_1, \zeta_3)$	$\mu_{\max} = \delta_{\max} = \frac{1}{16}, \rho_0 = \frac{5}{4}, \rho'_0 = \frac{1}{4}$	28	1,431, 687	31, 160	34, 14

Table 4: The eight covers of the Exotic component. The offset box is $\Phi = [-1, 1]^3$ for the square and pentagon and $\Phi = [0, 1] \times [-1, 1]^3$ along the arcs; the pairs are for $\sigma = 1$ and $\sigma = -1$, and the endpoint's two base radii are for $e < e_*$ and $e > e_*$. *Zooms* counts the zooms of one cover, including the seven sheared zooms of each crossing cover; *Cells* counts the terminal cells with a witness, and *Domain* those among them whose witness is an inequality of D (the fold for $g = k$ at the square, the tie inequality along the arcs). *Handed* counts the cells handed over by [Lemma 10.2](#).

References

- Douglas Adams. *The Hitchhiker's Guide to the Galaxy*. Pan Books, London, 1979. URL <https://search.worldcat.org/title/The-hitch-hiker%27s-guide-to-the-galaxy/oclc/856740544>.
- András Bezdek, Zhenyue Guan, Mihály Hujter, and Antal Joós. Cubes and boxes have Rupert's passages in every nontrivial direction. *The American Mathematical Monthly*, 128(6):534–542, 2021. doi: [10.1080/00029890.2021.1901461](https://doi.org/10.1080/00029890.2021.1901461).
- Ying Chai, Liping Yuan, and Tudor Zamfirescu. Rupert property of Archimedean solids. *The American Mathematical Monthly*, 125(6):497–504, 2018. doi: [10.1080/00029890.2018.1449505](https://doi.org/10.1080/00029890.2018.1449505).
- John H. Conway and Derek A. Smith. *On Quaternions and Octonions: Their Geometry, Arithmetic, and Symmetry*. A K Peters, Natick, MA, 2003.
- Albin Fredriksson. Optimizing for the Rupert property. *The American Mathematical Monthly*, 131(3): 255–261, 2024. doi: [10.1080/00029890.2023.2285200](https://doi.org/10.1080/00029890.2023.2285200).
- J. Garloff. Convergent bounds for the range of multivariate polynomials. In Karl Nickel, editor, *Interval Mathematics 1985*, volume 212 of *Lecture Notes in Computer Science*, pages 37–56. Springer, Berlin, 1986. doi: [10.1007/3-540-16437-5_5](https://doi.org/10.1007/3-540-16437-5_5). URL https://doi.org/10.1007/3-540-16437-5_5.
- Raj Gosain and Benjamin Grimmer. Some new insights from highly optimized polyhedral passages. arXiv:2509.08190v1 [math.OC], 2025. URL <https://doi.org/10.48550/arXiv.2509.08190>. Version 1, 9 September 2025.
- George W. Hart and Henri Picciotto. *Zome Geometry: Hands-on Learning with Zome Models*. Key Curriculum Press, Emeryville, CA, 2001. ISBN 978-1-55953-385-0. URL <https://www.zometool.com/content/ZomeGeometryText.pdf>.
- Richard P. Jerrard, John E. Wetzel, and Liping Yuan. Platonic passages. *Mathematics Magazine*, 90(2): 87–98, 2017. doi: [10.4169/math.mag.90.2.87](https://doi.org/10.4169/math.mag.90.2.87).
- David I. McCooey. Rhombicosidodecahedron. Visual Polyhedra, n.d. URL <https://dmccooey.com/polyhedra/Rhombicosidodecahedron.html>. Accessed 23 September 2026.
- Tom Murphy, VII. Some upsetting things about shapes (that we already knew). In *A Record of the Proceedings of SIGBOVIK 2025*, pages 342–368. Association for Computational Heresy, 2025. URL <https://sigbovik.org/2025/proceedings.pdf#page=346>.
- David Renshaw. Surprising news about potential Nopert 214. YouTube video, 2026a. URL <https://www.youtube.com/watch?v=XFeKRSN9l9w>. Published 16 September 2026.
- David Renshaw. Rupert.lean: Formalization of the Rupert problem for convex polyhedra. Software repository, 2026b. URL <https://github.com/dwrensha/Rupert.lean/tree/fb5226466d9c21edb7b755ba8302210db77a5cf9>. Commit fb5226466d9c (2026-09-10); accessed 23 September 2026.
- Jakob Steininger and Sergey Yurkevich. An algorithmic approach to Rupert's problem. *Mathematics of Computation*, 92(342):1905–1929, 2023. doi: [10.1090/mcom/3831](https://doi.org/10.1090/mcom/3831).
- Jakob Steininger and Sergey Yurkevich. A convex polyhedron without Rupert's property. arXiv:2508.18475v2 [math.MG], 2026. URL <https://doi.org/10.48550/arXiv.2508.18475>. Version 2, 28 January 2026.
- Jan Hendrik van Swinden. *Grondbeginfels der meetkunde*. Pieter den Hengst en Zoon, Amsterdam, second edition, 1816. doi: [10.3931/e-rara-60795](https://doi.org/10.3931/e-rara-60795). DOI identifies the ETH-Bibliothek digital facsimile.
- John Wallis. *De Algebra Tractatus; Historicus & Practicus*, volume 2 of *Opera Mathematica*. Theatrum Sheldonianum, Oxoniae, 1693. URL <https://digitale-sammlungen.gwl.de/resolve?PPN=851708439>.